Advanced Macroeconomics

Business Cycle Theory

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Chapter 1

Solution and Numerical Implementation of DSGE Models

1.1 Introduction to DSGE Models

A DSGE model is defined by three properties. **Dynamic**: agents solve intertemporal optimization problems. **Stochastic**: the economy is subject to unforeseen shocks. **General Equilibrium**: the model accounts for equilibrium in all markets simultaneously, as opposed to partial equilibrium models. In a competitive equilibrium, all market participants have maximized their objective functions and all markets clear.

1.1.1 How to Solve a DSGE Model

1. **Solve the maximization problems** of all agents in the economy and derive the resulting first-order conditions (FOCs) — by hand.

2. **Find the non-stochastic steady state**: impose all variables to be constant and all shocks to be zero — by hand or in Dynare.

3. **Log-linearize** the equilibrium conditions (FOCs and constraints) around the steady state — by hand or in Dynare.

4. **Solve the system of log-linearized conditions** — by hand or in Dynare.

5. **Analyze the model solution** via impulse response functions (IRFs), stochastic simulations, or by inspecting the derived analytical solutions.

Working with FOCs and Constraints Once we have solved the maximization problems and derived the first-order conditions (FOCs), we collect all technological constraints,

budget constraints, and resource constraints. By Walras' Law, only $n-1$ of the n budget/resource constraints are independent, so one is dropped. Finally, we specify the laws of motion of the exogenous variables (shock processes).

Steady State The non-stochastic steady state describes the state of the economy in the absence of shocks. To find it, we remove all time subscripts and set all shocks to zero, then solve the resulting static system of equations, expressing all endogenous variables as functions of model parameters alone.

Model Solution After collecting all equilibrium conditions, we log-linearize around the steady state. This yields a linear approximation that describes how the system responds whenever a shock perturbs it away from steady state. The log-linearized system can be written in state-space form as a second-order stochastic difference equation in s_t:

$$\Gamma \mathbb{E}_t s_{t+1} = \Phi s_t + \Psi s_{t-1} + \Omega \varepsilon_{t+1}, \tag{1.1}$$

where $\mathbb{E}_t$ denotes the expectations operator, s_t is the vector of all model variables, ε_{t+1} is the vector of structural shocks, and Γ, Φ, Ψ, Ω are coefficient matrices.

Reduction to a First-Order System Working directly with (1.1) is cumbersome. We therefore convert it into a first-order stochastic difference equation, which can be solved using standard linear-algebraic techniques. Following Blanchard and Kahn (1980), we construct the **stacked (Blanchard-Kahn) vector**

$$\mu_t \equiv \begin{bmatrix} s_{t-1} \\ s_t \end{bmatrix}, \tag{1.2}$$

where s_{t-1} collects the *predetermined* (backward-looking) variables and s_t collects the *non-predetermined* (forward-looking) variables. Accordingly, $\mu_{t+1} = \begin{bmatrix} s_t \\ s_{t+1} \end{bmatrix}$.

We can then rewrite (1.1) as:

$$A_0 \, \mathbb{E}_t \mu_{t+1} = A_1 \mu_t + B_0 \varepsilon_{t+1}, \tag{1.3}$$

where the matrices are defined by the Blanchard-Kahn construction:

$$A_0 = \begin{bmatrix} 0 & I \\ \Gamma & 0 \end{bmatrix}, \qquad A_1 = \begin{bmatrix} \Psi & \Phi \\ 0 & I \end{bmatrix}, \qquad B_0 = \begin{bmatrix} \Omega \\ 0 \end{bmatrix}. \tag{1.4}$$

Provided A_0 is invertible (i.e. has full rank), we can pre-multiply both sides by A_0^{-1} to

obtain the first-order system:

$$\boxed{\mathbb{E}_t \mu_{t+1} = A\mu_t + B\varepsilon_{t+1}}, \qquad A \equiv A_0^{-1}A_1, \quad B \equiv A_0^{-1}B_0. \tag{1.5}$$

This is a standard first-order stochastic difference equation in the stacked vector μ_t, which we solve using the Blanchard-Kahn method:

1. Partition μ_t into predetermined variables x_t and non-predetermined variables y_t.
2. Compute the Jordan decomposition $A = P\Lambda P^{-1}$, where Λ is the diagonal matrix of eigenvalues. This accounts for the possibility of repeated eigenvalues ($\lambda_1 = \lambda_2$).
3. Rearrange the problem into decoupled *unstable* and *stable* equations.
4. Combine the solutions of the unstable and stable subsystems.

The key requirement for a unique, non-explosive equilibrium is the **Blanchard-Kahn condition**: the number of eigenvalues of A outside the unit circle must equal exactly the number of non-predetermined variables in y_t.

1.1.2 Example: The New Keynesian Model

To make the preceding machinery concrete, we work through the three-equation New Keynesian model in full detail.

Step 1: The equilibrium conditions (FOCs). After solving the household and firm problems and substituting the Taylor rule into the Dynamic IS equation, the equilibrium of the model reduces to two log-linearised conditions and one shock process:

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma}\Big(\phi_\pi \pi_t + \phi_y \tilde{y}_t + v_t - \mathbb{E}_t \pi_{t+1}\Big), \tag{DIS}$$

$$\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t, \tag{NKPC}$$

$$v_t = \rho_v v_{t-1} + \varepsilon_t. \tag{shock}$$

Step 2: Classify every variable. Before writing the system in matrix form, we must ask of each variable: *is its value at time t already fixed before t arrives, or is it freely determined at t?*

- v_t — **predetermined.** It is fully pinned down by the past value v_{t-1} and the shock ε_t that hits at t. No agent chooses v_t; it simply arrives. Its initial condition is given by history and cannot be freely set.
- π_t — **non-predetermined.** From the NKPC, $\pi_t = \beta\, \mathbb{E}_t \pi_{t+1} + \kappa \tilde{y}_t$: inflation today depends on *expected future* inflation. It is forward-looking; no past decision pins down its value today.

- $\tilde{y}_t$ — **non-predetermined.** From the DIS, output today depends on expected future output. Also forward-looking.

We therefore have $q = 2$ non-predetermined variables, $(\tilde{y}_t,\, \pi_t)$, and one predetermined variable, v_t.

Step 3: State-space representation. Collecting the two FOCs and rearranging into the companion form $\mathbb{E}_t X_{t+1} = AX_t + Bv_t$, where $X_t \equiv (\tilde{y}_t,\, \pi_t)^\top$, yields the 2×2 matrix:

$$A = \Gamma_1^{-1}\Gamma_0, \qquad \Gamma_0 = \begin{bmatrix} 1 + \dfrac{\phi_y}{\sigma} & \dfrac{\phi_\pi}{\sigma} \\ -\kappa & 1 \end{bmatrix}, \qquad \Gamma_1 = \begin{bmatrix} 1 & \dfrac{1}{\sigma} \\ 0 & \beta \end{bmatrix}. \tag{1.6}$$

The two eigenvalues of A, call them λ_1 and λ_2, govern the dynamics.

Step 4: Stable and unstable equations. Via the Jordan decomposition $A = P\Lambda P^{-1}$, we change variables to $\tilde{X}_t = P^{-1}X_t$ and obtain two *decoupled* scalar equations:

$$\mathbb{E}_t \tilde{X}^{(i)}_{t+1} = \lambda_i\, \tilde{X}^{(i)}_t, \qquad i = 1, 2. \tag{1.7}$$

The general solution to each is $\tilde{X}^{(i)}_t = C_i\, \lambda_i^t$.

- If $|\lambda_i| < 1$: the mode is **stable** — perturbations in that direction vanish over time. ✓
- If $|\lambda_i| > 1$: the mode is **unstable** — perturbations grow without bound. The solution explodes.

Explosive paths violate the transversality condition (households cannot accumulate unbounded wealth; arbitrage pins down asset prices). Therefore, for every unstable eigenvalue we must impose $C_i = 0$, i.e. we must set the initial condition of that mode to exactly zero.

Step 5: Who provides the free initial conditions? Imposing $C_i = 0$ is a *constraint on an initial condition.* But:

- The initial conditions of **predetermined** variables (here v_0) are fixed by history — we cannot touch them.
- The initial conditions of **non-predetermined** variables (here $\tilde{y}_0$ and π_0) are free — they are determined endogenously by the equilibrium itself.

Each non-predetermined variable therefore provides exactly *one free initial condition* that can be used to kill one explosive path.

Step 6: The Blanchard-Kahn condition. For a unique, non-explosive equilibrium we need to eliminate *all* explosive paths using *all* available free initial conditions — no more, no less:

$$\boxed{r = q,} \tag{1.8}$$

where r is the number of eigenvalues of A outside the unit circle and q is the number of non-predetermined variables. In the NK model, $q = 2$, so we need exactly $r = 2$ unstable eigenvalues of A.

The three cases and their economic interpretations are:

Condition	Equilibrium	Interpretation
$r = q$	Unique saddle path	Determinate: model has a unique solution
$r > q$	No stable solution	TVC always violated; only explosive paths
$r < q$	Multiple equilibria	Sunspot equilibria: beliefs are self-fulfilling

Step 7: The Taylor principle. Whether A has two unstable eigenvalues depends on the monetary policy parameters ϕ_π and ϕ_y. One can show (Bullard and Mitra, 2002) that the necessary and sufficient condition for $r = 2$ is:

$$(\phi_\pi - 1)\kappa + \phi_y(1 - \beta) > 0. \tag{1.9}$$

A sufficient condition is $\phi_\pi > 1$: the central bank raises the *nominal* rate by more than one-for-one with inflation, so the *real* rate rises, providing a stabilising force. This is the **Taylor principle**. If instead $\phi_\pi < 1$ (passive policy), one eigenvalue moves inside the unit circle, leaving one free initial condition unused: any initial belief about π_0 becomes self-fulfilling and the model admits infinitely many equilibria.

Chapter 2

Optimal Monetary Policy

2.1 Optimal Monetary Policy in the New Keynesian Model

2.1.1 Monetary Policy Trade-offs and the Cost-Push Shock

In the basic New Keynesian model studied in Chapter 4, monetary policy faces no genuine trade-off: stabilizing inflation is equivalent to stabilizing the output gap, a result known as the *divine coincidence* (Blanchard and Galí, 2007). This result rests crucially on the assumption that the flexible-price equilibrium is itself efficient—that is, the only source of distortion in the economy is nominal price stickiness.

Lecture 5 departs from this assumption by introducing *real rigidities*: distortions that persist even when prices are fully flexible. These include fluctuations in desired markups, distortionary taxation, and any form of market power not corrected by an optimal subsidy. As a result, the natural level of output y_t^n—the output that would prevail under flexible prices—is no longer efficient. A wedge opens between y_t^n and the *efficient* level of output y_t^e, which corresponds to the first-best allocation under a social planner.

This wedge enters the New Keynesian Phillips Curve (NKPC) as an exogenous disturbance:

$$\pi_t = \beta E_t\{\pi_{t+1}\} + \kappa x_t + u_t, \qquad u_t \equiv \kappa(y_t^e - y_t^n), \tag{2.1}$$

where $x_t \equiv y_t - y_t^e$ is now the welfare-relevant output gap (distance from *efficient*, not natural, output). The term u_t is referred to as a **cost-push shock**. The associated Dynamic IS curve also updates:

$$x_t = E_t\{x_{t+1}\} - \frac{1}{\sigma}(i_t - E_t\{\pi_{t+1}\} - r_t^e), \qquad r_t^e \equiv \rho + \sigma E_t\{\Delta y_{t+1}^e\}, \tag{2.2}$$

where r_t^e is the *efficient* real interest rate associated with $y_t = y_t^e$.

What is a cost-push shock?

A cost-push shock captures anything that raises firms' marginal costs of production independently of aggregate demand. The canonical real-world example is an oil price spike: higher energy costs raise production costs for virtually all firms, who respond by raising prices. At the same time, higher costs depress output. The economy experiences *stagflation*—simultaneously rising inflation and falling output.

A positive cost-push shock ($u_t > 0$) shifts the NKPC upward: for any given level of the output gap, inflation is higher. This means the central bank cannot simultaneously achieve $\pi_t = 0$ and $x_t = 0$—the divine coincidence breaks down. Any policy that reduces inflation must accept a negative output gap, and vice versa. The welfare loss function:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\pi_t^2 + \vartheta x_t^2\right], \qquad \vartheta = \frac{\kappa}{\varepsilon}, \tag{2.3}$$

makes this trade-off precise: the central bank must choose how to distribute the welfare cost between inflation and output gap fluctuations.

2.1.2 Optimal Policy under Discretion

Under **discretion**, the central bank re-optimizes every period, treating private sector expectations as given. Because it makes no binding promises about the future, it cannot influence $E_t\{\pi_{t+1}\}$: the private sector knows that whatever the central bank says today, it will re-optimize tomorrow. The central bank therefore solves a purely static problem each period.

The first-order conditions yield the *targeting rule*:

$$x_t = -\frac{\kappa}{\vartheta}\pi_t, \tag{2.4}$$

which prescribes *leaning against the wind*: whenever inflationary pressure arises, the central bank contracts output until the marginal welfare cost of doing so equals the marginal gain from lower inflation. Combined with the NKPC, the equilibrium responses to an i.i.d. cost-push shock are:

$$\pi_t = \frac{\vartheta}{\vartheta + \kappa^2}\, u_t, \qquad x_t = -\frac{\kappa}{\vartheta + \kappa^2}\, u_t. \tag{2.5}$$

Both variables respond proportionally to the shock and return immediately to zero the following period, since u_t is i.i.d. and future expectations remain anchored. The associated nominal interest rate can be written as a feedback rule on inflation:

$$i_t = r_t^e + \varphi_\pi \pi_t, \qquad \varphi_\pi \equiv \frac{\sigma\kappa}{\vartheta}, \tag{2.6}$$

which delivers a unique equilibrium provided $\varphi_\pi > 1$ (the Taylor principle), requiring

$\sigma\kappa > \vartheta$.

The full stabilization after one period appears desirable, but conceals an important inefficiency. Because the central bank cannot use expectations, all adjustment must happen through current output compression. This is the **stabilization bias**: from the period after the shock onwards the economy is completely stabilized, even though a better policy—involving less current contraction and some use of expectations—exists. Commitment exploits precisely this channel.

2.1.3 Optimal Policy under Commitment

Under **commitment**, the central bank solves a forward-looking problem: it chooses an entire future path for π_t and x_t to minimize expected lifetime welfare losses subject to the sequence of NKPC constraints. The Lagrangian is:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} \left(\pi_t^2 + \vartheta x_t^2 \right) + \xi_t (\pi_t - \beta \pi_{t+1} - \kappa x_t - u_t) \right], \tag{2.7}$$

where ξ_t are the Lagrange multipliers on each period's NKPC. The first-order conditions with respect to x_t and π_t yield, after combining:

$$x_0 = -\frac{\kappa}{\vartheta} \pi_0, \tag{2.8}$$

$$x_t - x_{t-1} = -\frac{\kappa}{\vartheta} \pi_t, \qquad t \geq 1. \tag{2.9}$$

The appearance of x_{t-1} in (2.9) is the defining feature of commitment policy: it introduces **history dependence**. Current policy depends not only on current inflation but on where the economy has been. An equivalent way to state this is that optimal commitment policy involves **price-level stationarity**: defining $\hat{p}_t \equiv p_t - p_{-1}$, the targeting rule becomes $x_t = -(\kappa/\vartheta)\hat{p}_t$, meaning the central bank stabilizes the *price level* around its pre-shock path rather than just the inflation rate.

The solution for the price level and output gap takes the form:

$$\hat{p}_t = \delta \hat{p}_{t-1} + \delta u_t, \qquad x_t = -\frac{\kappa\delta}{\vartheta} x_{t-1} - \frac{\kappa\delta}{\vartheta} u_t, \tag{2.10}$$

where $\delta \in (0, 1)$ is determined by the model parameters. After a positive cost-push shock, inflation rises moderately on impact but then turns negative as the central bank honors its promise of future tightening, returning the price level to its original path. The output gap falls gradually and persistently rather than sharply and briefly.

Why commitment dominates discretion

The intuition is straightforward. Under commitment, the announcement of future tightening immediately lowers inflation expectations. Since current inflation depends on expected future inflation via the NKPC, lower expectations reduce current inflation without requiring as large an output compression today. The private sector does part of the stabilization work. Under discretion, expectations are anchored at zero regardless—the private sector knows full stabilization arrives next period—so all adjustment must occur through current output. Commitment achieves better outcomes along both dimensions simultaneously: smaller inflation deviation and smaller output contraction on impact. Several small deviations are preferred to one large one under the quadratic loss.

2.1.4 Credibility, Time Inconsistency, and Inflation Bias

Time inconsistency

The superiority of commitment policy comes with a fundamental caveat: it is **time inconsistent**. Once the cost-push shock has died out, the promised future contraction is no longer optimal ex post—the central bank has an incentive to renege. This is visible directly in the first-order conditions: the optimal plan formulated at time t is not optimal when period $t+1$ arrives, because the ξ_{t-1} term that enforces history dependence no longer appears in the re-optimized problem.

This is the classical result of **Kydland and Prescott (1977)**: a policymaker who cannot credibly bind future actions is unable to exploit the expectations channel. The equilibrium outcome converges to the inferior discretionary solution. Institutional arrangements—central bank independence, explicit inflation targets, transparent forward guidance—exist precisely to provide commitment devices that make forward-looking policy credible.

Steady-state distortions and inflation bias

Up to this point, the analysis assumed that while $y_t^e \neq y_t^n$ at cyclical frequencies, the steady state itself is efficient. If instead a permanent distortion exists—for example, monopolistic competition without a corrective employment subsidy—a wedge $\Phi > 0$ opens between the steady-state marginal rate of substitution and the marginal product of labor:

$$\text{MRS} \equiv -\frac{U_n}{U_c} = (1-\Phi)\,\text{MPN}, \qquad 1-\Phi = \mathcal{M}^{-1} < 1. \tag{2.11}$$

This modifies the welfare function: linear terms in $\hat{x}_t \equiv x_t - \bar{x}$ no longer vanish (where $\bar{x} < 0$ measures the steady-state inefficiency gap), and the loss function becomes:

$$\mathcal{W} = -E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\varepsilon}{2\lambda} \pi_t^2 + \frac{1}{2} \left(\sigma + \frac{\alpha + \varphi}{1-\alpha} \right) \hat{x}_t^2 - \Lambda \hat{x}_t \right], \tag{2.12}$$

where $\Lambda > 0$ reflects the welfare gain from raising output above its steady-state level toward the efficient level. The central bank now has an incentive to push $\hat{x}_t$ above zero.

Under discretion, this incentive feeds into the private sector's inflation expectations and, in turn, into actual inflation. The result is a permanent **inflation bias**: even in the absence of any shock, inflation is systematically above target. This is the Barro–Gordon (1983) result: the equilibrium features higher inflation than is socially optimal, because the private sector correctly anticipates the central bank's temptation to stimulate output.

Under commitment, the central bank can internalize this bias. The inflation bias converges to zero asymptotically: by committing credibly to future policy restraint, the bank removes the incentive for the private sector to build the bias into expectations. This asymptotic convergence underscores again the value of credible institutional frameworks.

2.1.5 Inflation Targeting, Price-Level Targeting, and Average Inflation Targeting

The theoretical distinction between discretion and commitment maps directly onto real-world monetary policy frameworks.

Under **inflation targeting**, the central bank announces a specific quantitative target for the inflation rate (e.g., 2 percent per year). After a shock pushes inflation above target, the bank returns inflation to 2 percent—but the higher past inflation is forgiven. "Bygones are bygones." The price level drifts permanently upward, with no offsetting deflation required.

Under **price-level targeting**, the bank targets the path of the price level rather than the inflation rate. A temporary shock that raises the price level must be reversed: the bank commits to a period of below-target inflation to bring the price level back to trend. This is formally equivalent to the commitment solution: history dependence is built directly into the policy framework.

In 2020, the Federal Reserve adopted **average inflation targeting** (AIT), committing to make up for periods of below-target inflation with periods of above-target inflation, averaged over a horizon. Depending on implementation, AIT can approximate price-level targeting. Crucially, in 2020 AIT represented a commitment to more accommodative future policy, given a decade of below-target inflation. Whether this commitment contributed to the subsequent inflation surge of 2021–2023 depends on the averaging horizon and the degree of credibility attached to the announced framework.

2.1.6 Should the Central Bank Respond to Supply Shocks?

The model's prescription is unambiguous: yes, monetary policy should respond to cost-push shocks. The quadratic welfare loss penalizes large deviations heavily, so distributing the adjustment across both inflation and output is preferred to absorbing it entirely in either

one. Optimal policy under discretion engineers a mild recession after an inflationary supply shock.

This prescription became highly relevant during the 2021–2023 energy price surge. **Bandera et al. (2023)** argued that monetary policy should "see through" such shocks on the grounds that households would passively accept lower real wages and firms lower markups for as long as energy prices remained elevated. The NK model is skeptical: if instead workers seek wage compensation or firms pass costs forward, inaction allows inflation expectations to become unanchored, creating a far worse trade-off later.

Two extensions complicate the prescription in opposite directions. First, **non-rational inflation expectations** (adaptive or extrapolative) amplify cost-push shocks: agents expecting today's inflation to persist raise their wage demands, feeding a wage-price spiral. This strengthens the case for early and forceful tightening. Second, **heterogeneous households**—in particular, hand-to-mouth consumers with high marginal propensities to consume—already suffer a large real income loss from higher energy prices. A contractionary interest rate policy amplifies their distress, weakening the aggregate demand channel and the case for tightening.

The appropriate response therefore depends on the persistence of the shock, the anchoring of expectations, and the distributional consequences of policy. The textbook model provides the normative benchmark; practical policy must weigh these additional considerations.

2.1.7 Optimal Monetary Policy at the Zero Lower Bound

A separate but related application of the commitment framework arises at the **zero lower bound** (ZLB) on nominal interest rates. When the natural rate of interest is negative—as occurred during the 2008–2009 financial crisis and its aftermath—the central bank cannot lower the nominal rate sufficiently to close the output gap, even at $i_t = 0$.

Recall from the IS equation, iterated forward:

$$x_t = -\frac{1}{\sigma} E_t \sum_{j=0}^{\infty} (i_{t+j} - \pi_{t+j+1} - r^e_{t+j}). \tag{2.13}$$

Future real interest rates matter as much as current ones. A commitment to keeping nominal rates low for longer—even after the ZLB ceases to bind—therefore stimulates current demand through the expectations channel. This is the logic of *forward guidance*: "lower for longer" reduces expected future real rates, supporting current output and inflation.

Under discretion, the central bank provides no such promise: it will raise rates as soon as conditions normalize. The private sector anticipates this, and current output contracts severely. Under commitment, the bank credibly promises to keep rates low even after

liftoff conditions are met, delivering a far smaller recession. The mechanism is exactly the same as in the cost-push shock case: the expectations channel allows better outcomes along both the inflation and output dimensions simultaneously.

A notable empirical challenge is the **forward guidance puzzle**: in the NK model, forward guidance is extremely powerful—a commitment to low rates far in the future has enormous effects on current output. Empirically, the effects are much more modest. This discrepancy suggests either that agents discount the future more heavily than the model implies, or that credibility of distant commitments is limited in practice.

2.1.8 Summary

The central insights of the lecture can be organized as follows. Once real rigidities introduce cost-push shocks, the divine coincidence breaks down and monetary policy faces a genuine trade-off between inflation and output stabilization. Discretionary policy resolves this trade-off period by period, with no use of the expectations channel: the result is stabilization bias and, in the presence of steady-state distortions, a permanent inflation bias. Commitment policy exploits the expectations channel by announcing credible future plans: the result is smaller deviations of both inflation and the output gap, price-level stationarity, and the asymptotic elimination of the inflation bias. The gains from commitment are only achievable if the central bank has sufficient credibility—an institutional problem, not merely a technical one. The same logic applies at the zero lower bound, where forward guidance is the commitment equivalent for a constrained instrument. Real-world frameworks such as inflation targeting, price-level targeting, and average inflation targeting represent different points on the spectrum from pure discretion to full commitment.

Chapter 3

Application of SOE-NK Models

3.1 Introduction

In this chapter, we look at two applications of SOE-NK models in particular, we look at fiscal policy with rule-of-thumb households and tariffs.

3.2 Fiscal Policy in SOE-NK

3.2.1 Revisiting the Closed Economy

In a closed economy, output satisfies $Y_t = C_t + G_t$. When the government increases consumption so that $G_t > G_{ss}$, empirical evidence suggests this typically raises private consumption $C_t > C_{ss}$, generating fiscal multipliers above unity.

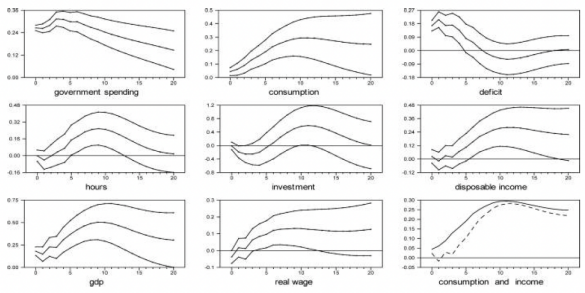


Figure 3.1: Impulse responses to a government spending shock. Source: Blanchard and Perotti (2002).

This finding is confirmed by Galí et al. (2007) among others, though with an important caveat: the consumption response tends to be modest, and some studies—notably Ramey (2011)—do not find a statistically significant increase at all.

Ricardian Households and the Limits of Fiscal Policy

Consider first the standard NK model with Ricardian households. Under perfect foresight, the household Euler equation is:

$$u'(C_t) = \beta(1 + r_t)\, u'(C_{t+1}).$$

Given r_t, this pins down C_t relative to C_{t+1}, and the entire consumption path is determined by the path of real interest rates.

To close the model, monetary policy must be specified. A standard Taylor-type rule sets the nominal rate as:

$$i_t = \bar{i} + \phi\pi_t,$$

where $\phi > 1$ ensures the Taylor principle holds and monetary policy is active. Instead of specifying a nominal rule, it is common in fiscal analysis to work directly with a *constant real rate rule* (Woodford 2011; Auclert et al. 2024):

$$r_t = r,$$

with the nominal rate then following residually from the Fisher equation $1 + i_t = (1 + r_t)(1+\pi_{t+1})$. This rule is the limiting case of the Taylor rule as $\phi \to 1^+$, and it is motivated on two grounds:

1. **Middle ground.** It lies between the zero lower bound ($\phi = 0$, where the CB cannot react) and active monetary policy ($\phi > 1$, where the CB aggressively offsets fiscal shocks). The real rate neither amplifies nor dampens fiscal transmission.

2. **Identification of pure fiscal effects.** By holding the real rate fixed, we isolate the transmission of fiscal policy itself from any endogenous monetary policy response.

Under the constant real rate rule, and normalising $\beta = 1/(1 + r)$, the Euler equation becomes:

$$u'(C_t) = u'(C_{t+1}),$$

which implies $C_t = C_{ss}$ for all t by stationarity. Consumption does not respond to the fiscal shock at all, and the fiscal multiplier is exactly unity: $dY_t = dG_t$.

Active Monetary Policy Makes Things Worse

With active monetary policy ($\phi > 1$), the situation deteriorates further. An increase in government spending raises inflation, which triggers a nominal—and real—interest rate increase. This induces Ricardian households to reduce current consumption in anticipation

of higher future returns, so that:

$$G_t \uparrow \implies \pi_t \uparrow \implies r_t \uparrow \implies C_t \downarrow .$$

The fiscal multiplier falls below unity. The underlying reason is Ricardian equivalence: a household receiving a transfer today, financed by taxes arbitrarily far in the future, will save the entire amount rather than consume it, rendering fiscal policy ineffective regardless of the horizon of repayment.

Rule-of-Thumb Households

The fundamental problem with Ricardian households is that they are fully forward-looking: any transfer today is saved to meet future tax obligations, leaving consumption unchanged. A tractable departure is to assume that a share $\lambda \in [0, 1]$ of households are *rule-of-thumb* (or hand-to-mouth).[1] They consume all of their current net income each period:

$$C_t^r = \begin{cases} w_t N_t - T_t & \text{(closed economy)} \\ \dfrac{P_{H,t}}{P_t} Y_t - T_t & \text{(open economy)} \end{cases}$$

In the open economy case, income is deflated by the CPI rather than the producer price index, since households consume the full consumption basket.

Aggregate consumption is a weighted average of the two types:

$$C_t = \lambda C_t^r + (1 - \lambda) C_t^o,$$

where C_t^o is the consumption of optimising (Ricardian) households, which continues to satisfy the Euler equation.

Closed-Economy Fiscal Multiplier

Under a constant real rate rule, $dC_t^o = 0$ as before. Linearising the rule-of-thumb consumption function gives:

$$dC_t^r = dY_t - dT_t,$$

so that aggregate consumption responds as:

$$dC_t = \lambda (dY_t - dT_t).$$

[1] These households have no access to financial markets—neither borrowing nor saving—and therefore cannot smooth consumption intertemporally.

Combined with goods market clearing $Y_t = C_t + G_t$, i.e. $dY_t = dC_t + dG_t$, we can solve for equilibrium output:

$$dY_t = \lambda(dY_t - dT_t) + dG_t \implies dY_t(1-\lambda) = dG_t - \lambda\, dT_t,$$

which yields:

$$\boxed{dY_t = \frac{dG_t - \lambda\, dT_t}{1-\lambda}.}$$

Whenever spending is at least partially deficit-financed ($dT_t < dG_t$), the multiplier exceeds unity. The mechanism is Keynesian: higher government spending raises current income, which raises consumption of rule-of-thumb households, which raises income further—a feedback loop whose strength is increasing in λ. Note that active monetary policy would dampen this through the interest rate channel, reducing $dC_t^o < 0$ and thus muting the overall multiplier.

3.2.2 Small Open Economy

We now turn to the small open economy case, following Druedahl et al. (2025), dropping the international risk-sharing assumption and allowing for household heterogeneity. The main result is Proposition 1, which generalises the closed-economy multiplier to the SOE:

$$dY = dG - (1-\alpha)M dT + (1-\alpha)R dr + (1-\alpha)M dY + \frac{2-\alpha}{1-\alpha}\alpha\eta\, dQ - \alpha M dQ,$$

where $dX = (dX_0, dX_1, \ldots)'$ denotes the full impulse response path of variable X, M is the matrix of intertemporal MPCs, and R is the matrix of consumption responses to real interest rates. Relative to the closed economy, three things change: channels 2–4 are attenuated by the home-bias factor $(1-\alpha)$, and two new channels driven by the real exchange rate Q appear.

Now it is important to define M in particular it represents the response of consumption to income, thus it is matrix of intertemporal MPCs. It is worth to notice that it is an average of the two types of households:

$$M = \lambda \underbrace{M^r}_{I} + (1-\lambda)\underbrace{M^o}_{0}$$

At the same time, R represents the matrix of responses of consumption to the interest rate: hand-to-mouth households have no access to financial markets, so $R^r = 0$, while Ricardian households respond normally, $R^o \neq 0$. In the open economy, the full output

decomposition is:

$$dY = \underbrace{dG}_{\text{1. Gov. cons.}} - \underbrace{(1-\alpha)M\,dT}_{\text{2. Taxes}} + \underbrace{(1-\alpha)R\,dr}_{\text{3. Interest rate}} + \underbrace{(1-\alpha)M\,dY}_{\text{4. Multiplier}}$$

$$+ \underbrace{\frac{2-\alpha}{1-\alpha}\,\alpha\eta\,dQ}_{\text{5. Exp. switching}} - \underbrace{\alpha M\,dQ}_{\text{6. Real income}} . \tag{3.1}$$

Under UIP and a fixed real interest rate, the real exchange rate remains constant, so $dQ = 0$ and channels 5 and 6 vanish. Under active monetary policy, however, a fiscal expansion triggers an interest rate rise that appreciates the exchange rate ($dQ < 0$). This activates two opposing forces: **expenditure switching** (domestic goods become relatively more expensive, reducing net exports and output) and a **real income effect** (cheaper imports raise households' purchasing power, boosting consumption and output). The net impact depends on the degree of openness α and the trade elasticity η.

The key difference between RANK and HANK lies in how these channels operate. In RANK, Ricardian households smooth consumption intertemporally, so the tax, multiplier, and real income channels are approximately zero. In HANK, hand-to-mouth households consume out of current income, amplifying all three channels. Table 3.1 summarises the sign and relative magnitude of each channel across the two frameworks.

Table 3.1: Fiscal transmission channels: RANK vs. HANK

	RANK		HANK
1. Government consumption	$+$	$=$	$+$
2. Taxes	≈ 0	$>$	$-$
3. Interest rate	$-$	$<$	$-$
4. Multiplier	≈ 0	$<$	$+$
5. Expenditure switching	$-$	$\leq$	$-$
6. Real income	≈ 0	$<$	$+$

Rows 2, 4, and 6 highlight the core HANK amplification: taxes reduce disposable income for hand-to-mouth households who immediately cut consumption (row 2 is negative and larger in magnitude), the spending multiplier is positive because current income feeds directly into demand (row 4), and the real income gain from exchange rate appreciation boosts consumption of constrained households (row 6). Rows 3 and 5 are negative in both models, but the interest rate channel is stronger in HANK because indebted households are more exposed to borrowing costs.

3.3 Tariffs

This section analyses the role of tariffs and their macroeconomic impact, with and without retaliation. The framework is close to **?** but departs from it in three key dimensions:

1. **Incomplete financial markets**: tariff shocks cannot be perfectly hedged across countries, so the balance of payments must be tracked explicitly.

2. **Imports used in the production of exports**: intermediate imports enter the production technology, capturing the role of global value chains and making the real effects of tariffs richer than in standard models.

3. **Downward nominal wage rigidity with flexible prices**: consistent with empirical evidence, this assumption implies full pass-through of tariffs to consumer prices, since firms can freely adjust prices while nominal wages cannot fall.

Because financial markets are incomplete, we must track the evolution of the net foreign asset position. Denoting by A_t net foreign assets, $\mathcal{E}_t$ the nominal exchange rate, i^* the world interest rate, and TB_t the trade balance, the balance-of-payments identity is:

$$A_t = \frac{\mathcal{E}_t}{\mathcal{E}_{t-1}}(1 + i^*)\, A_{t-1} + \mathcal{E}_t\, TB_t. \tag{3.2}$$

A key departure from the standard open-economy model is the production function, where imported intermediates M_t enter alongside labor N_t via a CES aggregator with elasticity η:

$$Y_t = \left[(1-\alpha)^{1/\eta} N_t^{\frac{\eta-1}{\eta}} + \alpha^{1/\eta} M_t^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \tag{3.3}$$

where $\alpha \in (0,1)$ governs the import intensity of production. A tariff on M_t therefore raises marginal costs directly, not only through expenditure switching.

Finally, the model imposes **downward nominal wage rigidity**[2]:

$$W_t \geq W_{t-1}. \tag{3.4}$$

Combined with flexible prices, this generates asymmetric dynamics: a tariff shock that would require a nominal wage cut instead produces unemployment, since wages are stuck at their previous level while firms reduce labor demand in response to higher costs.

Now we assume that foreign goods prices is given by $P_{f,t} = \varepsilon_t P_t^*$ we normalize $P_t^* = 1$ and asssume that the tariff $\tau_t > 0$ and impact foreign prices as $P_{f,t} = (1+\tau_t)\varepsilon_t$ so you have to pay more to import.

The paper explicitly distinguishes between the short-run and long-run effects of temporary tariffs. Monetary policy is assumed passive at period 0, $i_0 = i$, while at all other

[2] can rise but not fall

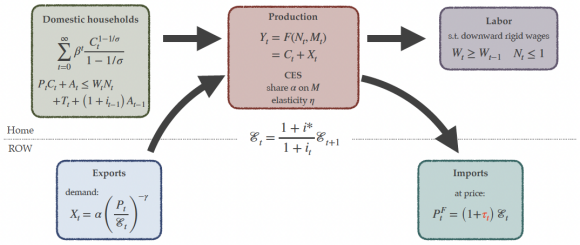


Figure 1: Summary of the spending flows in the baseline model

Figure 3.2

dates the central bank sets i_t to implement the full-employment allocation $N_t = \bar{N}$. The period-0 response of real GDP to a tariff shock $d\tau$ is:

$$d \log \text{GDP}_0 = -\alpha \left[(1-\alpha)\sigma + \alpha\gamma - \eta \right] d\tau, \tag{3.5}$$

where $\alpha \in (0,1)$ is the degree of openness, $\eta > 0$ is the elasticity of import substitution, $\gamma > 0$ is the elasticity of export demand, and $\sigma > 0$ is the elasticity of intertemporal substitution.

GDP falls on impact if and only if the bracket is positive:

$$\begin{aligned} d \log \text{GDP}_0 < 0 &\iff -\alpha \left[(1-\alpha)\sigma + \alpha\gamma - \eta \right] d\tau < 0 \\ &\iff (1-\alpha)\sigma + \alpha\gamma - \eta > 0 \\ &\iff (1-\alpha)\sigma + \alpha\gamma > \eta. \end{aligned}$$

The condition $(1-\alpha)\sigma + \alpha\gamma > \eta$ has a clear economic interpretation. The left-hand side captures the **demand compression** channels: intertemporal substitution away from domestic goods (weighted by $1-\alpha$) and the fall in export demand from trading partners (weighted by α). The right-hand side η captures the **import substitution** channel, through which higher tariffs redirect domestic spending toward home-produced goods, partially offsetting the contraction. GDP falls in the short run whenever the demand-reducing forces dominate import substitution. In highly open economies (large α) or with low substitutability (η small), the condition is more likely to hold and the recessionary effect of tariffs is stronger.

The condition $(1-\alpha)\sigma + \alpha\gamma > \eta$ has a clear economic interpretation. The left-hand side captures the **demand compression** channels: $(1-\alpha)\sigma$ reflects intertemporal substitution — when tariffs raise import prices today, agents postpone consumption to the future, depressing current demand — while $\alpha\gamma$ reflects export demand destruction, as trading partners reduce purchases of domestic goods. The right-hand side η captures the **expenditure switching** channel, through which higher tariffs redirect domestic spending toward home-produced goods, partially offsetting the contraction. GDP falls

in the short run whenever the combined effect of consumption postponement and export compression dominates expenditure switching, i.e. when agents prefer to delay rather than substitute domestically. Calibration reveals that tariffs are recessionary: a 10 percentage point increase in the import tariff reduces GDP by approximately 0.7%.

Alongside the output effect, tariffs alter the **trade balance**, defined as exports minus imports ($TB = X - M$). An **improvement** in the trade balance means the economy exports relatively more than it imports — net foreign demand for domestic goods rises. A **deterioration** means the opposite: the economy runs a larger deficit on its external account, importing more than it exports.

In the unilateral tariff case, the worsening of competitiveness caused by higher domestic prices reduces foreign demand for domestic exports. At the same time, the tariff makes imports more expensive, leading domestic agents to substitute toward home-produced goods and reduce import volumes. If the import compression effect dominates, the trade balance *improves* despite the loss of export competitiveness.

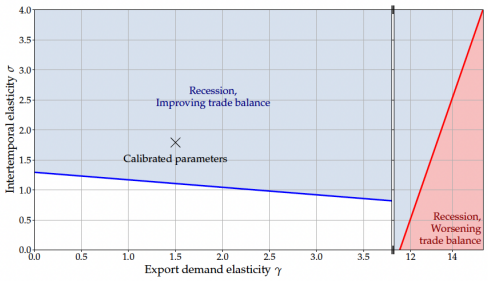


Figure 3: Unilateral tariff shock: Conditions for recession and improving trade balance

Figure 3.3: Impulse responses to a unilateral tariff shock.

Retaliation

When the foreign country retaliates by imposing a symmetric tariff, the trade balance dynamics reverse. Higher foreign tariffs directly reduce foreign demand for domestic exports ($X \downarrow$), while the domestic import compression channel remains active ($M \downarrow$). However, the export destruction effect now dominates, making a **deterioration** of the trade balance more likely: the economy loses foreign markets without a compensating gain in domestic demand. The result is a deeper recession than under unilateral tariffs, as both the income and the external demand channels contract simultaneously.

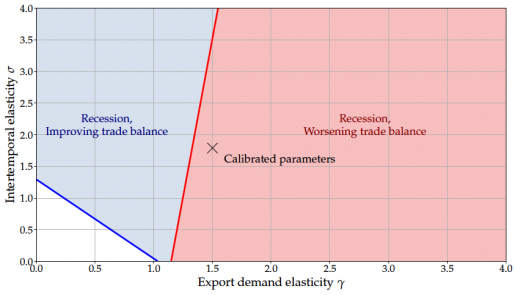


Figure 5: Retaliation: Conditions for recession and improving trade balance

Figure 3.4: Impulse responses to a tariff shock with foreign retaliation.

Chapter 4

Calibration and Estimation of DSGE Models

4.1 Introduction to Estimation of DSGE Models

Generally, DSGE Models are hard to identify and estimate for different reasons:

- **Likelihood Function too hard to optimise**
- **Large Number of Parameters and Few Variables**
- **Parameters that are hardly Identifiable using time series**
- **Model Specification**

4.2 Calibration

Calibration is one of the pioneering methods used to parametrize DSGE models. In the 1980s, RBC models were microfounded and their parameters set to replicate business cycle features using a data-driven quantitative approach (**??**).

Different sources of information can be used. Some parameters are taken directly from microeconometric studies, for example the coefficient of relative risk aversion or elasticity parameters. Proponents argue that since DSGE models are microfounded, micro-based estimates should transfer naturally. This is not always the case: the Frisch elasticity of labor supply is a clear counterexample, as macro models require values around 4 to match aggregate hours volatility, far above micro-estimates of 0.1–0.5. Other parameters are calibrated to long-run features of the data, for instance the discount factor is pinned down by the steady-state condition $r = \beta^{-1} - 1$, and the depreciation rate by matching the long-run capital-to-output ratio. Productivity shocks are recovered as the Solow residual and fitted to an AR(1) with $\rho_A \approx 0.95$ and $\sigma_\varepsilon \approx 0.007$.

RBC models calibrated in this way — using micro-evidence and long-run observations but no business-cycle statistics — proved able to quantitatively mimic the US business cycle, delivering strong out-of-sample performance.

- **Pros**: (i) does not require new empirical investigation; (ii) available information pins down parameter values or plausible intervals, leaving no free parameters to overfit the data (*"beware of economists with free parameters"* — Lucas); (iii) parameters retain realistic micro-level foundations.
- **Cons**: (i) combines estimates across different studies and contexts, which may be internally inconsistent; (ii) the model may match selected moments while failing others; crucially, calibration relies on **unconditional** moments, whereas **conditional** moments can have very different implications.

This is precisely where **Galí's (1999) critique** arises. Unconditional moments aggregate the effects of all shocks simultaneously. Denoting the variance of output as

$$\mathrm{Var}(y_t) = \sum_j \mathrm{Var}(y_t \mid \varepsilon_t^j), \tag{4.1}$$

the unconditional correlation between output and employment

$$\mathrm{Corr}(y_t, n_t) = \frac{\sum_j \mathrm{Cov}(y_t, n_t \mid \varepsilon_t^j)}{\sqrt{\mathrm{Var}(y_t)\,\mathrm{Var}(n_t)}} \tag{4.2}$$

is driven by whichever shock ε_t^j dominates the variance. If demand shocks dominate, the unconditional correlation will be positive even if the conditional correlation $\mathrm{Corr}(y_t, n_t \mid \varepsilon_t^A)$ conditional on technology shocks is negative. Using a structural VAR, Galí showed exactly this: conditioning on technology shocks, output rises but employment falls — contradicting the RBC prediction and demonstrating that matching unconditional moments is insufficient to validate the causal structure of the model.

4.3 GMM and SMM

Both methods estimate structural parameters by matching moments from the model to moments in the data. The key distinction: **GMM** matches analytical (theoretical) moments, while **SMM** matches moments computed from simulated model trajectories — useful when closed-form moments are unavailable (e.g., in non-linear models).

The objective is:

$$\min_\theta \; g(\theta)' \, W \, g(\theta), \qquad g(\theta) = \xi(\theta) - \gamma$$

where $\xi(\theta)$ are model-implied moments (or their simulated counterparts in SMM) and γ are the corresponding data moments. Estimation proceeds in two steps:

1. **First step**: $W = I$ (identity matrix). Yields a consistent but inefficient estimator $\hat{\theta}_1$.

2. **Second step**: $W = S^{-1}$, where S is a consistent estimate of the long-run variance of the moment conditions (e.g., Newey–West). Using $\hat{\theta}_1$ to construct S yields the asymptotically efficient estimator.

In practice: collect and detrend the data (e.g., via HP filter or quadratic detrending), select moments to match — e.g., the variance-covariance matrix and first-order autocorrelations of key series — then minimize. SMM adds a simulation step: moments are averages over H simulated paths, which inflates the asymptotic variance by a factor $(1 + 1/H)$.

- **Pros**: (i) computationally lighter than full-information methods; (ii) SMM handles non-linearities (e.g., zero lower bound on the nominal interest rate, occasionally binding constraints) that make the likelihood intractable.

- **Cons**: (i) results are sensitive to moment selection; (ii) identification can be weak; (iii) only unconditional moments are exploited.

4.3.1 Impulse Response Matching (Indirect Inference)

Rather than unconditional moments, this approach matches *conditional* model dynamics to empirical impulse response functions (IRFs). A structural VAR is estimated on the data to recover IRFs to shock(s) of interest; the model parameters θ are then chosen to minimize the distance between model-implied and SVAR-implied IRFs:

$$\min_{\theta} \left[\mathrm{IRF}_{\mathrm{model}}(\theta) - \widehat{\mathrm{IRF}}_{\mathrm{SVAR}}\right]' W \left[\mathrm{IRF}_{\mathrm{model}}(\theta) - \widehat{\mathrm{IRF}}_{\mathrm{SVAR}}\right]$$

This is especially powerful when studying a specific shock (e.g., monetary or fiscal policy). The key requirement is a credible identification scheme to recover $\widehat{\mathrm{IRF}}_{\mathrm{SVAR}}$ — the method inherits whatever identifying assumptions the SVAR imposes. A natural drawback is that the model fit for *other* shocks is left unvalidated.

Example: Jorgensen and Ravn (2022)

? apply IRF matching to estimate the effects of government spending shocks. The procedure has two stages.

Stage 1 — Empirical IRFs. A structural VAR is estimated on U.S. data (1966–2008) for nine variables: real government spending, real GDP, real private consumption, real non-residential investment, real tax revenues, real wages, the consumer price index, the nominal interest rate, and total factor productivity. A government spending shock is

identified from the VAR, and the IRF of each variable over the first 20 quarters is retained, yielding $9 \times 20 = 180$ moment conditions.

Stage 2 — DSGE estimation. A medium-scale New Keynesian DSGE model is estimated by minimizing the distance between the model's IRFs and the VAR-based IRFs. The following 13 parameters are estimated:

- **Household preferences**: risk aversion σ, inverse Frisch elasticity φ, habit formation θ.
- **Supply-side**: investment adjustment costs κ, capital depreciation δ, Rotemberg price and wage adjustment costs γ, γ_w, capital and technology utilization adjustment costs χ_2^K, χ_2.
- **Policy**: Taylor-rule coefficients on inflation ϕ_π and output gap ϕ_y, interest-rate smoothing ρ_r, persistence of government spending shocks ρ_G.

To assess the role of variable technology utilization, the authors estimate two models: a *baseline* model with endogenous technology utilization and a restricted model without it. Table 2 of their paper reports the estimates; a key finding is that without technology utilization, several parameters (e.g., σ, φ, γ_w) take implausibly large values with very wide standard errors, whereas the baseline model yields well-identified, economically reasonable estimates ($\sigma = 0.504$, $\varphi = 0.764$). The baseline model also fits the VAR evidence substantially better — in particular for output, consumption, and TFP — illustrating both the power and the disciplining role of IRF matching relative to unconditional moment matching.

4.4 Bayesian Estimation

Bayesian Maximum Likelihood is the most ambitious and computationally demanding estimation technique for DSGE models. Unlike GMM/SMM, which avoid the likelihood function altogether, the Bayesian approach attempts to maximize a simulated version of it — while circumventing its complications through the explicit use of *prior information*. This approach became feasible only in the early 2000s, with the landmark contributions of **??**, and has since become standard thanks to tools such as Dynare.

4.4.1 Priors

A **prior distribution** $p(\theta)$ encodes the researcher's beliefs about θ *before* observing the data. Prior knowledge can come from three sources:

- **Statistical restrictions**: e.g., a variance must be non-negative, a Calvo probability must lie in $[0, 1]$.

- **Basic knowledge of the economy**: e.g., risk-free rates are low, implying the discount factor β should be close to 1.

- **Evidence from previous econometric studies**: e.g., micro-studies of price-setting behavior find an average price duration of 3–4 quarters, implying a Calvo parameter of 0.67–0.75.

The choice of distributional family for each prior should respect the parameter's support:

- **Normal**: for parameters defined on $\mathbb{R}$ (e.g., Taylor rule coefficients). The mean is the best guess; the variance reflects uncertainty.

- **Beta**: for parameters defined on $[0, 1]$ (e.g., Calvo probabilities, habit formation). A Calvo parameter might receive a $\text{Beta}(0.70, 0.05)$ prior.

- **Inverse Gamma**: for standard deviations of shocks, consistent with the shocks themselves being normally distributed.

- **Uniform**: when little prior information is available; the data then dominates.

4.4.2 The Posterior and Bayes' Theorem

Let Y denote the observed data. By Bayes' theorem, the **posterior distribution** — the distribution of θ after observing the data — is:

$$\underbrace{p(\theta \mid Y)}_{\text{posterior}} = \frac{\underbrace{p(Y \mid \theta)}_{\text{likelihood}}\,\underbrace{p(\theta)}_{\text{prior}}}{\underbrace{p(Y)}_{\text{marginal likelihood}}}$$

The marginal likelihood $p(Y) = \int p(Y \mid \theta)\, p(\theta)\, d\theta$ is a normalizing constant that does not depend on θ, so for the purpose of estimation we work with the unnormalized posterior:

$$p(\theta \mid Y) \;\propto\; p(Y \mid \theta)\, p(\theta)$$

The posterior combines what the data say (likelihood) with what we believed beforehand (prior). As the sample size grows, the likelihood dominates and the posterior concentrates around the true parameter regardless of the prior.

4.4.3 Computing the Likelihood via the Kalman Filter

For a linearized DSGE model with state-space representation

$$s_{t+1} = A(\theta)\, s_t + B(\theta)\, \varepsilon_{t+1}, \qquad y_t = C(\theta)\, s_t + u_t$$

the likelihood $p(Y \mid \theta)$ can be evaluated recursively using the **Kalman filter**, which produces the sequence of prediction errors and their variances. This is a full-information approach: unlike GMM/SMM, Bayesian estimation exploits the *entire* joint distribution of the data, including both conditional and unconditional moments. An additional benefit is that the Kalman smoother yields estimated series for the structural shocks — those most likely to have generated the observed fluctuations.

4.4.4 The Metropolis–Hastings Algorithm

Since $p(\theta \mid Y)$ generally has no closed form for DSGE models, we characterize it numerically using the **Metropolis–Hastings (MH) algorithm**, a Markov Chain Monte Carlo (MCMC) method:

1. Start from an initial parameter vector $\theta^{(0)}$ (e.g., the posterior mode).
2. At iteration i, draw a candidate $\theta^* \sim q(\theta \mid \theta^{(i-1)})$ from a proposal distribution (typically a multivariate normal centered at $\theta^{(i-1)}$).
3. Compute the acceptance ratio:

$$\alpha = \min\left\{1,\ \frac{p(\theta^* \mid Y)}{p(\theta^{(i-1)} \mid Y)}\right\} = \min\left\{1,\ \frac{p(Y \mid \theta^*)\, p(\theta^*)}{p(Y \mid \theta^{(i-1)})\, p(\theta^{(i-1)})}\right\}$$

4. Accept $\theta^{(i)} = \theta^*$ with probability α; otherwise set $\theta^{(i)} = \theta^{(i-1)}$.

After a burn-in period, the chain $\{\theta^{(i)}\}$ is a correlated sample from the posterior. With sufficiently many draws (e.g., 10^6), the ergodic averages of the chain converge to the posterior moments: means, standard deviations, and credible intervals. The parameter estimate reported is typically the posterior **mode** or **mean**; inference is based on the full posterior distribution, not point estimates alone.

4.4.5 Advantages and Critique

Advantages:

- Full-information method: uses all moments of the data, both conditional and unconditional.

- Delivers a complete posterior distribution, enabling proper Bayesian inference (credible intervals, model comparison via marginal likelihoods).
- Priors rule out unreasonable parameter estimates from the outset, stabilizing the optimization.
- The Kalman filter provides estimated structural shock series.
- Treats parameters as random variables — a philosophically coherent framework when genuine uncertainty about parameter values exists.

Critique:

- Priors may prevent the data from speaking freely, particularly if they are tight. However, prior choices are explicit and transparent — and even frequentist researchers routinely reject unreasonable estimates.
- **Weak identification**: if the data contain little information about a parameter, the posterior will closely mimic the prior. This can be diagnosed by varying the prior and checking whether the posterior shifts accordingly; weakly identified parameters are typically calibrated instead.
- Requires at least as many structural shocks as observable variables to avoid *stochastic singularity* (a singular likelihood).
- Computationally demanding, though much less so with modern tools (Dynare, Julia-based packages) and parallel computing.

4.4.6 Assessing Prior Informativeness

A key diagnostic in Bayesian estimation is understanding how much the data actually update the prior — i.e., how *informative* the likelihood is about each parameter. The standard approach is to compare the prior and posterior distributions visually and quantitatively.

Graphical comparison. For each parameter, plot the prior $p(\theta_i)$ and the marginal posterior $p(\theta_i \mid Y)$ on the same graph. Three cases arise:

- **Posterior $\neq$ prior** (different location and/or smaller variance): the data are informative — the likelihood is shifting and/or tightening beliefs. This is the ideal outcome.
- **Posterior $\approx$ prior** (same shape and location): the data contain little information about θ_i; the parameter is *weakly identified*. The posterior is essentially just a rescaled prior.

- **Posterior at the boundary of the prior support**: a sign of potential misspecification or a prior that is too restrictive.

Formal sensitivity analysis. Repeat the estimation under alternative priors — e.g., shift the prior mean by one standard deviation, or double the prior variance — and check whether the posterior changes materially:

$$\text{If } p(\theta_i \mid Y) \text{ is robust to changes in } p(\theta_i) \implies \text{data are informative about } \theta_i$$

$$\text{If } p(\theta_i \mid Y) \text{ tracks } p(\theta_i) \implies \text{weak identification; consider calibrating } \theta_i$$

Practical rule. A useful summary statistic is the ratio of posterior to prior standard deviation:

$$r_i = \frac{\mathrm{SD}(\theta_i \mid Y)}{\mathrm{SD}(\theta_i)}$$

$r_i \ll 1$ means the data sharply updated the prior (highly informative); $r_i \approx 1$ means the posterior variance is essentially the prior variance — a red flag for weak identification. Parameters with $r_i \approx 1$ should either be calibrated or their identification source should be re-examined.

Chapter 5

From Structural VAR to new Approaches

5.1 SVAR Empirical Effects of Monetary Policy

After developing structural models such as the New Keynesian model[1], a natural next step is to confront them with the data and assess what empirical evidence tells us about the monetary policy transmission mechanism. As emphasized by Christiano et al. (1999b), monetary policy is endogenous to the business cycle: policymakers systematically respond to fluctuations in output, inflation, and other macroeconomic aggregates. It is therefore crucial to decompose observed monetary policy actions into two distinct components. The first is the *systematic component*, which captures the endogenous feedback rule of the central bank. The second is the *exogenous component*, identified as a monetary policy shock — an unpredictable deviation from the rule that is orthogonal to the current state of the economy.

Although monetary policy shocks are not of direct interest per se, as they account for only a modest fraction of macroeconomic fluctuations (Christiano et al., 1999b), they represent a powerful diagnostic tool for evaluating structural models. By comparing the model's theoretical impulse response functions to a policy shock with their empirical counterparts estimated from the data, one can assess the empirical plausibility of the model without confounding the exogenous shock with the endogenous systematic component of policy. This is precisely the spirit of the *Lucas program* applied to monetary economics.

[1]The reason why empirics is needed here relates to:

- **Causal identification** delivering credible identification.
- **Distinguish between theories**

5.1.1 Endogeneity Problem

The **endogeneity problem** is one of the most fundamental concerns in macroeconometrics. If the model fails to disentangle the effects attributable to different sources of variation, the resulting estimates will not recover true causal relationships. In the context of monetary policy, the problem is particularly acute: we wish to measure the causal effect of monetary policy on output, yet monetary policy is itself endogenously determined as a systematic response to output and other macroeconomic conditions. A naive regression of output on the policy instrument would therefore suffer from simultaneous equations bias, conflating the effect of policy on the economy with the effect of the economy on policy.

The solution is to isolate *exogenous* variation in monetary policy — that is, variation that is orthogonal to the current state of the economy and cannot be attributed to the central bank's feedback rule. This exogenous component, the monetary policy shock, provides the clean experiment needed for causal identification.

Once this shock is identified, the object of interest becomes the *impulse response function*: the dynamic response of macroeconomic aggregates such as output, prices, and employment to an exogenous monetary policy shock. Crucially, we are not measuring the effect of monetary policy *per se* — which is largely systematic and endogenous — but rather how the economy responds to an unexpected, exogenous deviation from the central bank's rule. This distinction is what gives the exercise its causal interpretation and its power as a diagnostic tool for evaluating structural models.

5.1.2 Structural VAR

The **Structural VAR (SVAR)** model bridges the gap between the statistical reduced-form representation and an economically meaningful one. The core problem with the reduced-form VAR is that innovations are correlated: $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \Omega \neq \mathbf{I}_m$, so a shock to one equation contaminates all others simultaneously, precluding a clean *ceteris paribus* interpretation. The SVAR resolves this by imposing enough structure to recover shocks that are (i) **orthogonal**, i.e. $\mathbb{E}[\zeta_t \zeta_t'] = \mathbf{I}_m$, and (ii) carry a specific **economic interpretation** (technology, demand, monetary policy, etc.).

The construction proceeds in two steps.

1. **Estimate the reduced-form VAR** by OLS to obtain consistent estimates of $\{\Phi_j\}_{j=1}^p$ and the innovation covariance $\widehat{\Omega} = T^{-1} \sum_t \widehat{\varepsilon}_t \widehat{\varepsilon}_t'$.

2. **Identify the structural shocks** by finding a matrix $\mathbf{H}$ such that

$$\varepsilon_t = \mathbf{H}\, \zeta_t, \qquad \mathbb{E}[\zeta_t \zeta_t'] = \mathbf{I}_m,$$

which requires $\Omega = \mathbf{H}\mathbf{H}'$. Since Ω has $m(m+1)/2$ free parameters while $\mathbf{H}$ has m^2, exactly $m(m-1)/2$ additional restrictions must be imposed for exact identification.

Linking Reduced-Form and Structural VAR

Pre-multiplying the reduced-form VAR by $\mathbf{B}_0 \equiv \mathbf{H}^{-1}$ gives the *structural form*:

$$\mathbf{B}_0\,\mathbf{y}_t = \sum_{j=1}^{p} \mathbf{B}_j\,\mathbf{y}_{t-j} + \zeta_t,$$

where $\mathbf{B}_j \equiv \mathbf{B}_0\Phi_j$ and $\zeta_t = \mathbf{B}_0\varepsilon_t$. The matrix $\mathbf{B}_0$ encodes the contemporaneous relationships among variables implied by economic theory, while off-diagonal zeros in $\mathbf{B}_0$ (or equivalently in $\mathbf{H}$) constitute the identifying restrictions.

From VAR to Structural MA(∞) Representation

Starting from the reduced-form VAR, write it in lag-operator notation as $\Phi(L)\,\mathbf{y}_t = \varepsilon_t$, where $\Phi(L) \equiv \mathbf{I}_m - \sum_{j=1}^{p} \Phi_j L^j$. Under covariance stationarity — equivalently, all roots of $\det \Phi(z) = 0$ lie strictly outside the unit circle — $\Phi(L)$ is invertible and we obtain the Wold (reduced-form) MA(∞):

$$\mathbf{y}_t = \Phi(L)^{-1}\varepsilon_t = \sum_{i=0}^{\infty} \mathbf{A}_i\,\varepsilon_{t-i},$$

where the dynamic multiplier matrices $\{\mathbf{A}_i\}$ satisfy the recursion

$$\mathbf{A}_i = \sum_{j=1}^{p} \Phi_j\,\mathbf{A}_{i-j}, \qquad i = 1, 2, \ldots,$$

with $\mathbf{A}_0 = \mathbf{I}_m$ and $\mathbf{A}_i = \mathbf{0}$ for $i < 0$. Substituting $\varepsilon_t = \mathbf{H}\zeta_t$ yields the **structural MA(∞) representation**:

$$\mathbf{y}_t = \sum_{i=0}^{\infty} \mathbf{A}_i\mathbf{H}\,\zeta_{t-i}, \qquad t = 1, 2, \ldots, T.$$

The matrix $\mathbf{A}_s\mathbf{H}$ governs the structural impulse responses at horizon s: its (k, j) element measures the response of variable k at time $t+s$ to a unit structural shock ζ_{jt}, holding all other shocks fixed. Formally, the **structural IRF** of the jth shock is

$$\frac{\partial\,\mathbf{y}_{t+s}}{\partial\,\zeta_{jt}} = \mathbf{A}_s\mathbf{H}\,\mathbf{e}_j,$$

where $\mathbf{e}_j$ is the jth column of $\mathbf{I}_m$. Plotting this as a function of $s = 0, 1, 2, \ldots$ yields the impulse response function (IRF). Because $\mathbf{A}_i \to \mathbf{0}$ as $i \to \infty$ under stationarity, the effect of any structural shock eventually dies out.

We now want to link the reduced-form solution to the structural VAR therefore we

proceed as follows:

$$\varepsilon_t = \boldsymbol{B}_0^{-1}\zeta_t$$

then we proceed:

$$E(\varepsilon_t\varepsilon_t') = \mathbf{B}_0^{-1}E(\zeta_t\zeta_t')\left(\mathbf{B}_0^{-1}\right)',$$

$$\Omega = \mathbf{B}_0^{-1}\Omega_\zeta\left(\mathbf{B}_0^{-1}\right)',$$

$$\Omega = \mathbf{B}_0^{-1}\left(\mathbf{B}_0^{-1}\right)'.$$

Now, recall that Ω is symmetric, so it has $m(m+1)/2$ free parameters, while $\mathbf{B}_0^{-1}$ has m^2 parameters. Therefore we need to impose $m^2 - m(m+1)/2 = m(m-1)/2$ restrictions, and it is here that identification comes in.

The standard approach is the **recursiveness assumption**: some variables do not react contemporaneously to others. This restricts $\mathbf{H}$ to be lower triangular, so that for $m = 3$:

$$\begin{bmatrix}\varepsilon_t^1\\ \varepsilon_t^2\\ \varepsilon_t^3\end{bmatrix} = \begin{bmatrix}h_{11} & 0 & 0\\ h_{21} & h_{22} & 0\\ h_{31} & h_{32} & h_{33}\end{bmatrix}\begin{bmatrix}\zeta_t^1\\ \zeta_t^2\\ \zeta_t^3\end{bmatrix}.$$

Variable 1 responds only to its own shock on impact, variable 2 responds to shocks 1 and 2 but not 3, while variable 3 — ordered last — responds to all shocks contemporaneously. Variables ordered earlier therefore react to later shocks only *with a lag*. This causal ordering must be justified by economic reasoning rather than statistical grounds. In the literature there exists different identification strategies and one of these is **Cholesky decomposition** of the variance covarianc matrix. Here we start defining a lower triangula $m \times m$ matrix $\boldsymbol{P}$ with positive main diagonal such that $\boldsymbol{P}\boldsymbol{P}' = \Sigma_\zeta$. Here $\boldsymbol{P}$ is not unique it changes given each different variable ordering therefore economic structure has important impact on choosing the ordering (robustness check control the fact that changing order the IRF structure does not changes). Essentially identification, using this scheme, is based on time.

5.1.3 Christiano, Eichenbaum and Evans (1999)

The central question of Christiano et al. (1999a) is: what are the dynamic effects of an exogenous monetary policy shock on the macroeconomy? This has direct implications for how powerful monetary policy is and how it should be conducted. A secondary contribution is methodological: how should one choose among competing identification schemes? Their answer is pragmatic — if no identification assumption delivers the qualitative features one would theoretically expect, reject the identification rather than the model (see Uhlig (2005b) for the sign restrictions approach).

The authors begin by clarifying the meaning of a **monetary policy shock**. Most central bank actions are endogenous responses to the state of the economy, so observed movements in the policy instrument confound cause and effect. The goal is to isolate the *unaccounted* variation — the component orthogonal to the Fed's systematic feedback rule — which is formalized as:

$$\underbrace{S_t}_{\text{CB instrument}} = \underbrace{f(\mathcal{I}_{-t})}_{\text{systematic response to information set}} + \underbrace{\sigma_s \, \varepsilon_t^s}_{\text{monetary policy shock}} .$$

CEE offer three interpretations of ε_t^s: (i) exogenous shifts in the Fed's preferences over unemployment and inflation, including shocks to individual FOMC members or to the aggregation of their views; (ii) strategic considerations that generate self-fulfilling expectational dynamics; and (iii) measurement error in the real-time data available to the Fed when setting policy.

The third interpretation carries an important caveat for identification. If the Fed reacts to a noisy real-time measure of x_t, the shock ε_t^s absorbs the measurement error, and the recursiveness assumption — that ε_t^s is orthogonal to the information set $\mathcal{I}_{-t}$ — only holds under restrictive conditions. In the empirically plausible case where measurement errors are correlated and the Fed reacts to lagged noisy data, recursiveness fails and the identified shock is contaminated, biasing the estimated impulse responses.

The caveat is that if you use recursiveness to identify the shock, you are implicitly assuming measurement error is not driving monetary policy actions. The caveat is that if you use recursiveness to identify the shock, you are implicitly assuming measurement error is not driving monetary policy actions.

Since $Q_V \cap Q_\tau$ contains more than one element, a normalization is needed for computation. The standard choice is the inverse of the lower triangular Cholesky factor of V. A natural concern is whether the ordering of variables within X_{1t} and X_{2t} matters. Result (iii) of Proposition 4.1 settles this: the monetary policy IRF is **invariant to that ordering**, since moving variables around within the block does not change what the Fed reacts to.

On identification

The recursiveness assumption therefore provides *just enough* restrictions to identify ζ_t^{MP} and nothing more. Non-policy shocks remain unidentified — their IRFs simply reflect the chosen normalization and carry no structural meaning. The SVAR is **partially identified** by design.

Since recursiveness does not uniquely pin down A_0, one still needs to choose the policy instrument S_t and the information set $\mathcal{I}_t$. CEE propose three benchmark schemes:

1. **FF shock**: $S_t = FF_t$, the federal funds rate. The information set $\mathcal{I}_t$ includes current

and lagged values of Y_t, P_t, $PCOM_t$ and lagged values of FF_t, NBR_t, TR_t, M_t.

2. **NBR shock**: $S_t = NBR_t$, non-borrowed reserves. The motivation is that innovations to NBR_t primarily reflect exogenous policy actions, while innovations to broader aggregates reflect money demand shocks.

3. **NBR/TR shock**: $S_t = NBR_t/TR_t$, following Strongin (1995). The argument is that total reserve demand is interest inelastic in the short run, so a policy shock only rearranges the composition of total reserves between borrowed and non-borrowed components.

In all three schemes $\mathbf{X}_{1t} = [Y_t,\ P_t,\ PCOM_t]'$, so the Fed is assumed to observe output, prices and commodity prices contemporaneously when setting policy.

After a contractionary shock, all three schemes deliver qualitatively similar results. The federal funds rate rises persistently and non-borrowed reserves fall, consistent with a strong liquidity effect. Real GDP declines in a **hump-shaped** pattern, with the peak contraction occurring roughly 4–6 quarters after the shock. The GDP deflator is flat for about six quarters before declining, though an initial rise — the **price puzzle** — is often observed. Monetary aggregates fall persistently.

The robustness of these findings across all three identification schemes is the main empirical contribution of the paper: the qualitative picture of how the economy responds to monetary policy does not depend on the precise choice of instrument.

Empirical evidence of Contractioanry Monetary Policy

Section 4.3 extends the benchmark results to a richer set of variables, providing a detailed picture of the transmission mechanism. The table below summarizes the response of key variables to a contractionary monetary policy shock.

Table 5.1: Response to a Contractionary Monetary Policy Shock

Variable	**Sign**	**Speed**	**Source**
Federal funds rate	+	Impact	CEE (1999)
Non-borrowed reserves	−	Impact	CEE (1999)
GDP / Output	−	Hump-shaped, peak ∼ 4–6Q	CEE (1999)
GDP deflator	≈ 0 then −	Slow, ∼ 6Q delay	CEE (1999)
Commodity prices	−	Fairly rapid	CEE (1999)
M1 / M2	−	Delayed	CEE (1999)
Unemployment	+	∼ 2Q delay	CEE (1996a)
Retail sales	−	Impact	CEE (1996a)
Corporate profits	−	Impact, sharp	CEE (1997a)
Manufacturing inventories	+	Impact	CEE (1996a)
Residential investment	−	Rapid, leads other components	Fisher (1997)
Equipment & durables	−	Delayed	Fisher (1997)
Real wages	−	Modest, delayed	CEE (1997a)
Bank deposits & loans	−	Deposits immediate, loans delayed	Bernanke-Blinder (19
Small firm credit	−	Faster than large firms	Gertler-Gilchrist (199
Business net borrowing	+ then −	Initially rises ∼1Y, then falls	CEE (1996a)
Exchange rate (USD)	Appreciation	Gradual	Eichenbaum-Evans (

In their analysis the analyse the effect of a contractionary monetary policy on different real variables, wage and profits. The results report that:

- **Unemployment**: rises after a delay about two quarters. Decline in employment reflects primarily increase in job destruction.
- **Investment components**: residential investment exhibit larger decline, followed by equipment, durables and structures.
- **Real wages**: Decline by modest amount more prominent in durable goods industry.
- **Borrowing and Lending activities**: persistent decline in volume of bank deposits, reflecting fall in the amount of securities held by banks. Eventually securities return to their pr-shock period.

Two findings stand out as particularly informative for theory. First, **real wages fall** after a contractionary shock, which contradicts models based purely on nominal wage stickiness — those models predict real wages should rise. Second, **profits fall sharply**, which contradicts models based purely on sticky prices without labor market frictions,

since those models predict profits should rise as firms charge above marginal cost. Together these results place tight restrictions on the class of models that can plausibly account for the monetary transmission mechanism. Finally they also report visible results of market imperfection given the asymmetric answer of different credit market players to contractionary monetary policy.

The Price Puzzle

The **price puzzle** refers to the empirical finding that following a contractionary monetary policy shock, the price level *rises* for several quarters before eventually falling — the opposite of what theory predicts.

The leading explanation, due to Sims (1992), is omitted variable bias. The Fed reacts to inflationary pressures observable in real time — commodity prices, forward-looking indicators — that are not included in the VAR. What looks like an exogenous rate hike is therefore partly an endogenous response to incoming inflation, and the VAR misattributes this to the shock. The standard fix is to include commodity prices ($PCOM_t$) in $\mathcal{I}_t$: once the Fed's forward-looking information is controlled for, the puzzle largely disappears.

The price puzzle matters beyond being an empirical curiosity. It is a direct challenge to the recursiveness assumption: if the Fed observes variables that the econometrician omits from the VAR, then the identified shock is not truly exogenous and the estimated IRFs are biased. This is precisely the same logic as the measurement error caveat of Section 2 — recursiveness fails whenever the Fed's information set is richer than what the model captures.

5.2 Alternative Approaches

We can now move from classical SVAR approach to new development in econometrics. In particular when using the classic Christiano et al. (1998) on recent, periods this gives rise to unclear effects on macroeconomic variables, implying that it has became much harder in recent decades to identify exogenous changes in policy rate. This effect could be explained by the necessity to update the econometric methodology and account for (i) institutional changes (ii) communication strategies of the CB (iii) policy rules introduced later than the classical SVAR literature.

5.2.1 Criticism of the Cholesky Decomposition

The cholesky Decomposition is more a mathematical/statistical device that allows for the identification of shocks. This rigidity does not enable the accounting for institutional timing and frequency of the monetary policy decisions, which is rigid (FED ECB meet regularly, while Cholesky treats every period as a possible date where a shock happens). Order matters; therefore, missordering could have effects on causality for example Christiano et al. (1998) preclude output and inflation to respond instantaneously to an MP shock, while this does not reflect the NK model.

5.2.2 Sign Restrictions: Uhlig (2005)

One of the first alternatives to recursive identification is the *agnostic* approach of Uhlig (2005a). Rather than imposing zero contemporaneous restrictions on the impact matrix, this methodology identifies structural shocks by requiring that impulse responses carry **economically plausible signs**, while remaining entirely silent on the variables of interest — in this case, output.

Concretely, Uhlig defines a contractionary monetary policy shock as any shock that, for $h = 0, 1, \ldots, H$ periods following the shock, satisfies:

$$\frac{\partial \pi_{t+h}}{\partial \varepsilon_t^{MP}} \leq 0, \qquad \frac{\partial i_{t+h}}{\partial \varepsilon_t^{MP}} \geq 0, \qquad \frac{\partial \text{NBR}_{t+h}}{\partial \varepsilon_t^{MP}} \leq 0,$$

where π_t denotes the price level, i_t the federal funds rate, and NBR_t non-borrowed reserves.[2] Crucially, **no restriction** is imposed on the response of output, which remains the object of inference.

[2]Non-borrowed reserves are reserves held by commercial banks at the Federal Reserve that were *not* obtained through discount window borrowing. A contractionary open market operation — the Fed selling Treasury securities — drains these reserves directly: banks purchase the securities by having their reserve accounts debited, reducing aggregate NBR and pushing the federal funds rate upward. NBR thus serves as the cleanest observable signal of an exogenous, supply-side policy tightening, as it is purged of the endogenous borrowing response that contaminates total reserves.

Identification via Rotation Matrices. The object of interest is the contemporaneous impact matrix $\boldsymbol{B}_0^{-1}$, which maps structural shocks $\boldsymbol{\varepsilon}_t$ to reduced-form innovations $\boldsymbol{u}_t$ via $\boldsymbol{u}_t = \boldsymbol{B}_0^{-1}\boldsymbol{\varepsilon}_t$. Let P denote the lower-triangular Cholesky factor of the reduced-form covariance matrix $\Sigma_{\boldsymbol{u}}$, so that $PP' = \Sigma_{\boldsymbol{u}}$. For any orthogonal matrix $Q \in \mathcal{O}(k)$ satisfying $QQ' = I_k$, we have:

$$(PQ)(PQ)' = PQQ'P' = PP' = \Sigma_{\boldsymbol{u}},$$

so $\widetilde{B}_0^{-1} = PQ$ is a valid candidate impact matrix for any such Q. The identification problem therefore reduces to selecting, from the continuum of orthogonal rotations Q, those that are consistent with the imposed sign restrictions.

Algorithm (Uhlig, 2005).

1. Draw a $k \times k$ matrix L with i.i.d. entries $L_{ij} \sim \mathcal{N}(0,1)$ and compute its QR decomposition $L = QR$, where $Q \in \mathcal{O}(k)$ is drawn uniformly from the Haar measure on the orthogonal group.

2. Form the candidate impact matrix $\widetilde{\boldsymbol{B}}_0^{-1} = PQ$ and compute the implied impulse response functions. **Retain** the draw if all sign restrictions are satisfied for $h = 0, \ldots, H$; **discard** it otherwise.

3. Repeat steps 1–2 a large number of times S to obtain a set $\mathcal{Q}^*$ of admissible rotations.

Posterior inference on impulse responses is then conducted over the retained draws, with point estimates typically reported as the **median** of the admissible set, and uncertainty summarised by 68% posterior credible bands.

Evaluation. The sign restriction approach offers several appealing properties relative to recursive identification, but also carries important limitations that must be acknowledged.

Advantages.

1. **Timing agnosticism.** Unlike the Cholesky decomposition, which precludes any within-period response of output and inflation to a monetary policy shock, sign restrictions are entirely *agnostic about the timing* of macroeconomic responses. Immediate reactions of output and inflation to a policy shock are explicitly permitted, making the approach consistent with the instantaneous transmission implied by New Keynesian models.

2. **Resolution of anomalies.** By directly encoding theoretical beliefs as restrictions on impulse response signs, the methodology eliminates well-known anomalies. Most notably, requiring prices to fall following a contractionary shock resolves the *price puzzle* — the counterintuitive finding, common in Cholesky SVARs, that prices rise after a tightening.

3. **Avoidance of arbitrary ordering.** The approach sidesteps the rigid, often economically indefensible zero restrictions implied by a particular Cholesky ordering, replacing them with weaker inequality restrictions that command broader theoretical consensus.

Disadvantages.

1. **Untestable theoretical priors.** The sign restrictions directly encode the researcher's theoretical beliefs, which may be wrong. Crucially, unlike zero restrictions — which can in principle be tested via over-identifying restrictions — sign restrictions are *not testable*: any draw from $\mathcal{Q}^*$ is by construction consistent with the data, so the restrictions impose structure without offering a falsification criterion.

2. **Set identification and shock conflation.** Sign restrictions only *partially* identify the structural parameters, yielding a set $\mathcal{Q}^*$ rather than a unique impact matrix. This produces wide posterior credible bands that can be difficult to interpret. More fundamentally, as emphasised by Wolf (2020), the admissible set $\mathcal{Q}^*$ may contain rotations dominated by non-monetary disturbances — demand or supply shocks — that happen to satisfy the sign restrictions by coincidence, effectively *conflating multiple structural shocks* and contaminating inference on the unrestricted variables.

5.2.3 Narrative Approach

The **Narrative Approach**, pioneered by Romer and Romer (2004a), constructs a measure of exogenous monetary policy shocks by reading and systematically coding historical documents — in particular, the records of Federal Open Market Committee (FOMC) meetings — rather than relying on purely statistical restrictions. The underlying idea is to isolate the component of policy decisions that is orthogonal to the central bank's own information set, thereby recovering a series of genuine policy surprises.

The procedure unfolds in three steps. First, for each FOMC meeting, one identifies the **intended change in the federal funds rate target** decided by the committee. Second, this intended change is regressed on the Federal Reserve's own internal *Greenbook* forecasts of output growth, inflation, and unemployment — variables that summarise the Fed's real-time information set at the time of the decision:

$$\Delta i_t^* = \alpha + \boldsymbol{\beta}' \boldsymbol{f}_t + \varepsilon_t,$$

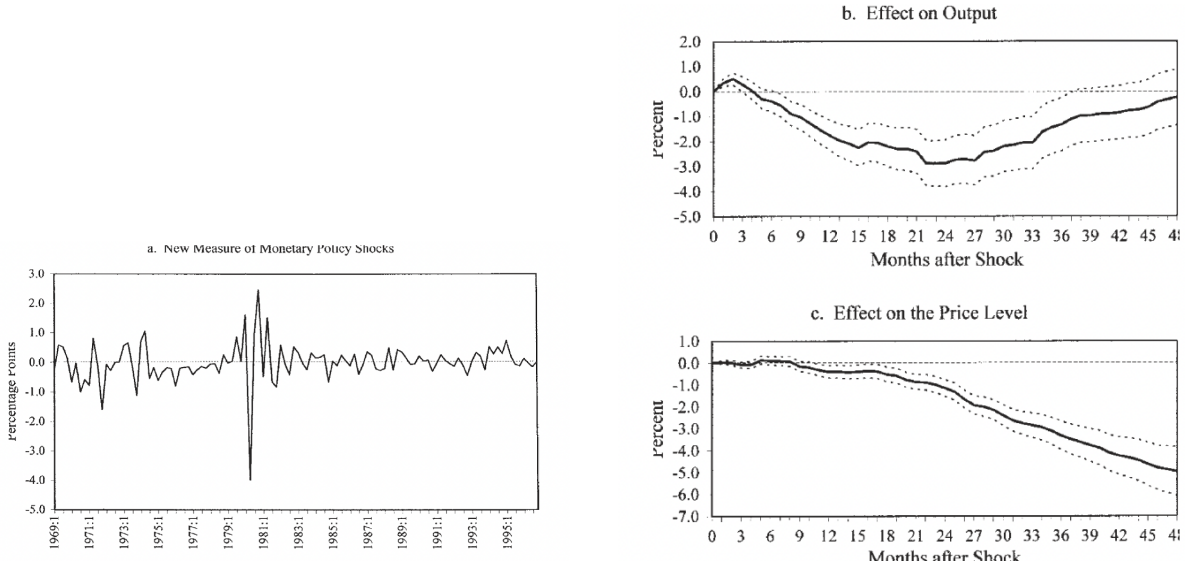
where Δi_t^* denotes the intended rate change at meeting t, $\boldsymbol{f}_t$ is the vector of Greenbook forecasts, and $\boldsymbol{\beta}$ captures the systematic, endogenous component of policy — i.e. the Fed's reaction function. Third, the **residuals** $\hat{\varepsilon}_t$ from this regression are retained as the measure of the monetary policy shock: the part of the policy decision that cannot be explained by

the Fed's own macroeconomic outlook, and hence reflects genuinely exogenous variation in the stance of policy.

An important restriction in Romer and Romer (2004a) is that the analysis is confined to **scheduled, regular FOMC meetings**. This is not merely a practical convenience but a substantive identification requirement: at scheduled meetings, the Fed prepares and presents Greenbook forecasts in advance of the decision, so the forecast and the policy action are directly comparable and cover the same information set. Inter-meeting emergency actions, by contrast, are taken in response to unforeseen shocks for which no corresponding Greenbook forecast exists, creating a fundamental disconnect between the information conditioning the forecast and the information driving the policy response. Including such observations would contaminate the residual $\hat{\varepsilon}_t$ with the very shock it was meant to isolate.

The appeal of this approach lies in its **direct use of institutional knowledge**: rather than imposing statistical assumptions on a reduced-form system, identification rests on the plausible economic argument that, conditional on the Fed's own forecasts, the remaining variation in policy decisions is exogenous to the state of the economy. The main limitation is the reliance on the **Greenbook forecasts as a sufficient statistic** for the Fed's information set — if the Fed acts on information not captured in $\boldsymbol{f}_t$, the residuals will remain endogenous, and the shock measure will be contaminated by anticipation effects.

$$\Delta ff_m = \alpha + \beta\, ffb_m + \sum_{i=-1}^{2} \gamma_i \widetilde{\Delta y}_{mi} + \sum_{i=-1}^{2} \lambda_i \left(\widetilde{\Delta y}_{mi} - \widetilde{\Delta y}_{m-1,i}\right) + \sum_{i=-1}^{2} \varphi_i \widetilde{\pi}_{mi} + \sum_{i=-1}^{2} \theta_i \left(\widetilde{\pi}_{mi} - \widetilde{\pi}_{m-1,i}\right) + \rho \widetilde{u}_{m0} + \varepsilon \tag{5.1}$$



(a) Shock measure plotted over time.

(b) Enter Caption.

Figure 5.1: Overall caption describing both figures.

Then they proceed either using the classical SVAR (using the shock residuals as the hock like from interest rate) approach or Local projection. As we see a contractionary monetary policy would lead output to drop and prices to decline, in line with the NK model and displaying no price puzzle.

5.2.4 High-Frequency Identification

Jarocinsky and Karadi (2020)

The core idea of high-frequency identification (HFI) is to exploit financial market data measured in a *narrow window* around policy announcements — typically 30 minutes — to isolate the causal effect of monetary policy. This approach has two key advantages:

- **Isolation of monetary policy shocks.** Within a half-hour window, it is unlikely that shocks unrelated to the central bank announcement systematically move asset prices. This dramatically reduces the contamination from other macroeconomic disturbances.
- **Resolution of the anticipation problem.** Financial markets continuously incorporate expected policy changes into prices. Consequently, only the *surprise* component of an announcement — the deviation from market expectations — moves futures prices upon announcement. Changes in, say, fed funds futures within the window thus measure the unexpected component of policy directly.

A key contribution of Jarociński and Karadi (2020) is to show that a single high-frequency interest rate surprise conflates *two* distinct structural shocks, which can be separated by examining the *joint* movement of interest rates and stock prices in the narrow window. The intuition follows directly from the standard asset pricing identity:

$$p_t = \frac{D_{t+1}}{1+r_t} + \frac{D_{t+2}}{(1+r_t)(1+r_{t+1})} + \cdots = \sum_{h=1}^{\infty} \frac{\mathbb{E}_t[D_{t+h}]}{\prod_{s=0}^{h-1}(1+r_{t+s})}$$

A **pure monetary policy tightening** raises r_t while leaving (or reducing) expected dividends $\{D_{t+h}\}$, unambiguously *reducing* p_t. Hence a monetary policy shock is identified by a **negative co-movement** between the interest rate surprise and the stock price surprise within the window. Conversely, if an announcement raises both interest rates *and* stock prices simultaneously, the rate increase alone cannot account for the stock market reaction. The only consistent interpretation is that the announcement also conveyed positive news about future fundamentals — higher expected dividends — that more than offsets the higher discount rate. This is the **central bank information shock**, identified by a **positive co-movement** between the two surprises.

Figure 5.2 makes this decomposition transparent. Quadrants II and IV (negative co-movement) correspond to announcements dominated by the monetary policy shock,

while Quadrants I and III (positive co-movement) reveal the presence of the information shock. Notably, around one-third of FOMC announcements between 1990 and 2016 fall in Quadrants I or III — a non-trivial share that, if ignored, biases the estimated effects of monetary policy. The March 20, 2001 episode is a canonical example: a larger-than-expected 50bp rate *cut* was accompanied by a stock market *decline*, consistent with the FOMC simultaneously signalling a deteriorating economic outlook.

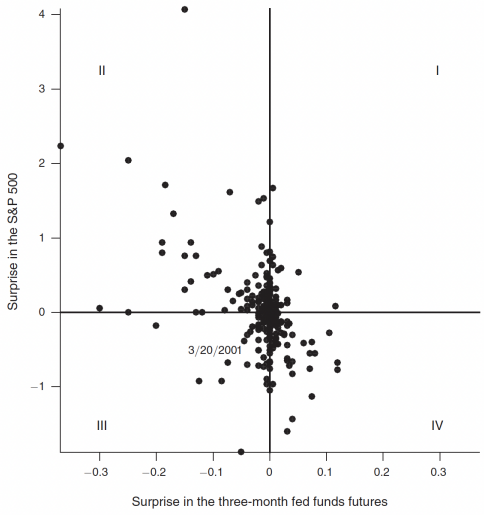


FIGURE 1. SCATTERPLOT OF INTEREST RATE AND STOCK PRICE SURPRISES

Notes: Change in the three-month fed funds futures and the S&P 500 index around FOMC announcements in percent. Each dot represents one FOMC announcement.

Figure 5.2: Scatterplot of surprises in the three-month fed funds futures (horizontal axis) and the S&P 500 (vertical axis) within a 30-minute window around FOMC announcements, 1990–2016. Quadrants II and IV exhibit the *negative* co-movement consistent with a pure monetary policy shock; Quadrants I and III exhibit *positive* co-movement, indicative of a central bank information shock. Source: Jarociński and Karadi (2020).

Formally, Jarociński and Karadi (2020) embed this logic in a Bayesian structural VAR. Letting $m_t = (\Delta \text{fff}_t,\ \Delta S\&P_t)^\top$ denote the high-frequency surprise vector, they impose: (i) an *exclusion restriction* — only the two announcement shocks affect m_t within the window; and (ii) *sign restrictions* on the contemporaneous impact of each shock on m_t, as summarised in the table below. No restrictions are imposed on the responses of monthly macroeconomic variables, yet the two identified shocks generate strikingly different impulse responses: the monetary policy shock contracts output and prices and tightens financial conditions, while the information shock expands both output and prices and eases credit spreads — consistent with the CB revealing positive news about future demand conditions. Disregarding the information component, as in the standard HFI of Gertler and Karadi (2015), attenuates the estimated effects of monetary policy and may account for the price puzzle documented in certain subsamples.

Results and Conclusion The purified monetary policy shock contracts output and prices and tightens financial conditions, consistent with standard theory. The central bank

information shock does the opposite: it raises output, prices, and eases credit spreads, consistent with the CB revealing positive news about future demand. Ignoring this distinction — as standard HFI does — attenuates the estimated effects of monetary policy, understates the price response (a likely source of the price puzzle), and overestimates interest rate persistence. Structurally, the purified shock raises the importance of financial frictions relative to nominal rigidities in monetary transmission. Results hold for the euro area, where the distortion is even more severe.

The broader lesson is that high-frequency interest rate surprises are not sufficient statistics for monetary policy shocks. At minimum, one additional asset price — here, the stock market — is needed to separate policy from information. Whether the information shock reflects superior CB forecasts materialising or a causal effect of communication on expectations remains an open question.

5.2.5 Miranda-Agrippino and Ricco (2021)

Miranda-Agrippino and Ricco (2021) address the same contamination problem as Jarociński and Karadi (2020), but from a different angle. Their starting point is that commonly used instruments for monetary policy shocks — both high-frequency surprises à la Gertler and Karadi (2015) and narrative measures à la Romer and Romer (2004c) — implicitly assume that either the central bank or market participants enjoy perfect information. Under dispersed and asymmetric information, this assumption fails: a policy rate hike is observationally equivalent to a true contractionary shock *or* to an endogenous response to stronger fundamentals that only the CB has observed. Conflating the two generates the well-known price and output puzzles.

A Model of Noisy and Asymmetric Information

To discipline the argument, they write down a simple signal-extraction model. Fundamentals follow an AR(1), $x_t = \rho x_{t-1} + \xi_t$, and neither private agents nor the CB observe x_t directly — both use a Kalman filter on private noisy signals. The CB sets the policy rate via a Taylor rule conditioned on its own forecast $F_{cb}x_t$:

$$i_t = \phi_0 + \phi_x' F_{cb,\underline{t}}\, x_t + u_t + w_{t|t-1},$$

where u_t is the true monetary policy shock and $w_{t|t-1}$ captures announced forward guidance. When agents observe i_t, they partially infer the CB's private forecast, generating a *signalling channel*: the rate announcement transfers information about ξ_t to the public. The key lemma of the model shows that the resulting high-frequency expectation revision — which is exactly what futures surprises measure — inherits three contaminating components: lagged expectation errors (due to slow information absorption), the CB's private signal

about fundamentals, and lags of past monetary policy shocks. This delivers three testable predictions: standard instruments should be (i) correlated with CB internal forecasts, (ii) serially correlated, and (iii) predictable by lagged macro variables. All three are confirmed empirically.

The Informationally Robust Instrument

Motivated by this evidence, MAR propose a two-step procedure to purify the high-frequency surprise. **Step 1** removes the CB information channel by projecting the raw FF4 surprise on Greenbook forecasts and their revisions for output, inflation, and unemployment:

$$FF4_m = \alpha_0 + \sum_{j=-1}^{3} \theta_j F_m^{cb} x_{q+j} + \sum_{j=-1}^{2} \vartheta_j \left[F_m^{cb} x_{q+j} - F_{m-1}^{cb} x_{q+j} \right] + MPI_m.$$

Since Greenbook forecasts directly proxy the CB's information set, projecting them out removes the endogenous component driven by the signalling channel. **Step 2** removes the slow absorption of information by private agents by filtering out the autoregressive component of the monthly aggregated residual:

$$\overline{MPI}_t = \phi_0 + \sum_{j=1}^{12} \phi_j \, \overline{MPI}_{t-j} + MPI_t.$$

The final residual MPI_t is orthogonal to both the CB's private information set and to its own past — the two sources of contamination identified by the model. Importantly, the information component is never structurally identified as a separate shock; it is treated as nuisance and captured entirely by the Greenbook projection. This differentiates MAR from Jarociński and Karadi (2020), who instead recover explicit IRFs for both shocks via sign restrictions but require stock price data and cannot vary the relative weight of the two components over time.

Results

Using MPI_t as an external instrument in a Bayesian SVAR-IV, a contractionary monetary policy shock is unambiguously recessionary: output falls on impact, prices decline, credit spreads widen, the dollar appreciates, and private sector expectations revise downward in line with deteriorating fundamentals. No price or output puzzle arises. The authors further show that these results are robust across model specifications — VAR, Local Projection, and their proposed Bayesian Local Projection (BLP) — with BLP optimally balancing the bias-variance trade-off between the two, delivering stable responses even in short samples and misspecified models.

The Zero Lower Bound

An important practical advantage of the MPI_t instrument is its validity at the **zero lower bound** (ZLB). When the federal funds rate is constrained at the effective lower bound — as was the case from December 2008 onwards — the standard policy rate ceases to be a meaningful measure of the monetary policy stance. Following Gertler and Karadi (2015), MAR address this by using the **one-year nominal Treasury rate** as the policy variable throughout, rather than the federal funds rate. The one-year rate incorporates the effects of forward guidance and unconventional monetary policy (e.g., asset purchase programs) and therefore remains informative about the overall stance of policy even when the short rate is pinned at zero. All impulse responses are normalized to a 100 basis point increase in this one-year rate, which ensures comparability across the full sample 1979–2014 including the ZLB period. This choice is also consistent with the maturity of the FF4 futures contracts used to construct the instrument, which settle on the average federal funds rate over the third calendar month ahead and therefore embed near-term forward guidance rather than just the spot policy decision. Consequently, the MPI_t instrument and the one-year rate together provide a coherent high-frequency identification scheme that remains valid both in conventional and unconventional monetary policy regimes.

Chapter 6

Fiscal Policy in DSGE and SVAR Models

6.1 Fiscal Policy in DSGE Models

We now turn to fiscal policy and its role within the **New Keynesian** framework. The central question is how government spending affects macroeconomic aggregates, and in particular whether fiscal interventions are an effective stabilization tool. To address this, we need both a theoretical framework and an empirical measure — the **fiscal multiplier**.

The fiscal multiplier is defined as the marginal change in output per unit increase in government spending:

$$\mathcal{M} = \frac{\Delta Y}{\Delta G}.$$

It can be measured (i) *on impact*, capturing the instantaneous response of output, or (ii) *over a horizon*, tracking the cumulative effect over time. The threshold value of one serves as the natural benchmark: a multiplier above unity implies a more than proportional increase in output — government spending crowds *in* private activity — while a multiplier below unity implies a less than proportional response, consistent with partial crowding out. Whether the multiplier exceeds or falls short of this threshold depends critically on the state of the economy, the monetary policy regime, and the degree of nominal and real frictions, as we discuss below.

When analysing fiscal policy in macroeconomic models, government spending is typically treated as **wasteful** in the sense that it enters neither the household utility function nor the production technology directly. Combined with the assumption that government consumption is exogenous, this would ordinarily imply that the optimal level of public spending is zero. The natural question is therefore: *why introduce government spending at all?* The answer lies precisely in its indirect effects — even wasteful government expenditure affects the economy through its impact on **production capacity**, labor supply decisions, and the intertemporal budget constraint of households via the tax financing requirement.

6.1.1 Fiscal Policy in the Real Business Cycle Model: Baxter and King (1993)

Baxter and King (1993) study fiscal policy within a quantitatively disciplined neoclassical model, providing one of the first rigorous analyses of fiscal multipliers in a dynamic general equilibrium setting. Their main findings are:

1. A **permanent** increase in government spending can generate both short-run and long-run multipliers exceeding one, with substantially larger effects than a temporary change of equal size. The permanence of the shock matters because it triggers a stronger negative wealth effect, inducing households to supply more labor and accumulate more capital.

2. The **financing decision** — whether spending is financed via lump-sum taxes, distortionary taxes, or debt — is quantitatively more important than the direct resource cost of the government purchase itself. Distortionary financing significantly dampens the multiplier relative to lump-sum financing.

3. **Public capital** has important spillover effects on private output and investment. When government spending augments productive capacity — as in infrastructure — the multiplier can be substantially larger than under purely wasteful expenditure.

Model

We start setting up the model with the utility function and the production function, as:

$$U_t = \log C_t + \theta_L \log L_t + \Gamma\left(G_t^B, K_t^G\right).$$

we see here we have separability so when considering the FOCS for consumption and leisure are unaffected by the government component.

the **Production Technology** is instead given by:

$$Y_t = A K_t^{\theta_K} N_t^{\theta_N} \left(K_t^G\right)^{\theta_G}.$$

with a law od motion for capital as:

$$K_{t+1} = (1-\delta)K_t + I_t$$

we assume her that $\theta_K + \theta_N = 1$ and we consider $\theta_G = 0$, specifying for the moment that governement spending does not affect production directly however we do have some indirect effects through labour suply and wealth ($\theta_G = 0$ pure demand drain).

The representative household faces the following constraints:

$$L_t + N_t \leq 1,$$

$$C_t + I_t \leq (1 - \tau_t)Y_t + TR_t.$$

TR_t is lump-sum transfers or taxes, and τ_t is the income tax (the representative household owns the firm).

Aggregate resource constraint:

$$Y_t = C_t + I_t + G_t,$$

$$G_t = G_t^B + I_t^G.$$

Government runs a balanced budget (no debt):

$$\tau_t Y_t = G_t + TR_t.$$

Equilibrium Conditions

The household maximises the Lagrangian

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t + \theta_L \log L_t + \Gamma(G_t^B, K_t^G) + \lambda_t \Big((1 - \tau_t)Y_t + TR_t - C_t - I_t \Big) \right],$$

subject to the technology, capital accumulation, and time endowment constraints:

$$Y_t = F(K_t, N_t, K_t^G), \qquad K_{t+1} = (1 - \delta)K_t + I_t, \qquad L_t + N_t = 1.$$

Since $N_t = 1 - L_t$, a rise in leisure reduces labor input, so $\partial Y_t/\partial L_t = -\partial Y_t/\partial N_t$. The three optimality conditions are as follows.

Consumption. The FOC with respect to C_t gives

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Longrightarrow \quad \frac{1}{C_t} = \lambda_t,$$

so the multiplier equals the marginal utility of consumption.

Leisure–Labor. The FOC with respect to L_t, combined with $\partial Y_t/\partial L_t = -\partial Y_t/\partial N_t$, yields the standard intratemporal condition:

$$\frac{\theta_L}{L_t} = \lambda_t (1 - \tau_t) \frac{\partial Y_t}{\partial N_t},$$

equating the marginal utility of leisure to the after-tax marginal product of labor weighted

by the shadow value of wealth.

Capital Accumulation. Substituting $I_t = K_{t+1} - (1-\delta)K_t$ into the budget constraint, K_{t+1} appears negatively at t and positively at $t+1$ through both production and undepreciated capital. The FOC with respect to K_{t+1} gives the intertemporal Euler equation:

$$\lambda_t = \beta\, E_t\left[\lambda_{t+1}\left((1-\tau_{t+1})\frac{\partial Y_{t+1}}{\partial K_{t+1}} + 1 - \delta\right)\right],$$

which equates the current shadow cost of investing one unit to the discounted expected return from holding that unit of capital net of depreciation and taxes.

On the firm side, we obtain (under perfect competition) that:

$$\frac{\partial Y_t}{\partial N_t} = W_t, \quad \frac{\partial Y_t}{\partial K_t} = R_t$$

The authors calibrate the model to some **great ratios**, so they calibrate them directly to postwar U.S. averages and then choose parameters so that the steady state reproduces those ratios. For example in steady state $L = 0.8$ and $N = 0.2$ and tehy then set $s_k = \frac{qK}{Y} = \theta_k = 0.42$ adn $s_n = \frac{wN}{Y} = \theta_n = 0.58$.

Labor Market Equilibrium. To make the elasticity unit-free and express it as a percentage evaluation, we compute

$$\frac{\partial L_t}{\partial w_t}\cdot\frac{w_t}{L_t},$$

which is the elasticity of leisure with respect to the wage. Combining labor supply with labor demand, from the intratemporal optimality condition we have:

$$L_t = \frac{\theta_L}{\lambda_t(1-\tau_t)w_t}.$$

Computing the elasticity:

$$\frac{\partial L_t}{\partial w_t}\cdot\frac{w_t}{L_t} = -\left(\frac{\partial L_t}{\partial w_t}\right)\cdot\left(\frac{w_t}{L_t}\right) = -\frac{\theta_L}{\lambda_t(1-\tau_t)w_t^2 L_t}\cdot w_t\cdot L_t\cdot\frac{\lambda_t(1-\tau_t)w_t}{\theta_L} = -1,$$

so a 1% higher wage implies 1% lower leisure.

To evaluate the effect on labor supply, we differentiate the time constraint $N_t + L_t = 1$ with respect to w_t and multiply both sides by w_t/L_t:

$$\frac{\partial N_t}{\partial w_t} + \frac{\partial L_t}{\partial w_t} = 0 \quad\Longrightarrow\quad \frac{\partial N_t}{\partial w_t}\cdot\frac{w_t}{L_t} + \frac{\partial L_t}{\partial w_t}\cdot\frac{w_t}{L_t} = 0.$$

Since the second term equals -1, we obtain:

$$\frac{\partial N_t}{\partial w_t}\cdot\frac{w_t}{L_t} = 1.$$

Multiplying both sides by L_t/N_t gives the **Frisch elasticity** of labor supply:

$$\text{Frisch} \equiv \frac{\partial N_t}{\partial w_t} \cdot \frac{w_t}{N_t} = \frac{L_t}{N_t}.$$

Calibrating with $L_t = 0.8$ and $N_t = 0.2$ yields a Frisch elasticity of 4, which is considerably higher than microeconomic estimates typically suggest, representing a well-known tension between RBC calibrations and labor supply evidence at the individual level.

Permanent Spending Shock Financed by Lump-Sum Taxation

A permanent increase in G_t financed by lump-sum taxes reduces household lifetime wealth. With Ricardian households, this triggers a **negative wealth effect**: consumption falls on impact and households substitute leisure for labor, increasing hours worked. The rise in labor supply raises the marginal product of capital, stimulating investment and triggering capital accumulation via

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Over time, the joint expansion of capital and labor increases productive capacity, partially offsetting the initial consumption decline.

This dynamic generates the key asymmetry between short-run and long-run multipliers. On impact the multiplier is approximately 0.8 — below unity since consumption falls before investment responds. In the long run it rises to 1.16, exceeding unity once capital has fully accumulated. The result underlines that **permanence** of the shock and the **capital accumulation channel** are the key drivers of multipliers above one, even in a frictionless neoclassical framework.

However, the entire mechanism here works in a **strange way** in particular, we have that since households become poorer, the $N \uparrow$ and $Y \uparrow$. So basically, we are making agents poorer and forcing them to work more to increase output, which does not make sense.

Here the crucial element is the **Frisch elasticity** in fatc using the calibration in Baxter and King (1993), we would produce a multiplier of 1.16. Using microeconometric evidence, we end up with a much lower multiplier of 0.49. It turns negative when considering distortionary taxation.

Temporary Spending Shock Financed by Lump-Sum Taxation

Still under the assumption of $\theta_G = 0$, we now consider a temporary shock of 4 years where G increases by a four years ump sum tax. Here, the effect is much lower than before.

Prductive Government spending

Here we assume that $\theta_G > 0$ so we allow for higher public investment to work as a productivity shock, obtaining therefore high values of fiscal multipliers for $\theta_G = \{0.01, 0.05, 0.10\} \implies$

$\mathcal{M} = \{1.45, 2.466, 4.12\}$, however, we see how the RBC models have clear problems in reproducing empirical evidence.

6.1.2 Fiscal Policy in the New Keynesian Model

The New Keynesian framework improves upon the neoclassical model by introducing nominal and real frictions that allow it to better replicate short-run macroeconomic dynamics. The key friction here is **sticky prices**: firms cannot freely adjust their prices in response to demand shifts, at least in the short run.

The transmission of a government spending shock $G_t \uparrow$ works as follows. The increase in public demand directly raises **aggregate demand**. Firms face higher demand for their products and would optimally like to raise prices, but price stickiness prevents immediate adjustment. Constrained from raising prices, firms instead respond by **increasing production**, which requires more labor. Labor demand rises, pushing wages up. The rise in wages increases households' disposable income, which further stimulates private consumption, amplifying the initial demand impulse.

This mechanism highlights a fundamental difference from the neoclassical model: in the New Keynesian framework, **output is demand-determined in the short run**. The fiscal multiplier can therefore exceed one, since the initial spending impulse propagates through demand complementarities rather than being offset by crowding out of private consumption via a negative wealth effect. The size of the multiplier depends critically on the degree of price stickiness, the response of monetary policy, and the persistence of the fiscal shock.

Despite successfully reproducing several features of the business cycle, the standard New Keynesian model tends to generate **low fiscal multipliers**. The reason is that households behave in a Ricardian fashion — internalising the government's intertemporal budget constraint and smoothing consumption according to the permanent income hypothesis. A tax-financed spending increase therefore triggers a negative wealth effect that partially offsets the demand stimulus.

To address this, Galí et al. (2007) introduces **agent heterogeneity** into the New Keynesian framework. A share λ of households are **rule-of-thumb** (or hand-to-mouth) consumers: they have no access to financial markets and simply consume their current disposable income each period. The remaining share $1 - \lambda$ are **Ricardian optimizers** who smooth consumption intertemporally via the Euler equation and also own the firms, determining investment decisions. This two-agent structure breaks the full Ricardian offset: when government spending rises and wages increase, rule-of-thumb households — whose consumption moves one-for-one with current income — immediately consume more, amplifying the aggregate demand response and generating larger fiscal multipliers than the representative agent model predicts.

Ricardian Households $(1-\lambda)$

The economy is populated by two types of households and a continuum of firms operating under monopolistic competition.

Ricardian households have full access to asset markets and maximise lifetime utility:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t^o - \frac{N_t^{o\,1+\varphi}}{1+\varphi} \right],$$

subject to the budget constraint

$$P_t(C_t^o + I_t^o + B_t^o) = W_t P_t N_t^o + R_t^k P_t K_t^o + \mathcal{B}_t^o + D_t^o,$$

where $\mathcal{B}_t^o$ denotes bond holdings and D_t^o are dividends from firm ownership. Capital evolves according to

$$K_{t+1} = (1-\delta)K_t + \left[1 - S\left(\frac{I_t}{I_{t-1}}\right)\right] I_t,$$

where $S(\cdot)$ captures investment adjustment costs with $S'' > 0$.

Rule-of-Thumb Households (λ)

Rule-of-thumb households have no access to financial markets and therefore cannot smooth consumption. They maximise period-by-period utility:

$$\max\ U_t \quad \text{s.t.} \quad P_t C_t^r = W_t N_t^r - P_t T_t^r,$$

so their consumption is determined entirely by current disposable labor income. In addition, under perfectly competitive labor markets, both types supply labor at the common wage:

$$W_t \ell = \frac{W_t}{C_t^r}.$$

Firms

Firms operate under monopolistic competition. Each period they reset their price with probability $1-\theta$, keeping the previous price with probability θ. Knowing this, the optimal reset price equals the expected marginal cost over all future periods in which the current price remains in place. Log-linearised, this yields the standard **New Keynesian Phillips Curve**:

$$\pi_t = \beta E_t[\pi_{t+1}] + \lambda_p mc_t.$$

Monetary and Fiscal Policy

The central bank follows a **Taylor rule** with no output gap and satisfying the Taylor principle ($\phi_\pi > 1$):

$$i_t = r + \phi_\pi \pi_t, \qquad \phi_\pi > 1.$$

Government spending follows an **AR(1)** process:

$$G_t = \rho_G G_{t-1} + \varepsilon_t^G,$$

and fiscal policy is governed by a **tax rule**:

$$t_t = \phi_\tau b_t + \phi_G G_t,$$

where b_t denotes the debt level. The budget is not required to balance period by period — deficits are permitted in the short run — but debt must return to its steady state in the long run.

Now the solution of the model lead to the following lnearized euler equation which we can write as:

$$c_t = E_t[c_{t+1}] - \sigma(r_t - E_t[\pi_{t+1}]) - \Theta_n E_t[\Delta n_{t+1}] + \Theta_\tau E_t[\Delta t^r_{T+1}]$$

Here we have that $\Theta_n E_t[\Delta n_{t+1}]$ is the expected change in labour income, so if the labour is expected to change, then the rule of thumb will start spending more. Then $\Theta_\tau E_t[\Delta t^r_{T+1}]$ represents expected taxation, where if it rises then the rule of thumb household will start to consume more.

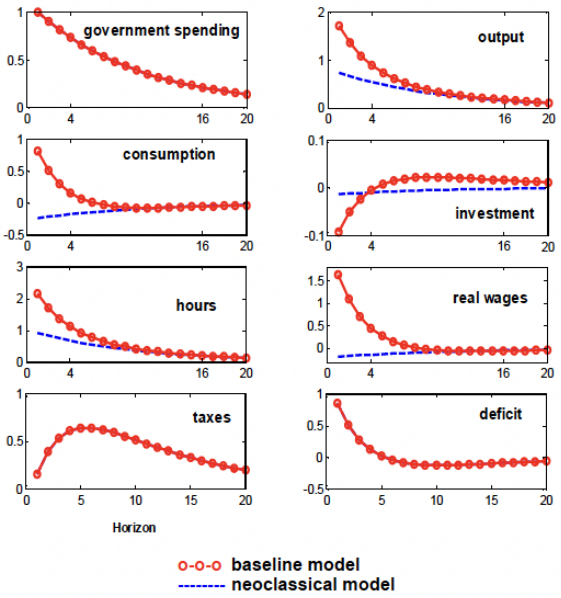
the parameters $\sigma, \Theta_{n,\tau}$ are related to specific labour market structures. Here we can see why the Ricardian equivalence breaks down, $n \uparrow \implies c \uparrow$ and also delayed taxation produces higher consumption. As $\lambda \to 0, \Theta_{n,\tau} \to 0$ we go back to the standard version.

Government Spending shock

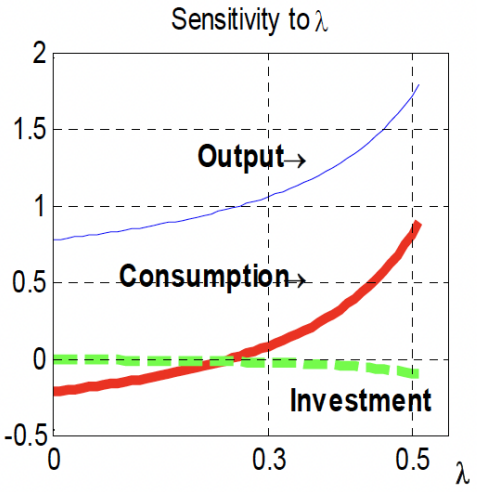
Now that the model is set, we can observe and study the effects of government spending shocks with and without limited asset participation and focus on the model with imperfect competitive labour markets.

$\uparrow G \Rightarrow \uparrow$ Aggregate Demand $\Rightarrow$ (with sticky prices) P_i too low $\Rightarrow \uparrow Y_i \Rightarrow \uparrow N_i \Rightarrow \uparrow H_i \Rightarrow \uparrow W_i^R$

Here, the rule of thumb households are the ones that increase consumption and lead to higher levels of C since they do not care about their future level of income will decrease. Intuitively, the λ would control the total effect of consumption.



(a) Impulse responses to a government spending shock: baseline Galì et al. model (red, circles) versus the neoclassical flex-price model with no rule-of-thumb households (blue, dashed).



(b) Impact multipliers for output, consumption, and investment as a function of the share λ of rule-of-thumb households.

Figure 6.1: Galì et al. (2007): model comparison and sensitivity analysis. The baseline model features sticky prices and a share λ of rule-of-thumb households; the neoclassical benchmark has flexible prices and only Ricardian agents.

Figure 6.1a compares the impulse responses of the two models following a government spending shock. The key difference lies in the response of **consumption** and **real wages**. In the neoclassical model, consumption falls on impact due to the standard negative wealth effect on Ricardian households. In the baseline model, by contrast, sticky prices prevent firms from raising prices immediately, so they expand production and increase labor demand, pushing real wages up. Rule-of-thumb households, whose consumption moves one-for-one with current labor income, therefore consume *more*, generating a positive consumption response absent in the neoclassical benchmark. Hours and output both rise more strongly in the baseline, while investment falls slightly due to the crowding-out effect of higher government spending.

Figure 6.1b shows the sensitivity of impact multipliers to the share λ of rule-of-thumb households. Three results stand out. First, the **output multiplier** is increasing in λ and exceeds unity for $\lambda \gtrsim 0.3$, since a larger share of hand-to-mouth consumers amplifies the aggregate demand response. Second, **consumption** switches from negative to positive as λ rises, reflecting the growing importance of non-Ricardian behavior. Third, **investment** remains negative throughout and becomes more negative at high λ, consistent with crowding out of private capital. These results highlight that the degree of household heterogeneity is a key determinant of the size and sign of fiscal multipliers.

6.2 Empirical Evidence in SVAR Models

The identification of structural shocks in SVARs relies on timing (Cholesky) restrictions that encode assumptions about within-period decision-making.

- **Monetary policy**: the central bank observes current output and inflation before setting the interest rate. Hence, interest rates are ordered *last*: they respond contemporaneously to output and inflation, but transmit to the real economy only with a lag.
- **Fiscal policy**: government spending is assumed to be predetermined relative to output — budgets are set in advance and cannot respond to output within the same period. Hence, government spending is ordered *first*: it affects output contemporaneously, but does not react to it.

Causal identification of fiscal policy shocks is substantially harder than for monetary policy. The raw correlation between government spending and output conflates several distinct mechanisms:

- **Automatic stabilizers**: tax revenues fall and transfers rise mechanically during recessions, even without any discretionary policy action.
- **Discretionary countercyclical policy**: governments may deliberately increase spending precisely when output is low, inducing a negative correlation between spending and output that has nothing to do with the causal multiplier.
- **Procyclical austerity**: conversely, governments may cut spending in downturns to contain deficits, again confounding the fiscal-output relationship.

This is a fundamental **endogeneity problem**: G_t and Y_t are jointly determined, so OLS — or a naive Cholesky ordering — does not recover the causal effect of an exogenous spending increase. If governments systematically spend more in bad times, the estimated multiplier will be *downward biased*, understating the true expansionary effect of fiscal policy. Identifying a genuine fiscal *shock* therefore requires isolating variation in G_t that is orthogonal to the state of the business cycle — a much stricter requirement than the within-period timing restriction sufficient for monetary policy.

Here the main point is that $Cov(Y_t, G_t) \neq \frac{\partial Y_t}{\partial G_t}$ so we want to disentangle this effect and find or identify shocks that are exogenous to fiscal policy so not state dependent.

Now possible identification strategies should be grounded on how the fiscal policy is transmitted in the real economy in particular we have:

- **Long inside lags**: takes time before new legislations is passed.
- **Short Outside lags**: Immediate impact on GDP.

6.2.1 Blanchard and Perotti (2002)

Now the key identification assumption that Blanchard and Perotti (2002) propose is that we do not have effects from discretionary from fiscal policy within lag, so it takes more than one lag (quarter) before legislation is passed and takes action so no effect of fiscal policy on output within lag (this assumption is true whenever we consider quarterly data).

Here we introduce the VAR model used by the authors so we get:

$$Y_t = A(L, q)Y_{t-1} + U_t$$

where we define $Y_t = [T_t, G_t.X_t]'$ is a vector of variables where T_t is tax revenues net of transfer payments, G_t is government spending and investment, X_t is GDP. We then identify $U_t = [t_t, g_t, x_t]'$ which is a vector of estimated residuals. Now we ant to obtain a direct mapping from the estiated residuals to structural shocks therefore, we consider $U_t = \boldsymbol{B}_0^{-1} E_t$ where $E_t[e_t^t, e_t^g, e_t^x]'$ so to obtain:

$$t_t = a_1 x_t + a_2 e_t^g + e_t^t,$$
$$g_t = b_1 x_t + b_2 e_t^t + e_t^g,$$
$$x_t = c_1 t_t + c_2 g_t + e_t^x.$$

We now move to formal identification. The reduced-form VAR residuals u_t are related to the underlying structural shocks ε_t through a linear mapping:

$$u_t = B\varepsilon_t,$$

where B is the impact matrix to be identified. Since the structural shocks are assumed orthogonal ($\mathbb{E}[\varepsilon_t \varepsilon_t'] = I$), the covariance matrix of the residuals satisfies $\Sigma = BB'$. This system is under-identified without further restrictions — Σ has $n(n+1)/2$ free parameters, while B has n^2, leaving $n(n-1)/2$ restrictions to be imposed.

The key parameters are a_1 and b_1, the **within-period elasticities** of taxes and government spending with respect to the output residual u_t^y. Each captures two conceptually distinct forces:

- **Automatic stabilizers**: output fluctuations mechanically affect fiscal variables — tax revenues move with income and profits, while transfers such as unemployment benefits move counter-cyclically, independently of any policy decision.
- **Discretionary response**: policymakers may also adjust taxes or spending within the period in response to an exogenous output movement, reflecting active stabilization policy.

The identification strategy of Blanchard and Perotti (2002) exploits the fact that **discretionary fiscal policy cannot be enacted within a quarter** — legislative and

implementation lags make it implausible that governments respond to output innovations within the same period. This means a_1 and b_1 capture *only* automatic stabilizer effects, which can be calibrated externally from institutional information (e.g., the tax code elasticity with respect to GDP). Once a_1 and $b_1 = 0$ are pinned down this way, the remaining structural shocks — the truly exogenous, discretionary fiscal impulses — are identified as the residual variation left after purging the automatic component. Having obtained a_1 and b_1 from external calibration, we can construct **cyclically-adjusted residuals** that strip out the automatic stabilizer component:

$$t'_t = u^t_t - a_1 u^x_t, \qquad g'_t = u^g_t - b_1 u^x_t = u^g_t,$$

where the second equality reflects the assumption $b_1 = 0$ — government spending does not respond automatically to output within a quarter. By construction, t'_t and g'_t are orthogonal to the structural output shock ε^x_t, since we have already removed everything that co-moves mechanically with output. Crucially, however, they remain correlated with the discretionary fiscal shocks ε^t_t and ε^g_t — they contain the exogenous policy variation we care about.

This makes them valid **instruments**: they satisfy both relevance (correlated with fiscal residuals) and the exclusion restriction (orthogonal to ε^x_t). We can therefore estimate the output equation:

$$u^x_t = c_1 t'_t + c_2 g'_t + \varepsilon^x_t,$$

recovering the contemporaneous effects of fiscal shocks on output.

The remaining ordering ambiguity. One identification problem remains: t'_t and g'_t are still mutually correlated, since both contain discretionary shocks that may respond to each other within the period. To resolve this, Blanchard and Perotti impose either $a_2 = 0$ (spending does not respond to tax shocks within the quarter) or $b_2 = 0$ (taxes do not respond to spending shocks within the quarter).

The key insight is that **this choice is inconsequential for the main results**. Whether taxes or spending is ordered first affects only the correlation between the two fiscal shocks themselves — not their joint effect on output. Since the object of interest is the response of output and other macro variables to each fiscal shock, and since the two orderings produce nearly identical impulse responses in practice, the identification is robust to this residual ambiguity.

From this identification we can obtain some results that can be summarized as follows:

- 1 dollar increase in taxation leads to a GDP decrease of 0.75 dollars, with a decline that is quick and persistent.
- An increase in government spending leads to an increase of GDP of 0.8 on impact

rising to 1.25 dollars after 3-4 years.

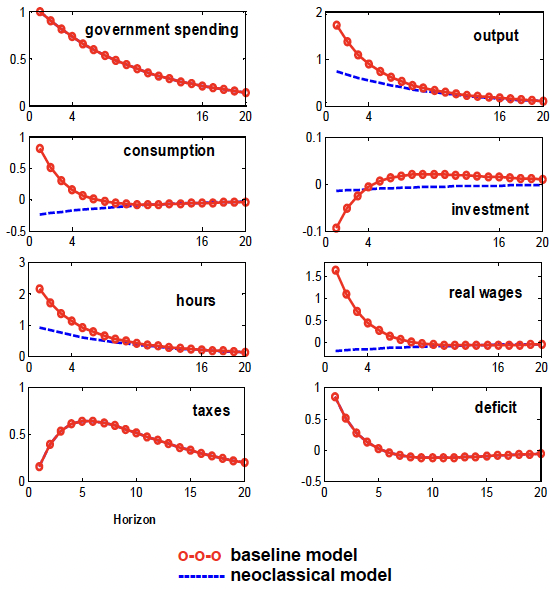
- Private consumption increases significantly and persistently after spending shock (not as predicted from the RBC and NK classical) and declines after a tax shock.
- Investment and net exports both decline aster a spending shock.

The authors also verify the entire framework with some robustness checks by revising subsample stability (not very stable) and other values of a_1 (same shape but different multiplier range 0.5 to 1.05)

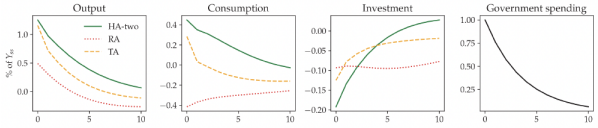
Critiques

We can now introduce some relevant critiques that have been raised against the strategy proposed by Blanchard and Perotti (2002) in particular we have:

- **Antipation effects**: in this paer the shock hits wne it can be observed in national accounts however private agents learn about the change in fiscal policy well before that date.
- **Fast Policy**: like covid 19 could lead to policy processes that are very fats therefore violating the identification assumption.



(a) IRFs to a government spending shock: NK model with rule-of-thumb households (baseline, red) vs. neoclassical RBC model (blue). The NK-RoT model generates a positive consumption response — breaking the neoclassical crowding-out result — driven by the wealth effect on rule-of-thumb households who consume their current labor income. Real wages and hours both rise, while investment crowds out only modestly. The neoclassical model instead predicts falling consumption and real wages, as the negative wealth effect from higher future taxes dominates.



(b) IRFs to a government spending shock: standard NK model with heterogeneous agents featuring two-asset households (HA-two, green), representative-agent Ricardian households (RA, red dotted), and two-agent model with rule-of-thumb households (TA, orange dashed). Output and consumption responses are largest under HA-two, reflecting stronger hand-to-mouth dynamics and general equilibrium amplification. The RA model delivers the weakest consumption response and the largest investment crowding-out, consistent with standard NK predictions. The TA model lies between the two, illustrating how the share of non-Ricardian households governs the size of the fiscal multiplier.

Figure 6.2: Impulse responses to a government spending shock across model specifications. The key takeaway across both panels is that departures from full Ricardian equivalence — whether through rule-of-thumb behavior or richer household heterogeneity — are necessary to match the empirically observed co-movement of consumption and government spending.

6.2.2 Narrative Approach

As discussed in the previous section, one response to the identification problems of the BP approach is to exploit **external, non-statistical information** to construct measures of exogenous fiscal shocks. Rather than relying on orthogonality conditions derived from the data, the narrative approach searches for historical episodes where changes in government spending were driven by events plausibly unrelated to the current state of the economy.

Ramey and Shapiro (1998)

Ramey and Shapiro (1998) examine weekly editions of *Business Week* and identify large, discrete increases in US defense spending triggered by three major geopolitical events: the Korean War (1950), the Vietnam War buildup (1965), and the Carter-Reagan defense expansion (1980). The identifying assumption is that these military buildups were driven by geopolitical considerations orthogonal to the domestic business cycle, providing a source of plausibly exogenous variation in government spending.

Ramey (2011)

Ramey (2011) extends this approach by constructing a narrative measure of **defense spending news** — capturing the present discounted value of expected future military spending at the time of each episode — and including it as an instrument in the VAR. The news variable is ordered *first* in the system, before government spending, output, and other endogenous variables. This ordering ensures the identified shock is not contaminated by contemporaneous responses of the economy, interpreting it as a genuine surprise to agents.

The approach yields a **positive output multiplier**, broadly consistent with BP. However, the response of private consumption is **negative** — in line with the neoclassical wealth effect — whereas BP finds a positive consumption response. This discrepancy is consequential and remains unresolved in the literature.

Limitations. The approach faces several important criticisms:

- **Few identified episodes**: the entire identification rests on three or four historical events, raising concerns about statistical power and out-of-sample relevance.
- **External validity**: the shocks are driven by US military spending during the Cold War era, making it unclear whether the estimated multipliers generalize to peacetime fiscal expansions or other economies.
- **Composition**: military spending is a specific and arguably atypical component of G. Whether its multiplier is informative about the effects of, say, infrastructure investment or social transfers is questionable.

- **Anticipation**: as discussed above, even narrative dates may themselves be anticipated if geopolitical tensions build gradually before a formal announcement, leaving the approach vulnerable to the same fiscal foresight critique it was designed to address.

6.2.3 Auerbach and Gorodnichenko (2012)

Auerbach and Gorodnichenko (2012) estimate a **Smooth Transition VAR (STVAR)** where VAR parameters vary continuously with the state of the business cycle via a smooth weighting function $F(z_t) \in [0, 1]$, with z_t a measure of GDP growth. This allows the fiscal multiplier to be state-dependent rather than a fixed parameter. The key finding is that multipliers are large in recessions (1–1.5) and small in expansions (0–0.5), suggesting fiscal stimulus is most effective precisely when the economy has slack — consistent with models featuring binding credit constraints or a high share of hand-to-mouth households during downturns.

6.2.4 The Fiscal Price Puzzle: Jørgensen and Ravn (2022)

A robust empirical finding across identification strategies is that government spending expansions are accompanied by a **fall** in the price level — directly contradicting the standard prediction of any demand-driven model, where fiscal expansion raises aggregate demand and hence prices.

Jørgensen and Ravn (2022) document that this *fiscal price puzzle* is robust across sample periods, identification strategies, and price indices. Crucially, the price drop co-exists with increases in output, consumption, **and total factor productivity (TFP)** — suggesting the mechanism is supply-side rather than demand-side. They propose a New Keynesian model augmented with **endogenous TFP** through variable technology adoption and utilization: a government spending shock raises demand, inducing firms to adopt and utilize technology more intensively, which raises TFP, lowers marginal costs, and puts downward pressure on prices. The model successfully accounts for the joint response of output, consumption, TFP, and the price level — resolving the puzzle within a coherent NK framework.

Chapter 7

New Empirical Methods in Macroeconomics

We now introduce a new methodology to re-evaluate macroeconomic questions: **Local Projections** (LP), proposed by Jordà (2005).

7.1 Local Projections

So far we have used VAR models as our main tool for empirical macroeconomic analysis, imposing economic structure via the SVAR to compute impulse response functions (IRFs). The SVAR approach involves two independent procedural steps:

- **Estimation** of the reduced-form VAR by OLS;
- **Identification**: imposing structure (e.g. Cholesky decomposition, sign restrictions, external instruments) to recover structural shocks and derive IRFs.

Local Projections alter only the *estimator* used to compute IRFs, leaving the identification strategy entirely unchanged. LP is *not* an identification strategy. Whether one uses Cholesky identification, narrative restrictions, sign restrictions, or external instruments, those choices are made prior to and independently of the decision to use LP or VAR-based IRFs. The two dimensions — identification and IRF estimation — are orthogonal.

7.1.1 Motivation: Why Local Projections?

The VAR approach to IRFs. Recall that from the estimated VAR coefficient matrices $\{\hat{\Phi}_j\}_{j=1}^{p}$ one can invert the system to obtain a Wold moving-average (MA) representation. The structural IRF of y_t at horizon s to the j-th structural shock is

$$\frac{\partial y_{t+s}}{\partial \zeta_{jt}} = A_s H e_j,$$

where e_j is a selector vector with a 1 in the j-th position and zeros elsewhere, $H = B_0^{-1}$ maps reduced-form residuals to structural shocks via $\varepsilon_t = B_0^{-1}\zeta_t$, and the dynamic multiplier matrices A_s satisfy the recursion

$$A_i = \sum_{j=1}^{p} \Phi_j A_{i-j}, \quad i = 1, 2, \ldots, \qquad A_0 = I_m, \quad A_i = \mathbf{0} \text{ for } i < 0.$$

Conceptually, a VAR-based IRF is the difference between two multi-step forecasts: one conditional on the shock ζ_{jt} having occurred, and one without it. The estimated autoregressive structure is "rolled forward" period by period to produce the full IRF path. An important implication is that to estimate a VAR with p lags one needs as few as $p + 1$ time periods, yet the model can extrapolate the effect of a shock to an arbitrarily distant horizon. This parsimony is both a strength and a weakness.

The bias–variance trade-off. Imposing the autoregressive structure of the VAR yields smooth, low-variance IRFs. However, if the VAR is misspecified — due, for instance, to an incorrect lag order or omitted variables, which is ubiquitous in practice — misspecification is *compounded* at each step of the recursion: it enters the innovations and is reinforced at every subsequent horizon, leading to IRFs that are biased in a way that grows with the forecast horizon. This is the fundamental drawback of the VAR approach to IRF estimation.

The LP solution. Local Projections address this problem by estimating each point of the IRF directly and independently via a separate regression of $y_{i,t+s}$ on the shock of interest. This breaks the multiplicative recursive structure: misspecification at horizon s does not propagate to horizon $s + 1$. The trade-off is increased variance — LP-based IRFs tend to be more jagged and less smooth than VAR-based ones since much less structure is imposed — but bias from model misspecification does not accumulate. LP thus occupies a different point on the bias-variance frontier: *less biased, higher variance* than the VAR.

7.1.2 The Local Projection Estimator

The LP estimator at horizon s is obtained from the regression

$$y_{i,t+s} = \theta_{i,s}\, \zeta_{jt} + Z_t'\gamma_s + v_{i,t+s}, \tag{7.1}$$

where:

- $y_{i,t+s}$ denotes the value of variable i observed s periods after time t;
- ζ_{jt} is the structural shock of type j at time t, which still requires a set of identifying assumptions to recover — the LP regression takes the shock as given, it does not

produce it;

- $\theta_{i,s}$ is the response coefficient of variable i at horizon s, measuring the effect of a one-unit structural shock on $y_{i,t+s}$;
- Z_t is a vector of control variables observed at time t, typically including lags of the endogenous variables and other predetermined controls;
- γ_s is the corresponding coefficient vector at horizon s;
- $v_{i,t+s}$ is the regression error, collecting all factors affecting $y_{i,t+s}$ not captured by the shock or the controls.

Equation (7.1) is estimated separately for each horizon $s = 0, 1, \ldots, H$. The IRF of variable i to shock j is then the sequence of OLS estimates

$$\left\{\hat{\theta}_{i,s}\right\}_{s=0}^{H}.$$

An LP-based IRF of H horizons thus consists of $H + 1$ separate regressions, each with a different left-hand-side variable but run on the same sample.

7.1.3 Choice of Controls

A natural baseline for Z_t includes a constant, time trends, and period dummies. In macroeconomic applications one additionally includes lagged values of the endogenous variables of interest — output, inflation, and the interest rate, for instance — much as one would in a VAR. Further controls (unemployment, commodity prices, stock prices, etc.) can be added if deemed important. In that sense, LP with a rich control set is not conceptually far from a VAR.

Indeed, Plagborg-Møller and Wolf (2021) show that, when the lag structure of the controls is sufficiently rich, LP and VAR estimate *the same* IRFs in population (i.e. they share the same estimand). The two methods therefore differ only in their finite-sample properties: LP sacrifices efficiency (higher variance) to gain robustness to misspecification (lower bias).

7.1.4 LP in Practice

LP has become increasingly popular over the last decade, in large part because of a parallel shift in identification strategies away from Cholesky-type recursive restrictions toward more credible external instruments. Narrative measures of monetary policy surprises (e.g. Romer and Romer, 2004b) or fiscal policy surprises (e.g. Auerbach and Gorodnichenko, 2012; Ramey, 2011) can be plugged directly into (7.1) as ζ_{jt} without the need to first estimate a full VAR system. LP is therefore particularly attractive when a credible shock

series is available externally: one need not commit to the full dynamic structure of a VAR merely to identify the shocks.

7.2 Regional Variation

Macroeconomic empirical work is increasingly moving toward data-rich environments that disaggregate aggregate time series and exploit new sources of variation — in particular, regional variation across states or metropolitan areas. The aggregate time series available to macroeconomists is short, and the information it contains on long-debated questions may be close to exhausted. Regional data, by contrast, offers a large cross-sectional dimension that can be combined with time-series variation, enabling sharper identification.

This shift is partly a response to the so-called *credibility revolution* in microeconometrics.[1] While not all aspects of this revolution carry over directly to a macroeconomic context — general equilibrium effects, monetary policy responses, and the absence of a clean control group are all complications — macroeconomists have widely adopted its tools and standards.

A prominent strand of this literature exploits regional variation in federal defense spending across U.S. states and Metropolitan Statistical Areas (MSAs) to estimate the causal effect of government spending shocks. The key observation is that when the federal government allocates defense expenditure, these funds do not flow uniformly across regions: some areas host large military bases or defense contractors, others do not. This heterogeneity in exposure can be used to estimate the effect of one region receiving an additional dollar of government spending *relative* to other regions, holding aggregate conditions fixed. Pioneered by Nakamura and Steinsson (2014), this approach has been extended and refined by Auerbach et al. (2020) and Larsen (2023), among others.

Following Nakamura and Steinsson (2014), the baseline estimating equation takes the form of a panel LP:

$$\frac{Y_{i,t+h} - Y_{i,t}}{Y_{i,t}} = \alpha_i + \gamma_{t+h} + \beta_h \frac{G_{i,t} - G_{i,t-1}}{G_{i,t-1}} + \lambda X_{i,t} + \varepsilon_{i,t+h}, \tag{7.2}$$

where $Y_{i,t}$ is the outcome of interest (output, employment, house prices, etc.) in region i at time t, so the left-hand side is the h-period cumulative growth rate of Y; $G_{i,t}$ is federal defense spending directed to region i, with its one-period growth rate $(G_{i,t} - G_{i,t-1})/G_{i,t-1}$ entering as the main regressor; α_i is a region fixed effect absorbing permanent cross-sectional heterogeneity; γ_{t+h} is a time fixed effect that differences out aggregate shocks

[1] In the 1980s and 1990s, empirical macro-micro work relied heavily on structural models where identification was often indistinguishable from functional-form assumptions — in effect, curve fitting. The credibility revolution, associated with Card, Angrist, Imbens and others, reoriented applied work around natural experiments and quasi-random variation to identify causal effects on outcomes, setting a higher bar for identification that macroeconomists have increasingly sought to meet.

common to all regions; $X_{i,t}$ is a vector of additional controls with coefficient λ; and $\varepsilon_{i,t+h}$ is the error, clustered at the region level to account for serial correlation induced by the overlapping structure of the dependent variable when $h > 0$. The parameter of interest is β_h: the effect of a one-percentage-point increase in local spending at t on the outcome h periods later. Collecting $\{\hat{\beta}_h\}_{h=0}^{H}$ traces out a regional IRF, in direct analogy with the LP framework of equation (7.1).

OLS estimation of (7.2) is, however, likely to be biased. There is ample evidence that federal military contracts tend to flow disproportionately toward regions experiencing adverse economic conditions, inducing a negative correlation between the spending regressor and the error term that attenuates OLS estimates of β_h toward zero. To address this endogeneity, Nakamura and Steinsson (2014) and subsequent authors adopt an IV strategy based on the **Bartik** (1991) **shift-share** instrument, constructed as

$$Z_{i,t} = \bar{s}_i \cdot \frac{\Delta G_t^N}{G_{t-1}^N}, \tag{7.3}$$

where $\Delta G_t^N / G_{t-1}^N \equiv (G_t^N - G_{t-1}^N)/G_{t-1}^N$ is the growth rate of *national* defense spending at time t — the **shift** — and $\bar{s}_i$ is region i's pre-determined average share of federal military spending relative to local GDP — the **share**. The instrument therefore interacts an aggregate shock, common to all regions, with a region-specific measure of baseline exposure. Its validity rests on two conditions. First, **relevance**: regions with a higher baseline defense share $\bar{s}_i$ receive proportionally larger spending increases when the national budget G_t^N expands, ensuring a strong first stage. Second, **exogeneity**: the national component $\Delta G_t^N / G_{t-1}^N$ is driven by geopolitical considerations and federal budget decisions orthogonal to the economic conditions of any single region, while $\bar{s}_i$ is pre-determined and thus uncorrelated with current regional shocks. The identifying assumption is therefore that, conditional on fixed effects and controls, $Z_{i,t}$ affects regional outcomes only through its effect on local spending.

The regional approach, while offering cleaner identification than aggregate time-series methods, measures something conceptually distinct from the standard aggregate fiscal multiplier. Two systematic forces drive a wedge between the regional and aggregate estimates, both pushing the regional multiplier *upward*.

Tax financing externality. Government spending must ultimately be financed by taxes. At the aggregate level, a spending increase ΔG requires a present-value-equivalent tax rise, reducing households' disposable income and partially offsetting the stimulus. In the regional setting this offset is nearly absent: when region i receives ΔG_i in additional defense spending, the associated tax burden is spread across all regions proportionally to their income shares, so region i bears only $\Delta G_i / N \approx 0$ of the cost while capturing the full demand benefit. The negative wealth effect of taxation is effectively *externalized* to the

rest of the country, inflating the local multiplier relative to the aggregate one.

Absence of monetary policy response. At the aggregate level, a fiscal expansion raises output and inflation, prompting the central bank to raise interest rates and crowd out private spending. In the regional setting, monetary policy responds to *national* aggregates: an increase in defense spending in a single state is too small to trigger a Federal Reserve reaction, so the real interest rate faced by local households and firms is unchanged and the crowding-out channel is entirely absent.

The open-economy relative multiplier. For these two reasons, β_h in (7.2) is not the standard closed-economy aggregate multiplier but what Nakamura and Steinsson (2014) call an **open-economy relative multiplier**: it captures the response of region i to a spending increase *relative* to other regions, holding the aggregate monetary stance fixed and externalizing the fiscal cost. Under additional structural assumptions — typically a two-region general-equilibrium model — one can map this regional estimate back to the aggregate multiplier by reintroducing the tax and monetary policy channels analytically.

7.3 Estimation of the Phillips Curve

For decades, macroeconomists have sought to estimate the slope of the Phillips curve using aggregate time-series data, progressively exhausting the information that such data can credibly provide. A central unresolved question in this literature is whether the Phillips curve has *flattened* over time — that is, whether the sensitivity of inflation to labor market slack has declined.

To fix ideas, consider the New Keynesian Phillips Curve (NKPC):

$$\pi_t = \beta E_t \pi_{t+1} - \kappa \left(u_t - u_t^n\right) + v_t, \tag{7.4}$$

where π_t is inflation, $E_t\pi_{t+1}$ is expected inflation one period ahead, $(u_t - u_t^n)$ is the unemployment gap, v_t is a cost-push shock, and $\kappa > 0$ governs the slope: for a given deviation of unemployment from its natural rate, a larger κ produces larger movements in inflation.

Figures 7.1 and 7.2 illustrate the empirical puzzle motivating this literature. U.S. unemployment has continued to fluctuate substantially over the post-war period, yet inflation became remarkably stable after the early 1980s — with the notable exception of the post-pandemic episode. If the variance of $(u_t - u_t^n)$ has not fallen, then stable inflation must reflect a smaller κ, i.e. a flatter Phillips curve. Formally, the question is whether κ has declined over time, and if so, by how much.

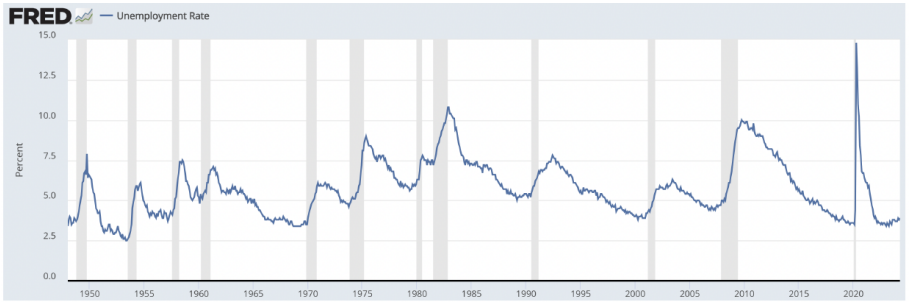


Figure 7.1: U.S. unemployment rate, 1950–present. Shaded areas denote NBER recessions.

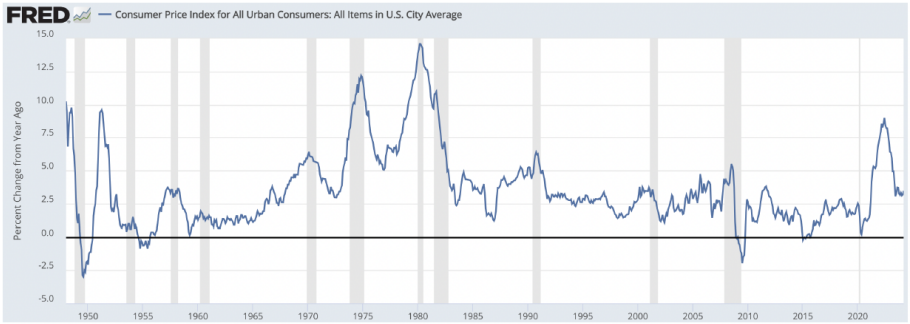


Figure 7.2: U.S. CPI inflation (year-on-year), 1950–present. Inflation stabilized markedly after the Volcker disinflation of the early 1980s.

Estimating κ from aggregate data is, however, complicated by the behavior of long-run inflation expectations $E_t\pi_{t+\infty}$. Solving (7.4) forward under the assumption that unemployment follows an AR(1) process yields:

$$\pi_t = -\psi\,\tilde{u}_t + E_t\pi_{t+\infty} + \omega_t, \tag{7.5}$$

where $\tilde{u}_t$ denotes the unemployment deviation from its long-run level, $\psi > 0$ is a composite slope parameter, and ω_t is an error term. Two problems arise immediately. First, if $E_t\pi_{t+\infty}$ fluctuates substantially over time — as it did during and after the Volcker disinflation — it acts as a confounder: movements in long-run expectations are absorbed into the error term and, if correlated with $\tilde{u}_t$, bias the OLS estimate of ψ. Second, large variation in $E_t\pi_{t+\infty}$ reduces the signal-to-noise ratio in regressions of π_t on $\tilde{u}_t$, weakening identification. Both problems suggest that estimates of κ from aggregate data may be unreliable, providing the motivation for a regional approach.

Hazell et al. (2022) propose to exploit regional variation across U.S. states to estimate the slope of the Phillips curve, with an important theoretical contribution: they formally relate the regional slope parameter to the aggregate one, so that the regional estimate can be used to make statements about the aggregate Phillips curve. The key insight is that long-run inflation expectations $E_t\pi_{t+\infty}$ — the confounder that plagues aggregate estimation — are constant across states at any given point in time. This follows directly from the fact that U.S. states form a monetary union: the Federal Reserve sets a single

monetary policy for the entire country, and long-run inflation expectations are anchored by that common regime rather than by state-level conditions. Crucially, $E_t\pi_{t+\infty}$ therefore has no cross-sectional dimension — it varies only over time, not across states. In a panel regression that includes time fixed effects γ_t, this term is completely absorbed and partialled out:

$$\pi_{i,t} = -\psi\,\tilde{u}_{i,t} + \underbrace{E_t\pi_{t+\infty}}_{\text{absorbed by } \gamma_t} + \omega_{i,t}, \tag{7.6}$$

eliminating the source of bias that makes aggregate estimation unreliable. What remains is purely the within-period, cross-state variation in inflation — variation that, by construction, cannot be driven by aggregate expectation shifts and must instead reflect differences in regional labor market conditions, allowing a clean identification of κ.

The Hazell et al. (2022) Model

Hazell et al. (2022) construct a two-region New Keynesian model set in a monetary union, so that both regions share the same monetary and fiscal policy. Labour is immobile across regions, firms in both the tradeable and non-tradeable sectors face Calvo (1983) price rigidities, and financial markets between regions are complete. The model delivers two Phillips curves with the same slope coefficient κ. The aggregate Phillips curve is:

$$\pi_t = \beta E_t\pi_{t+1} - \kappa\,\hat{u}_t + v_t,$$

and the regional Phillips curve for non-tradeable goods inflation in the home region is:[2]

$$\pi^N_{Ht} = \beta E_t\pi^N_{H,t+1} - \kappa\,\hat{u}_{Ht} - \lambda\hat{p}^N_{Ht} + v^N_{Ht}. \tag{7.7}$$

The fact that κ is identical in both equations is the key result: it implies that estimating the regional Phillips curve for non-tradeable inflation consistently recovers the same slope parameter that governs the aggregate relationship, providing a direct bridge between the regional empirical strategy and aggregate policy-relevant inference.

The results deliver two main findings. First, the Phillips curve is positively sloped throughout the sample — κ and ψ are positive and precisely estimated — but remarkably flat: even pre-1990, $\kappa \approx 0.01$, implying that a one-percentage-point rise in unemployment reduces non-tradeable inflation by only about one basis point per quarter. Second, the curve did flatten after 1990, with κ roughly halving across all specifications, but the decline is modest in absolute terms since the curve was already flat before 1990. Critically, comparing columns (1)–(2) with (3)–(4) reveals that the apparent drop in κ is dramatically

[2] $\hat{p}^N_{Ht}$ is the percentage deviation of the non-tradeable goods price in region H relative to the overall regional CPI from its steady-state value, capturing the relative price of non-tradeables within the region. It enters with a negative sign since, when non-tradeable goods are already relatively expensive, further inflationary pressure in that sector is dampened.

Table 7.1: Has the Phillips Curve Flattened? (Hazell et al., 2022)

	Lagged Unempl. IV Without Time FE		Lagged Unempl. IV With Time FE		Tradeable Demand IV With Time FE	
	Pre-1990	Post-1990	Pre-1990	Post-1990	Pre-1990	Post-1990
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Estimates of κ					
κ	0.0278	0.0002	0.0107	0.0050	0.0109	0.0055
	(0.0025)	(0.0017)	(0.0080)	(0.0040)	(0.0062)	(0.0028)
	Panel B: Estimates of ψ					
ψ	0.449	0.009	0.198	0.090	0.422	0.332
	(0.063)	(0.025)	(0.113)	(0.057)	(0.232)	(0.157)

larger without time fixed effects: without them, κ falls from 0.028 to essentially zero, whereas with time fixed effects the decline is far more contained. This confirms the theoretical argument above — the apparent steepness of the pre-1990 Phillips curve in aggregate data was driven by the confounding of long-run inflation expectations during the Volcker disinflation, not by a genuinely steep structural relationship. Once expectations are absorbed by time fixed effects, the Phillips curve was always flat, and the Volcker disinflation is better understood as working through a reduction in $E_t\pi_{t+\infty}$ rather than through exploitation of a steep Phillips curve.

Appendix A

Log-Linearization

Log-Linearization

Log-linearization approximates a nonlinear equilibrium condition by a linear equation in percentage deviations from the deterministic steady state. Let a bar denote steady-state values and define

$$\hat{x}_t \equiv \frac{X_t - \bar{X}}{\bar{X}} \approx \ln X_t - \ln \bar{X},$$

where the approximation $\frac{X_t - \bar{X}}{\bar{X}} \approx \ln X_t - \ln \bar{X}$ follows from a first-order Taylor expansion of $\ln X_t$ around $\bar{X}$:

$$\ln X_t \approx \ln \bar{X} + \frac{1}{\bar{X}}(X_t - \bar{X}) = \ln \bar{X} + \frac{X_t - \bar{X}}{\bar{X}},$$

so that $\hat{x}_t = \ln X_t - \ln \bar{X}$ holds to first order. Equivalently, writing $X_t = \bar{X}\, e^{\hat{x}_t}$ and applying the first-order Taylor expansion of the exponential around zero,

$$e^{\hat{x}_t} \approx 1 + \hat{x}_t \quad \text{for } \hat{x}_t \text{ small},$$

gives $X_t \approx \bar{X}(1 + \hat{x}_t)$, confirming that $\hat{x}_t$ measures the percentage deviation of X_t from its steady-state value $\bar{X}$.

Step 1: Solve for the steady state. Set all shocks to zero and impose $X_t = \bar{X}$ for all t. Solve the resulting algebraic system. Steady-state ratios (e.g. $\bar{C}/\bar{Y}$) become the coefficients in the linearized equations.

Step 2: Substitute $X_t = \bar{X}(1 + \hat{x}_t)$. Replace every variable in the equation using

$$X_t = \bar{X}\, e^{\hat{x}_t} \approx \bar{X}(1 + \hat{x}_t).$$

Step 3: Expand and cancel steady-state terms. Multiply out all products. Terms involving only steady-state values reproduce the steady-state equation, which equals zero by Step 1; cancel them.

Step 4: Drop second-order terms. Discard any product of two or more hat-variables (e.g. $\hat{x}_t\hat{y}_t \approx 0$). This is the linearization approximation.

Step 5: Collect and simplify. What remains is a linear equation in $\{\hat{x}_t\}$ with steady-state ratios as weights.

Useful Rules

The following results follow directly from Steps 2–4 and can be applied mechanically.

$$X_t^{\alpha} \approx \bar{X}^{\alpha}(1 + \alpha\hat{x}_t) \tag{A.1}$$

$$X_t Y_t \approx \bar{X}\bar{Y}(1 + \hat{x}_t + \hat{y}_t) \tag{A.2}$$

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}}{\bar{Y}}(1 + \hat{x}_t - \hat{y}_t) \tag{A.3}$$

$$X_t^{\alpha} Y_t^{\beta} \approx \bar{X}^{\alpha}\bar{Y}^{\beta}(1 + \alpha\hat{x}_t + \beta\hat{y}_t) \tag{A.4}$$

Rule (A.4) is particularly useful for Cobb–Douglas expressions: in log-deviation form, $\widehat{X_t^{\alpha}Y_t^{\beta}} = \alpha\hat{x}_t + \beta\hat{y}_t$.

Alternative: Log-Then-Taylor Method

For multiplicative equations it is often faster to take logs of both sides first. Use the approximation

$$\ln X_t \;\approx\; \ln \bar{X} + \hat{x}_t,$$

which follows from a first-order Taylor expansion of $\ln X_t$ around $\bar{X}$. After logging, steady-state log-levels cancel by the same argument as Step 3, leaving a linear equation in hat-variables directly, with no need to expand products explicitly.

Example: Resource Constraint

Consider $Y_t = C_t + \delta K_t$. Substituting Step 2:

$$\bar{Y}(1 + \hat{y}_t) = \bar{C}(1 + \hat{c}_t) + \delta\bar{K}(1 + \hat{k}_t).$$

The steady-state terms satisfy $\bar{Y} = \bar{C} + \delta\bar{K}$ and cancel, leaving

$$\bar{Y}\hat{y}_t = \bar{C}\hat{c}_t + \delta\bar{K}\hat{k}_t.$$

Dividing by $\bar{Y}$:

$$\hat{y}_t = \frac{\bar{C}}{\bar{Y}}\hat{c}_t + \frac{\delta\bar{K}}{\bar{Y}}\hat{k}_t.$$

Example: Cobb–Douglas Production

Consider $Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$. Taking logs:

$$\ln Y_t = \ln A_t + \alpha \ln K_t + (1-\alpha)\ln L_t.$$

Applying $\ln X_t \approx \ln \bar{X} + \hat{x}_t$ to each term and cancelling steady-state levels (which satisfy the same equation at the steady state):

$$\hat{y}_t = \hat{a}_t + \alpha\hat{k}_t + (1-\alpha)\hat{l}_t.$$

Appendix B

Statistical Models

B.1 Vector Autoregressive (VAR) Models

Reduced-Form VAR

Let $\mathbf{y}_t$ be an $m \times 1$ vector of jointly determined variables. A pth-order Gaussian VAR is

$$\mathbf{y}_t = \mathbf{c} + \sum_{j=1}^{p} \Phi_j \mathbf{y}_{t-j} + \varepsilon_t, \qquad t = 1, 2, \ldots, T, \tag{B.1}$$

where $\{\Phi_j\}_{j=1}^{p}$ are $m \times m$ coefficient matrices, $\mathbf{c}$ is a constant vector, and $\varepsilon_t \sim iid\mathcal{N}(\mathbf{0}, \Omega)$ is the vector of reduced-form innovations with $\mathbb{E}(\varepsilon_t \varepsilon_t') = \Omega$.

Estimation

Stacking regressors as $\mathbf{X}_t = \left[1,\, \mathbf{y}_{t-1}',\, \ldots,\, \mathbf{y}_{t-p}'\right]'$ and defining $\Pi = [\mathbf{c}\ \Phi_1 \cdots \Phi_p]$, the model becomes $\mathbf{y}_t = \mathbf{X}_t \Pi' + \varepsilon_t$. The ML estimator coincides with OLS:

$$\widehat{\Pi} = \left(\sum_{t=1}^{T} \mathbf{X}_t' \mathbf{X}_t \right)^{-1} \left(\sum_{t=1}^{T} \mathbf{X}_t' \mathbf{y}_t \right) = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}' \mathbf{y}, \qquad \widehat{\Omega} = \frac{\widehat{\varepsilon}' \widehat{\varepsilon}}{T}. \tag{B.2}$$

Wold (MA) Representation

If $\mathbf{y}_t$ is covariance stationary, inverting (B.1) yields the infinite moving-average representation

$$\mathbf{y}_t = \sum_{i=0}^{\infty} \mathbf{A}_i \, \varepsilon_{t-i}, \tag{B.3}$$

where the dynamic multiplier matrices $\mathbf{A}_i$ satisfy the recursion $\mathbf{A}_i = \sum_{j=1}^{p} \Phi_j \mathbf{A}_{i-j}$ with $\mathbf{A}_0 = \mathbf{I}_m$ and $\mathbf{A}_i = \mathbf{0}$ for $i < 0$. The element (i, j) of $\mathbf{A}_s$ gives the response of variable i at horizon s to a unit innovation in variable j at t.

Structural VAR and Identification

Since reduced-form innovations ε_t are correlated, structural identification requires a mapping to orthogonal structural shocks ζ_t:

$$\varepsilon_t = \mathbf{H}\zeta_t, \qquad \mathbb{E}(\zeta_t\zeta_t') = \mathbf{I}_m, \tag{B.4}$$

so that $\Omega = \mathbf{H}\mathbf{H}'$. Because Ω has $m(m+1)/2$ free parameters while $\mathbf{H}$ has m^2, at least $m(m-1)/2$ additional restrictions must be imposed. A standard approach is *recursive identification* (Cholesky decomposition), setting $\mathbf{H} = \mathbf{P}$ where $\mathbf{P}$ is the lower-triangular matrix satisfying $\mathbf{P}\mathbf{P}' = \Omega$. This imposes a causal ordering: variable j does not respond contemporaneously to shocks in variables ordered after it.

Structural IRFs

Given $\mathbf{H}$, the structural impulse response of all variables to the jth shock at horizon s is

$$\frac{\partial \mathbf{y}_{t+s}}{\partial \zeta_{jt}} = \mathbf{A}_s\mathbf{H}\mathbf{e}_j, \tag{B.5}$$

where $\mathbf{e}_j$ is the jth column of $\mathbf{I}_m$. Plotting this as a function of s yields the impulse response function (IRF).

Recursive Identification: Christiano, Eichenbaum and Evans (1999)

In the baseline monetary policy SVAR with $\mathbf{y}_t = [y_t, \pi_t, i_t]'$, recursive identification imposes that output and inflation do not respond to the monetary policy shock ζ_t^{MP} within the period, while the central bank can react contemporaneously to both:

$$\begin{bmatrix} \varepsilon_t^y \\ \varepsilon_t^\pi \\ \varepsilon_t^i \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} \zeta_t^1 \\ \zeta_t^2 \\ \zeta_t^{MP} \end{bmatrix}. \tag{B.6}$$

This timing assumption — monetary policy's *inside lag* is short while its *outside lag* is long — is economically defensible at monthly or quarterly frequencies, though recent high-frequency evidence (Buda et al., 2025) questions whether real activity truly has zero contemporaneous response.

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Advanced Macroeconomics (Ex2)

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March 2026

Introduction

We start the exercise by writing down the model and specifying the equation characterizing it.

$$\pi_t = \beta E_t\{\pi_{t+1}\} + \kappa \tilde{y}_t, \qquad 0 < \beta < 1, \qquad \kappa > 0, \tag{1}$$

$$\tilde{y}_t = -\frac{1}{\sigma}\left(i_t - E_t\{\pi_{t+1}\} - r_t^n\right) + E_t\{\tilde{y}_{t+1}\}, \qquad \sigma > 0, \tag{2}$$

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \phi_y \tilde{y}_t^n, \qquad \rho > 0, \qquad \phi_\pi > 1, \qquad \phi_y > 0. \tag{3}$$

- **NKPC curve**: The New Keynesian Phillips Curve describes the dynamics of inflation in a forward–looking setting with nominal rigidities. In particular, current inflation depends on expected future inflation and on the level of economic activity, typically proxied by the output gap. The forward–looking component reflects firms' price–setting behaviour under Calvo pricing: since firms anticipate future marginal costs and demand conditions when adjusting prices, higher expected inflation at time $t+1$ leads to higher inflation already at time t.

  The second component captures the effect of real economic activity on inflation. When the output gap increases, demand for goods rises relative to the economy's productive capacity, which pushes marginal costs upward. Firms responding to higher marginal costs adjust their prices, generating inflationary pressure. Therefore, positive deviations of output from its natural level translate into higher current inflation.

- **IS curve**: The first equation, namely the IS curve, describes the intertemporal dynamics of the output gap and is derived from the Euler equation of the representative household. It states that the output gap at time t depends positively on the expected output gap at time $t+1$ and negatively on the deviation of the ex–ante real interest rate from the natural rate, i.e. $i_t - E_t\{\pi_{t+1}\} - r_t^n$. The forward–looking component reflects optimal consumption smoothing: if agents expect higher economic activity in the future, they increase consumption today, which raises current demand and therefore the output gap.

  The second term captures the role of monetary policy through the real interest rate gap. When the nominal interest rate is set above the level consistent with the natural rate, the ex–ante real interest rate increases and households have an incentive to postpone consumption, reducing aggregate demand and the output gap. The parameter σ represents the inverse of the intertemporal elasticity of substitution (equivalently the coefficient of relative risk aversion under CRRA preferences) and governs how strongly output responds to changes in the real interest rate: a larger σ implies weaker intertemporal substitution and therefore a smaller response of the output gap to real interest rate differentials.

- **Taylor Rule**: The last equation represents the Taylor rule describing how the central bank sets the nominal interest rate in response to macroeconomic conditions. In particular, the policy rate reacts positively to both inflation and the output gap. When the output gap increases and the economy becomes overheated, the central bank raises the nominal interest rate in order to dampen aggregate demand.

  Similarly, when inflation rises above its target level, the central bank increases the policy rate to reduce consumption and investment incentives. By raising the nominal interest rate, the ex–ante real interest rate increases, encouraging households to postpone consumption and firms to delay investment, thereby reducing demand pressures and stabilizing inflation and output around their equilibrium levels.

Exercise 1

In this first part, we explore the New Keynesian model and analyse how changes in ϕ_π affect the behaviour of the main macroeconomic variables and their dynamics over time, with particular focus on the resulting impulse response functions. The parameter ϕ_π governs the responsiveness of monetary policy to inflation deviations and therefore plays a key role in shaping the stability and transmission of shocks in the economy.

We begin from a baseline calibration where $\beta = 0.98$, implying a high degree of patience and therefore a strong weight placed on expected future inflation in agents' decisions. The risk aversion parameter is set to $\sigma = 2$, which corresponds to a commonly used calibration in the literature and implies a moderate degree of intertemporal substitution. Finally, we set the responsiveness of the Taylor rule to output gap deviations to $\phi_y = 0.2$, capturing a relatively mild reaction of monetary policy to fluctuations in real economic activity. Finally, we vary the parameter ϕ_π and first consider the case in which it is set equal to 0.5. In this scenario the model exhibits instability, as monetary policy does not respond sufficiently to increases in inflation. When the policy response is too weak, an increase in inflation is not counteracted by a sufficiently strong rise in the nominal interest rate, leading to a decline in the real interest rate and further stimulating demand. This mechanism amplifies inflationary pressures and may generate a self-reinforcing inflationary spiral.

$$k(\phi_\pi - 1) + \phi_y(1 - \beta) > 0$$

For this reason, monetary policy must satisfy the so-called Taylor principle, according to which the central bank should respond more than one-for-one to inflation. By setting $\phi_\pi > 1$, an increase in inflation leads to a sufficiently strong rise in the nominal interest rate, which raises the real interest rate and dampens aggregate demand, thereby stabilizing inflation and the overall dynamics of the system.

We now recall the relations involving the shock to our economy from both output and inflation size:

$$y_t = E_t\{y_{t+1}\} - \frac{1}{\sigma}\left(r_t - E_t\{\pi_{t+1}\}\right) - e_{z,t}$$

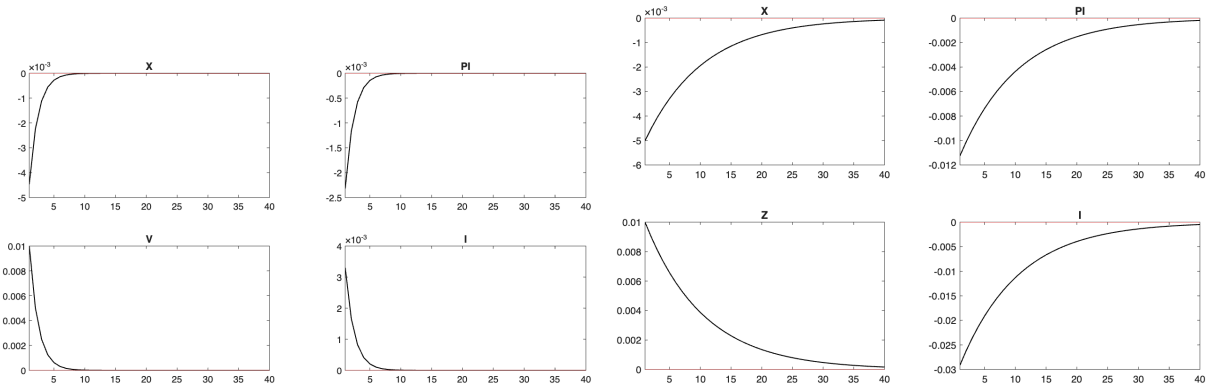
$$\pi_t = \beta E_t\{\pi_{t+1}\} + \frac{(\sigma+\eta)(1-\beta\omega)(1-\omega)}{\omega} y_t$$

$$r_t = \phi_\pi \pi_t + \phi_y y_t + e_{v,t}$$

$$e_{z,t} = \rho_z e_{z,t-1} + \varepsilon_{z,t}$$

$$e_{v,t} = \rho_v e_{v,t-1} + \varepsilon_{v,t}$$

In particular, assuming that $\beta = 0.98$, therefore assuming a high discount factor, points to persistent memory of past dynamics. Now, assuming $\phi_\pi = 0.5$, we get that the Taylor rule is not bigger than zero. In particular, in this case, the CB is not very responsive to inflation and, therefore, cannot keep inflation under control. Now, considering $\phi_\pi = 2.5$ the Taylor rule is indeed met and therefore our system can provide the impulse response.



(a) Orthogonalized shock to e_v, inflation (b) Orthogonalized shock to e_z ouput

Figure 1: Impulse response functions of the New Keynesian model to structural shocks.

Here we report the impulse responses of the variables to orthogonalized shocks, namely a monetary policy shock and a technology shock. From the first set of graphs we observe that the shock to v_t decays geometrically over time due to the autoregressive structure of the process, $v_t = \rho_v v_{t-1} + e_{v,t}$, with $\rho_v = 0.5$ and innovation variance given by the calibration (stderr = 0.01). As expected for an AR(1) process with coefficient smaller than one, the effect of the shock gradually vanishes as the system converges back to its steady state.

The monetary shock affects inflation negatively because it effectively acts as a contractionary monetary policy disturbance entering the Taylor rule. A positive innovation in v_t raises the nominal interest rate, which increases the ex–ante real interest rate. This reduces current consumption and investment through the intertemporal substitution channel, lowering aggregate demand and therefore the output gap. The decline in economic activity reduces marginal costs for firms, which in turn leads to a fall in inflation through the New Keynesian Phillips curve. Consequently, both output and inflation decrease on impact before gradually returning to equilibrium.

Now, looking at the second figure, we are imposing a positive output shock that decays more slowly

since we have $\rho_z = 0.9$. On impact, it generates an effect that is negative corresponding to a decrease in output gap therefore with consequential negative impact on inflation and investments.

When ω is small, stickiness is lower, assuming a positive production shock. Lower marginal costs imply that firms want to lower prices, which increases demand and, consequently, output. When $\omega \uparrow$, we have more price stickiness, therefore it will be more difficult for firms to adjust their price, hence this will have a much more persistent effect on the negative output gap, which will be maintained for a while.

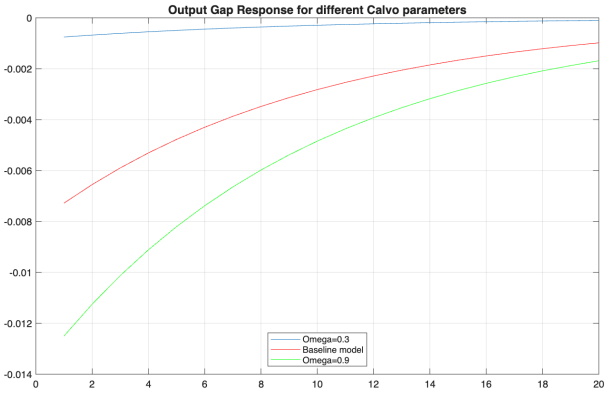


Figure 2: Caption

Exercise 2

Listing 1: New NK MODEL - BASELINE VERSION WITH 6 TYPOS

```
1
2  var y pi r ez ev ;
3  varexo epsz epsv;
4
5  parameters betta sigma omega eta phipi phiy rhoz rhov stdev_ez stdev_ev;   %
       --> phiy missing
6
7  betta=0.98; %Discount factor
8  omega=0.75;  %Calvo parameter price stickiness
9  phipi = 1.5;   %Taylor rule: feedback on expected inflation                 -->
       "; MISSING!"
10 phiy = 0.2;   %Taylor rule: feedback on output gap
11 eta=1;       %Labor supply elasticity
12 sigma=2;     %CRRA parameter in consumption
13 rhoz=0.8;  %persistence of technology shock                                 %
       --> rhoz >1 super persistent
14 rhov=0.5;      %persistence of monetary policy shock
15 stdev_ez=0.01; %std.dev. of tech shock
16 stdev_ev=0.01; %std.dev. of MP shock
17
18 model(linear);
19 y = y(+1)-(1/sigma)*(r-pi(+1))-ez;
20 pi = betta*pi(+1)+((sigma+eta)*(1-betta*omega)*(1-omega)/omega)*y;          %
       --> "; MISSING!"
21 r = phipi*pi+phiy*y+ev;                                                    %
       --> r(+1) replace dwith r
22 ez = rhoz*ez(-1)+epsz;
23 ev = rhov*ev(-1)+epsv;
24 end;
25
26
27 steady;
28
29 shocks;
30 var epsz; stderr stdev_ez;                                                 %
       was written ez instead of epsz
31 var epsv; stderr stdev_ev;
32 end;
33
34 check;
35
36 stoch_simul(order=1, irf=20);
```

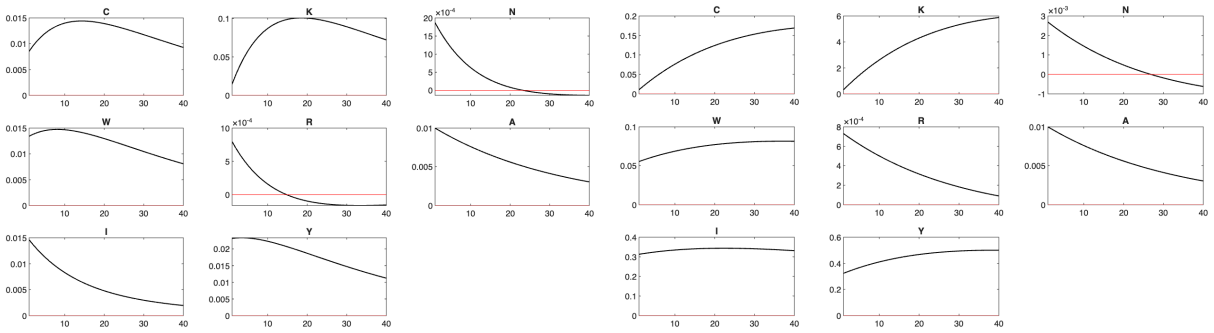
Exercise 3 and 4

In this exercise we are assuming a different functional form for the $f(K, L)$ in particular we assume a Cobb-Douglas with *constant elasticity of substitution* so that the ratio of substitution between capital and labor stays constant to any level of those two variables, this is going to have implication in our model. In particular we can recognize that as $\eta \to 1$ we converge to the cobb douglas function:

$$Y = A_t K^{\alpha} L^{1-\alpha}$$

This can be illustrated by comparing the CES specification for $\eta = 1.7$ with the near–Cobb–Douglas case $\eta = 1.00001$. The latter is used because the CES production function is only well defined at $\eta = 1$ as a limit, while setting $\eta = 1$ exactly would generate terms such as $\eta/(\eta - 1)$ and $1/(\eta - 1)$, which are not numerically admissible. Hence, $\eta = 1.00001$ provides a convenient approximation to the standard Cobb–Douglas case.

Starting from the baseline model, we first compute the steady state under the near–Cobb–Douglas specification and use it as an initial reference point for the CES economy. We then consider a higher value of η, namely $\eta = 1.7$, which implies a greater elasticity of substitution between capital and labor. Since this represents a non-negligible departure from the benchmark case, we use steady-state values obtained from an intermediate calibration as initial values in Dynare and then re-estimate the model under the new parameterization. In this way, we can study how moving away from the Cobb–Douglas benchmark affects the dynamics of the model and the impulse responses to technology shocks.



(a) Impulse responses with $\eta = 1.00001$.

(b) Impulse responses with $\eta = 1.7$.

Figure 3: Comparison of impulse responses to a technology shock under alternative CES elasticities of substitution.

For $\eta > 1$, capital and labor become good substitutes in the production process. In this case, when a positive technology shock hits the economy, firms can adjust the relative use of capital and labor more easily in response to changes in marginal productivity. This greater flexibility allows firms to reallocate inputs more efficiently, amplifying the immediate response of output and factor prices. As a consequence, the effects of the shock tend to be larger in magnitude and can persist for longer, since firms are less constrained in adapting their production decisions to the new technological environment.

Finally we observe the effects of difference in magnitude and persistent of the effects under the two different functions.

Exercise 5

In the last exercise we finish by considering the New Keynesian framework and assessing an effect of a technology shock. Assuming the shock to follow an AR(1) with no persistent, namely $a_t = \varepsilon_t \implies$

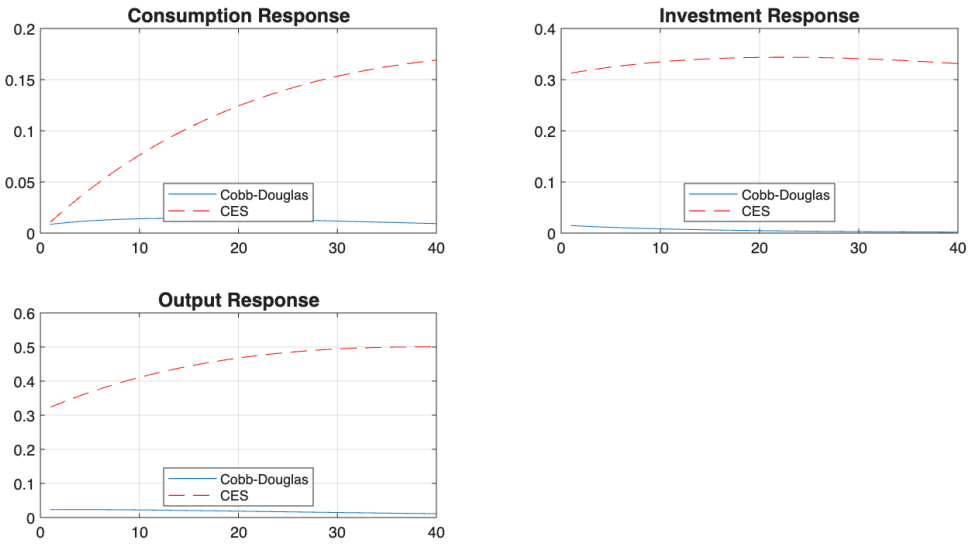


Figure 4

$E_t[a_{t+1}] = 0$. Once we have that we can retrieve the New Keynesian equations derived and consider the following two initial guesses:

$$\pi_t = ba_t,$$
$$\tilde{y}_t = ca_t.$$

We conjecture linear policy functions because the model has been log-linearized around the steady state. In a first-order approximation, equilibrium variables can be expressed as linear functions of the state variables. Since the only state variable in the system is the technology shock a_t, inflation and the output gap can be written as proportional to a_t. The coefficients b and c are then determined by substituting these conjectures into the equilibrium conditions of the model.

We proceed by substituting the terms in the equations so:

$$ca_t = -\frac{1}{\sigma}(\rho + \phi_\pi kc + \phi_y ca_t + \phi_y \psi_y a_t - \rho + \sigma\psi_y a_t), \quad E[\pi_{t+1}] = E[\tilde{y}_{t+1}] = 0$$
$$ba_t = kca_t \implies b = kc$$

therefore we now find c as follows:

$$c(\sigma + k\phi_\pi\phi_y) = -\psi_y(\phi_y + \sigma)$$
$$c = -\frac{\psi_y(\phi_y + \sigma)}{(\sigma + k\phi_\pi\phi_y)}$$

Now we can plug it in $\tilde{y}_t = -\frac{\psi_y(\phi_y+\sigma)}{(\sigma+k\phi_\pi\phi_y)}a_t$ and then directly obtain b as:

$$b = \frac{k\psi_y(\phi_y + \sigma)}{(\sigma + k\phi_\pi\phi_y)}$$

therefore $\pi_t = -\frac{k\psi_y(\phi_y+\sigma)}{(\sigma+k\phi_\pi\phi_y)}a_t$. We observe that both inflation and the output gap respond negatively to a positive technology shock. This occurs because an improvement in technology raises the economy's potential level of output. As potential output increases, actual production does not immediately adjust by the same magnitude, which leads to a decline in the output gap.

The reduction in the output gap implies that firms face lower marginal costs. As production becomes more efficient due to the technological improvement, firms can produce the same level of output with fewer inputs. Lower marginal costs translate into weaker price pressures, which in turn leads to a decline in inflation.

As given by the text we know that $y_t^n = \psi_y a_t$ and $\tilde{y} = y_t - y_t^n$ therefore we can find that:

$$\begin{aligned} y_t &= \underbrace{\psi_y a_t}_{y_t^n} - \frac{\psi_y(\phi_y + \sigma)}{(\sigma + k\phi_\pi + \phi_y)} a_t \\ &= \left(1 - \frac{(\phi_y + \sigma)}{(\sigma + k\phi_\pi + \phi_y)}\right) \psi_y a_t \\ &= \psi_y a_t \underbrace{\frac{k\phi_\pi}{\sigma + k\phi_\pi + \phi_y}}_{<1} \end{aligned}$$

A positive technology shock increases productivity and therefore raises the level of real output, as firms can produce more efficiently due to lower marginal costs. If both natural output and actual output adjusted by the same magnitude, the output gap would remain unchanged.

However, in the presence of price stickiness, actual output cannot adjust as quickly as natural output. Firms would like to reduce prices immediately in response to the lower marginal costs generated by the technology improvement, but nominal rigidities prevent them from doing so instantaneously. As a consequence, output under flexible prices (natural output) increases more rapidly than output in the sticky-price equilibrium.

Since the output gap is defined as the deviation of actual output from natural output, this slower adjustment implies a temporary decline in the output gap following the technology shock.

$$\begin{aligned} n_t &= \frac{1}{1-\alpha}(y_t - a_t) \\ &\Leftrightarrow n_t = \frac{1}{1-\alpha}\left(\frac{\kappa\phi_\pi}{\sigma + \kappa\phi_\pi + \phi_y}\psi_{ya} a_t - a_t\right) \\ &\Leftrightarrow n_t = \frac{1}{1-\alpha}\left(\frac{\psi_{ya}\kappa\phi_\pi}{\sigma + \kappa\phi_\pi + \phi_y} - 1\right) a_t \\ &\Leftrightarrow n_t = \frac{1}{1-\alpha}\left(\frac{\psi_{ya}\kappa\phi_\pi}{\sigma + \kappa\phi_\pi + \phi_y} - \frac{\sigma + \kappa\phi_\pi + \phi_y}{\sigma + \kappa\phi_\pi + \phi_y}\right) a_t \\ &\Leftrightarrow n_t = \frac{(\psi_{ya} - 1)\kappa\phi_\pi - \sigma - \phi_y}{(1-\alpha)(\sigma + \kappa\phi_\pi + \phi_y)}\, a_t. \end{aligned}$$

Here we see that $\psi_y > 1$; however, we need to know the size of the parameters to claim whether the response to unemployment to such a shock is positive or negative. In particular:

$$k\phi_\pi(\psi_y - 1) > \sigma + \phi_y$$

Generally, we assume $\sigma \approx 2$ and $\phi_y < 1$ to be low, while monetary authorities are generally highly responsive to inflation with $\phi_\pi > 1$. However, this is not enough to determine the direction of n_t after a technology shock. Assuming, however, that those parameters are actually following the inequality expressed above, ϕ_π ceteris paribus, plays an important role in increasing unemployment since $mc \downarrow$, but to do that we need labour, therefore $n \uparrow$.

1 RBC Model

1.1 Dynare Implementation of the Baseline RBC Model

The following discusses the Dynare code for the baseline RBC model block by block.

Variable and Parameter Declarations.

```
var C K N W R A I Y;
varexo eA;
parameters betta varphi delta alpha rhoA;
```

The `var` command declares the eight endogenous variables of the model: consumption (C), capital (K), labour (N), real wage (W), rental rate of capital (R), total factor productivity (A), investment (I), and output (Y). The `varexo` command declares the single exogenous shock ε_A, which drives fluctuations in TFP. Finally, `parameters` lists the five structural parameters, calibrated as follows:

$\beta = 0.98$	(household discount factor)
$\varphi = 2$	(inverse Frisch elasticity of labour supply)
$\delta = 0.05$	(quarterly capital depreciation rate)
$\alpha = 0.33$	(capital share in production)
$\rho_A = 0.97$	(persistence of the TFP process)

Note that Dynare does not accept `beta` as a parameter name since it is a reserved keyword, hence the spelling `betta`.

Model Block.

```
model;
1/C = betta*((1/C(+1))*(R(+1)+1-delta));
N^varphi = W/C;
R = alpha*A*(K(-1))^(alpha-1)*N^(1-alpha);
W = (1-alpha)*A*(K(-1))^alpha*N^(-alpha);
K = I+(1-delta)*K(-1);
Y = C+I;
Y = A*(K(-1))^alpha*N^(1-alpha);
log(A) = rhoA*log(A(-1))+eA;
end;
```

The `model` block contains the eight equilibrium conditions of the model. In Dynare, `(+1)` denotes $\mathbb{E}_t[\cdot]$ (expected next period) and `(-1)` denotes a one-period lag. The eight equations correspond to:

1. **Euler equation** (household intertemporal optimality):

$$\frac{1}{C_t} = \beta\,\mathbb{E}_t\left[\frac{1}{C_{t+1}}\left(R_{t+1} + 1 - \delta\right)\right]. \tag{4}$$

   The household equates the marginal utility of consuming today to the discounted expected marginal utility of the return on saving.

2. **Labour supply** (household intratemporal optimality):

$$N_t^{\varphi} = \frac{W_t}{C_t}. \tag{5}$$

The marginal disutility of labour equals the consumption-deflated real wage (the utility value of the wage).

3. **Capital demand** (firm FOC for capital):

$$R_t = \alpha A_t K_{t-1}^{\alpha-1} N_t^{1-\alpha}. \tag{6}$$

The rental rate equals the marginal product of capital. Note that capital is predetermined: firms choose K_{t-1} before period t begins, hence the `K(-1)` notation.

4. **Labour demand** (firm FOC for labour):

$$W_t = (1-\alpha) A_t K_{t-1}^{\alpha} N_t^{-\alpha}. \tag{7}$$

The real wage equals the marginal product of labour.

5. **Capital accumulation**:

$$K_t = I_t + (1-\delta) K_{t-1}. \tag{8}$$

Next period's capital stock equals investment plus undepreciated existing capital.

6. **Resource constraint**:

$$Y_t = C_t + I_t. \tag{9}$$

Output is split between consumption and investment (closed economy, no government).

7. **Production function** (Cobb-Douglas):

$$Y_t = A_t K_{t-1}^{\alpha} N_t^{1-\alpha}. \tag{10}$$

8. **TFP process** (AR(1) in logs):

$$\ln A_t = \rho_A \ln A_{t-1} + \varepsilon_t^A, \qquad \varepsilon_t^A \sim \mathcal{N}(0, \sigma_A^2). \tag{11}$$

Technology follows a stationary first-order autoregressive process. Writing the process in logs ensures $A_t > 0$ at all times.

Steady-State Block.

```
steady_state_model;
A = 1;
R = (1/betta)+delta-1;
N = (((1-alpha)/(R/alpha-delta))*R/alpha)^(1/(1+varphi));
K = (R/(A*alpha))^(1/(alpha-1))*N;
I = delta*K;
Y = A*K^alpha*N^(1-alpha);
C = Y-I;
W = C*N^(varphi);
```

```
end;
```

The `steady_state_model` block provides the analytical steady state derived earlier. Dynare uses these values as the point around which the model is log-linearised. The solution follows the sequential order established above:

1. Normalise $A = 1$.
2. Solve R from the Euler equation (depends on parameters only).
3. Solve N from the combined FOCs and labor supply condition.
4. Solve K from the capital FOC given R and N.
5. Recover I, Y, C, W sequentially from the remaining equations.

Providing an analytical steady state (rather than letting Dynare solve it numerically) improves speed and numerical accuracy, and is strongly recommended.

Steady State and Determinacy Check.

```
steady;
check;
```

The `steady` command instructs Dynare to compute (or verify) the steady state. The `check` command then computes the eigenvalues of the linearised system and verifies the Blanchard-Kahn condition. In the RBC model we have one non-predetermined variable (C_t) and one predetermined variable (K_t), so we require exactly one eigenvalue outside the unit circle for a unique determinate equilibrium.

Shock Specification.

```
shocks;
var eA; stderr 0.01;
end;
```

The `shocks` block specifies the standard deviation of the TFP shock: $\sigma_A = 0.01$, i.e. a one-standard-deviation shock raises $\ln A_t$ by one percentage point.

Simulation.

```
stoch_simul(order=1, irf=40);
```

Finally, `stoch_simul` solves and simulates the model. The option `order=1` requests a first-order (linear) approximation around the steady state — equivalent to the log-linearisation performed by hand. The option `irf=40` instructs Dynare to compute and plot impulse response functions over 40 periods following a one-standard-deviation shock to ε_t^A.

Advanced Macroeconomics (Ex3)

Filippo Nardoni

March 2026

Introduction

Exercise 1

In the first exercise, we have downloaded the data and used the used a matlab code do transform variables. In particular we have demeaned and took the difference of particular variables.

In the model we specify the following equations:

$$x_t = E_t[x_{t+1}] - \frac{1}{\sigma}\left(r_t - E_t[\pi_{t+1}] + e_{z,t}\right) \tag{1}$$

$$\pi_t = \beta E_t[\pi_{t+1}] + \left((\sigma+\eta)(1-\beta\omega)\frac{1-\omega}{\omega}\right)x_t + e_{w,t} \tag{2}$$

$$r_t = \phi_\pi \pi_t + \phi_x x_t + e_{v,t} \tag{3}$$

$$e_{z,t} = \rho_z e_{z,t-1} + \varepsilon_{z,t} \tag{4}$$

$$e_{v,t} = \rho_v e_{v,t-1} + \varepsilon_{v,t} \tag{5}$$

$$e_{w,t} = \rho_w e_{w,t-1} + \varepsilon_{w,t} \tag{6}$$

Here we have introduced e_w which is the cost push shock. After running the program, we let our MATLAB code identify what functions:

```
// Now let the estimation begin!
// These variables must be in the datafile (which they are in our case)
varobs x_obs, pi_obs, r_obs;

estimated_params;
// Parameter Name, Prior_shape, Mean, Standard error,
// Parameter 3 (if applicable), Parameter 4 (if applicable)
// Prior_shape: BETA_PDF, GAMMA_PDF, NORMAL_PDF, INV_GAMMA_PDF

// Structural parameters
omega,   BETA_PDF,   0.7, 0.05;
eta,     NORMAL_PDF, 1,   0.2;
phipi,   NORMAL_PDF, 1.5, 0.1;
phix,    BETA_PDF,   0.2, 0.1;

// Shock parameters
```

```
// Ensure that they are not above 1.
rhoz,    BETA_PDF,    0.9, 0.05;
rhov,    BETA_PDF,    0.9, 0.05;
rhow,    BETA_PDF,    0.9, 0.05;

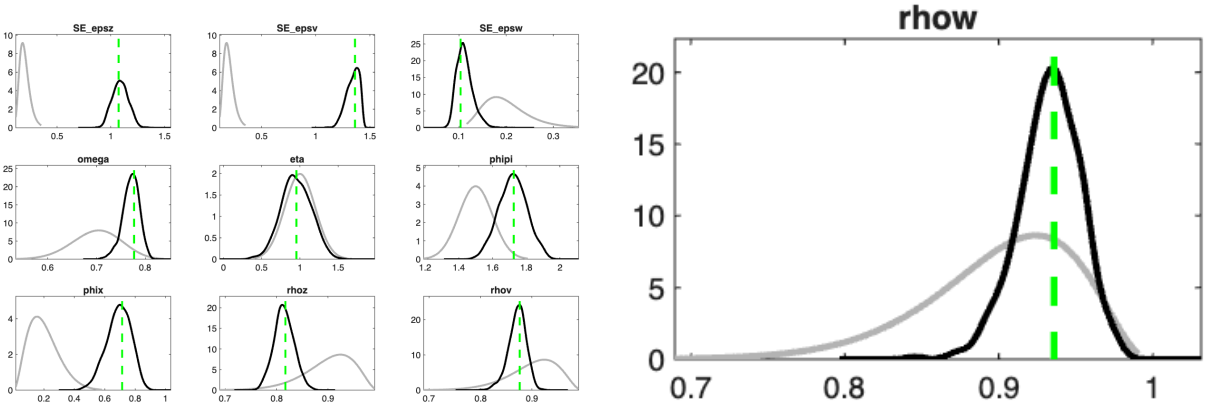
// See slide 33 from lecture 6 for this choice of distribution.
stderr epsz, INV_GAMMA2_PDF, 0.2, 0.05;
stderr epsv, INV_GAMMA2_PDF, 0.2, 0.05;
stderr epsw, INV_GAMMA2_PDF, 0.2, 0.05;

end;

estimation(datafile=usdatafile, mode_check, mode_compute=1,
           mh_replic=5000, mh_nblocks=2,
           mh_jscale=0.65, mh_drop=0.2) x pi r;
```

Basically, we ensure that our variables are the data, and we start the estimation by imposing prior distributions on the structural and shock parameters. In particular, we are willing to find:

$$p(\theta|x) = p(x|\theta)p(\theta)$$



(a) Priors vs posteriors (set 1)

(b) Priors vs posteriors (set 2)

Figure 1: Comparison between prior and posterior distributions of the estimated parameters.

In Figure 1, we observe a clear discrepancy between prior and posterior distributions for several parameters. This indicates that the information introduced through the likelihood significantly updates the prior beliefs, leading to posterior distributions that are more tightly centered and data-driven. In particular, for the shock processes, the posterior distributions of the standard deviations suggest values exceeding one for e_v and e_z, highlighting a higher level of volatility than initially assumed in the priors. Moreover, the persistence parameters of the shocks are estimated to be slightly lower than the prior mean of 0.9, indicating less persistence in the data relative to prior beliefs. Finally, the estimated value of ϕ_x is larger than its prior mean, suggesting a stronger response of the policy rule to the output gap than initially imposed. Recalling the ω Calvo parameter, we have this to be close to one reported in the literature, 0.77, which corresponds to a price duration of $1/(1-0.77) \approx 4.35$, which corresponds to 4 quarters or slightly more than one year. The Blanchard-Kahn condition is met since $\phi_\pi > 1$ therefore the model runs.

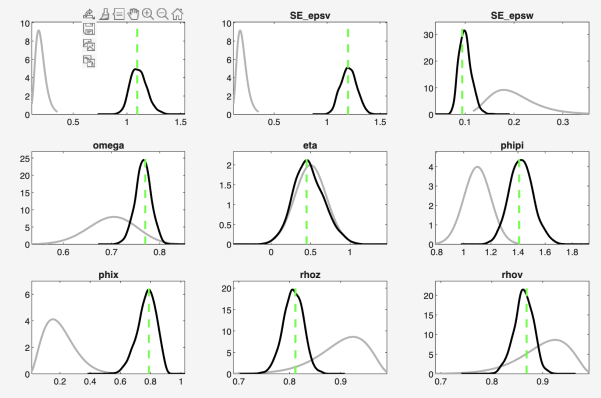
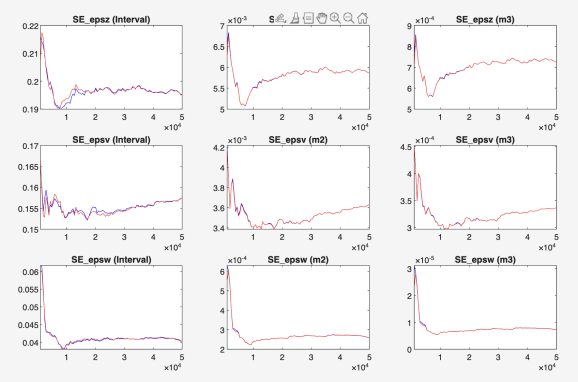
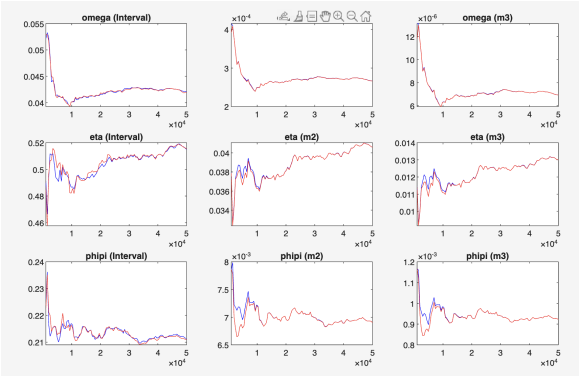


Figure 2: Enter Caption

We now consider an alternative prior specification, setting the mean of η to 0.5 and ϕ_π to 1.1. From the figure, we observe that the data provide little information about ω, as the posterior distribution closely overlaps with the prior. This suggests that the likelihood is relatively uninformative for this parameter, so that $p(\theta \mid x) \approx p(\theta)$. In contrast, the monetary policy parameter ϕ_π is tightly concentrated around $\phi_\pi \approx 1.45$ in the posterior distribution. This indicates that the data are informative for this parameter, leading to a posterior that departs from the prior and reflects the information contained in the likelihood.



(a) First image



(b) Second image

Figure 3: Enter overall caption

We see that increasing $M = 5,000 \rightarrow M = 50,000$ does not lead to any improved estimation. So basically results are identical.

0.1 Including Covid Period

In this exercise, we re-run the entire code, including the code period. We modify one parameter, namely *stderr epsv,* $INV_GAMMA2_PDF, 0.2, 0.10,$ *changing*$0.05 \rightarrow 0.10$*, since the error relates to hessian inverts therefore we modify a parameter to let this work.*

Exercise 2

We have run the code and we see that the program has elapsed in 1.463883 *seconds which is very fast.*

We then proceed and find that the standard deviation of sigma_a estimated is 0.0027*. We then find that the standard deviation of output is* 0.0131 *which is equal to the one set targeted in the data. this allows us to consider that with one parameter and one moment we can match the data exactly.*

After allowing for two parameter estimation we find that the estimated technology shock is 0.7795 *which is different from the one produced in the data, while the sigma_a is* 0.0059

Advanced Macroeconomics (Ex4)

Filippo Nardoni

May 2026

Introduction

In this exercise we study the estimation of impulse response functions (IRFs) via two complementary approaches—Local Projections (LP) and Structural Vector Autoregressions (SVAR)—and compare their finite-sample behaviour.

Local Projections, introduced by Jorda2005, estimate the IRF horizon-by-horizon by running a separate OLS regression for each variable $i = 1, \dots, N$ and each horizon $h = 0, 1, \dots, H$, yielding a total of $N \times (H + 1)$ regressions. Each regression takes the form

$$y_{i,t+h} \;=\; \alpha_{i,h} \;+\; \beta_{i,h}\,\varepsilon_t \;+\; \boldsymbol{\gamma}_{i,h}'\mathbf{X}_t \;+\; u_{i,t+h}, \qquad h = 0, 1, \dots, H, \quad i = 1, \dots, N, \tag{1}$$

where ε_t is the structural shock of interest, $\mathbf{X}_t = (y_{1,t-1}, \dots, y_{N,t-p})'$ collects p lags of all variables as controls, and $\beta_{i,h}$ is the parameter of interest—the response of variable i to the shock at horizon h. Tracing $\{\beta_{i,h}\}_{h=0}^{H}$ over the horizon directly delivers the IRF without requiring the inversion of any moving-average representation.

A key practical issue is that the error term $u_{i,t+h}$ follows an MA(h) process under the null for $h \geq 1$, inducing serial correlation that must be corrected via heteroskedasticity- and autocorrelation-consistent (HAC) standard errors, typically Newey–West with at least h lags.

Exercise 1

Part (a)

The analysis is carried out using the script `c02_show_lplinchol.m`. Before running any estimation, it is essential to ensure that the working directory is set correctly: all auxiliary functions must reside in the same folder—or on the MATLAB search path—so that they can be located and called without errors at runtime.

The core estimation is performed by the function `estimateLP.m`, invoked as follows:

```
LP = estimateLP(data, h, p, c_case, [], alpha);
```

The arguments are defined as follows.

- `data`: the matrix of observables on which the local projections are estimated.
- `h`: the maximum horizon H, determining how many step-ahead regressions are run.
- `p`: the lag order, i.e. the number of lags of $\mathbf{X}_t$ included as controls in equation (1).
- `c_case`: an integer flag selecting the deterministic specification—typically 1 for a constant only, or 2 for a constant and a linear trend.

- `[]`: a placeholder for an optional argument (passed empty here).
- `alpha`: the significance level $\alpha \in (0, 1)$ used to construct the pointwise confidence bands around the estimated IRFs.

Part (b)

We can store the results and plot them using some figures therefore we can see visually what follows.

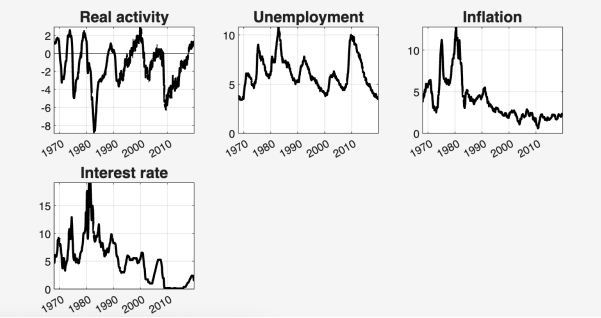


Figure 1: Series plotted over time

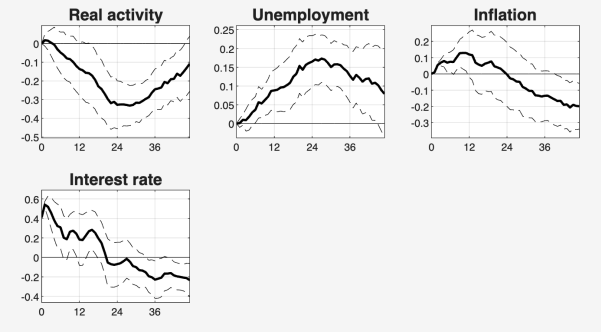


Figure 2: Local projections

We observe that the local projections we obtain are similar to those derived from the SVAR. However, we also notice—and later confirm—that the point estimates tend to be less smooth, which is a well-known characteristic of local projection methods.

Part (c)

In this section, we first work with the non-detrended data and then incorporate a trend into the local projection code. The results we obtain are presented as follows.:

The solution requires two modifications to the main LP script.

The first change replaces the variable list at line 11 with

```
vars = {'IP', 'Unemp', 'CPI', 'FFR'};
```

so that the system now comprises Industrial Production, Unemployment, the Consumer Price Index, and the Federal Funds Rate—a standard four-variable specification for the empirical analysis of monetary policy transmission. The previous variable list is simply commented out.

The second change sets `c_case` $= 2$ in line 10, replacing the constant-only specification (`c_case` $= 1$) with a constant-plus-linear-trend specification. This is warranted when the variables enter the system in levels—as is the case for the price level—since omitting a trend from a trending series would induce a misspecification bias in all $N \times (H + 1)$ local-projection regressions.

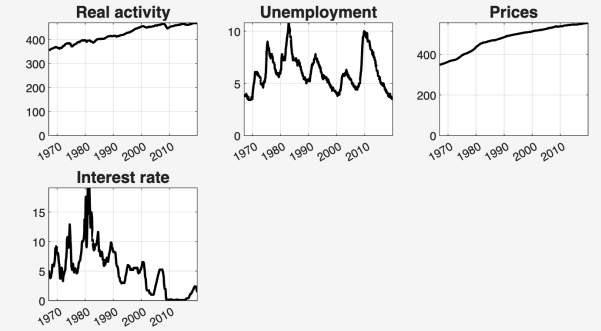


Figure 3: Time series

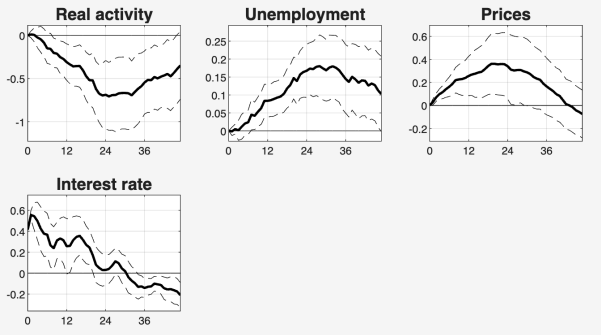


Figure 4: Local projection

Regarding the economic interpretation, it is important to distinguish between the *price level* (CPI, p_t) and *inflation* ($\pi_t = \Delta p_t$), as the two objects yield related but distinct impulse responses. Denoting by $\widehat{\mathcal{I}}_{p,h}$ and $\widehat{\mathcal{I}}_{\pi,h}$ the estimated IRFs of the price level and inflation at horizon h, respectively, it follows by the telescoping identity that

$$\widehat{\mathcal{I}}_{p,h} \;=\; \sum_{j=0}^{h} \widehat{\mathcal{I}}_{\pi,j}, \tag{2}$$

so the price-level IRF is the cumulative sum of the inflation IRF. Consequently, $\widehat{\mathcal{I}}_{p,h}$ attains its maximum at the horizon h^* where the inflation IRF changes sign, i.e. $\widehat{\mathcal{I}}_{\pi,h^*} \approx 0$ and $\widehat{\mathcal{I}}_{\pi,h} < 0$ for $h > h^*$. The observation that the price-level response peaks at approximately $h^* = 24$ months is therefore fully consistent with—and indeed implied by—the inflation response turning negative around the same horizon.

Exercise 2

Figure 5 displays the estimated impulse response functions to a contractionary monetary policy shock for four variables—real activity (industrial production), unemployment, inflation, and the interest rate—over a horizon of $H = 48$ months, with $p = 12$ lags and 90% pointwise confidence bands. The solid blue line reports the VAR (Cholesky) estimates, while the dashed red line reports the LP (Cholesky) estimates.

Qualitative agreement at short horizons. The two methods produce broadly consistent point estimates over the first 10–15 months across all four panels. Real activity falls on impact, unemployment rises gradually, inflation exhibits a modest initial increase before reverting, and the interest rate jumps on impact and decays monotonically—all of which are consistent with the conventional transmission mechanism of a monetary tightening.

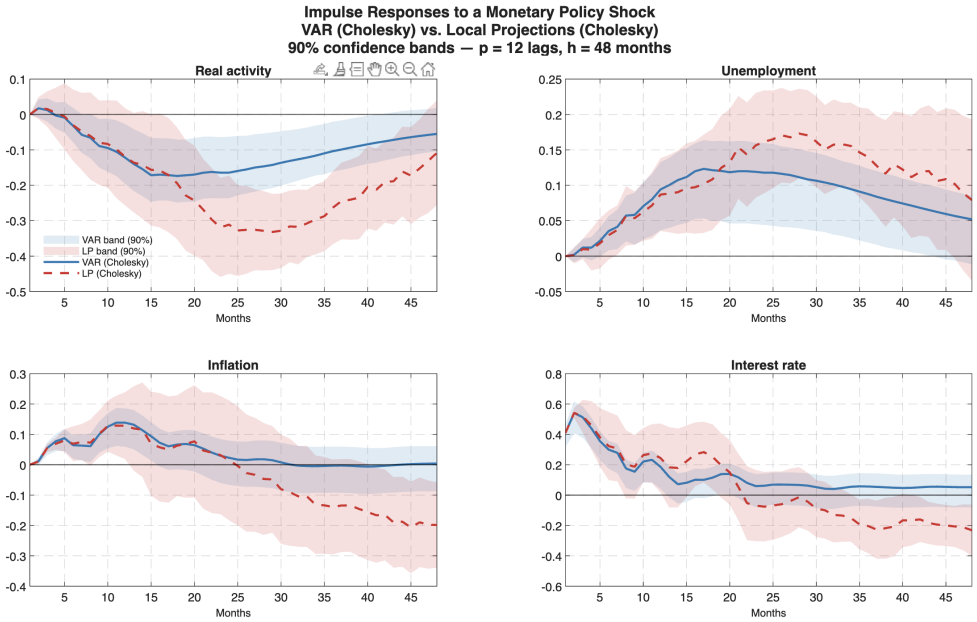


Figure 5: Comparison of LP and SVAR IRF

Divergence at longer horizons. Beyond $h \approx 20$ months, the two estimators diverge noticeably. For real activity, the VAR response reverts toward zero by month 48, whereas the LP response remains substantially more negative and shows no signs of mean-reversion. A similar pattern appears for unemployment, where the LP estimate continues to rise at long horizons while the VAR estimate peaks and declines. For inflation, the LP produces a pronounced and persistent negative response after month 20—consistent with the price-level argument in equation (2)—while the VAR estimate hovers near zero. For the interest rate, the LP oscillates and turns negative after month 25, while the VAR decays smoothly.

Efficiency loss of LP at long horizons. A salient feature of the figure is that the LP confidence bands (pink shading) widen considerably relative to the VAR bands (blue shading) as h increases. This reflects the fundamental efficiency trade-off between the two approaches. The VAR imposes the cross-equation restrictions implied by the VAR(p) structure—namely that the IRF at horizon h is a deterministic function of the same Np^2 reduced-form parameters estimated once—yielding tighter inference. LP, by contrast, runs a separate OLS for each (i, h) pair without exploiting any cross-horizon restrictions, and the accumulated estimation uncertainty compounds as h grows, resulting in wider and increasingly uninformative bands at long horizons.

Bias–variance trade-off. The comparison therefore illustrates the classical bias–variance trade-off between the two estimators. The VAR is more *efficient* under correct specification, but its IRFs inherit any misspecification in the lag structure or the shock identification. LP is *robust* to such misspecification—it only requires the shock ε_t to be correctly identified at each horizon—at the cost of higher variance, particularly in long-horizon inference.

Advanced Macroeconomics: Business Cycles

Take-home Exam — Summer 2025

Exam no. XXXX

June 12, 2026

Abstract

AI Assistance Statement. Claude (Anthropic) was used as a structuring and verification assistant during the preparation of this exam. All mathematical derivations, proofs, and economic interpretations are the author's own work; the tool was used to verify logical steps and assist with formatting. All MATLAB code is reported in the Appendix.

Part I — The Phillips Curve

Question 1

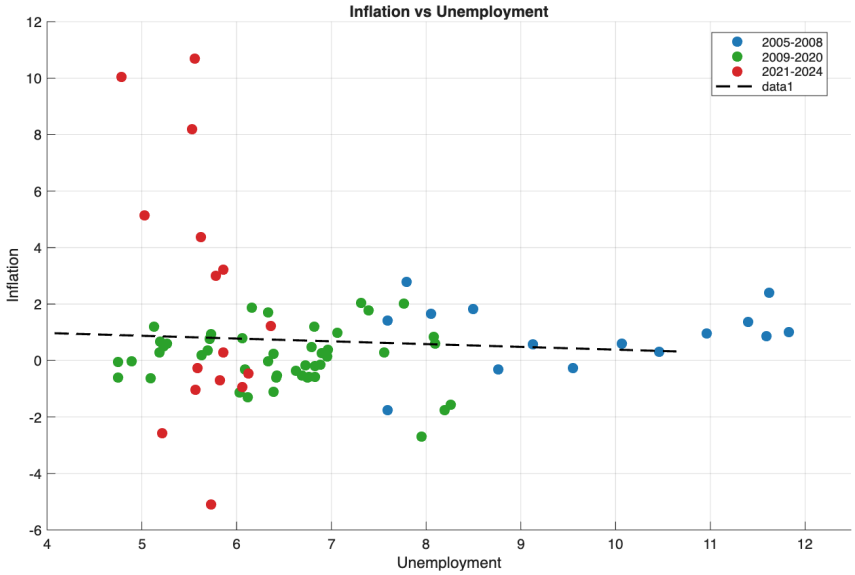


Figure 1: Quarterly PPI inflation against the unemployment rate, Germany 2005Q1–2024Q4. Points are coloured by sub-period; the dashed line is the full-sample OLS fit.

Figure 1 plots quarterly PPI inflation against the unemployment rate. We split the sample into three sub-periods—before the global financial crisis (2005–2008), the post-crisis low-inflation era (2009–2020), and the post-Covid period (2021–2024)—since the monetary-policy regime and institutional setting plausibly differ across them.

A textbook Phillips Curve predicts a downward-sloping relation, yet the unconditional data display, if anything, only a very weak negative association: the full-sample fitted line is nearly flat, and within the pre-2021 sub-periods the slope is close to zero or even mildly positive. The apparent negative tilt is driven almost entirely by the post-Covid cluster, where inflation spiked sharply while unemployment remained low—an episode the European Central Bank countered with a series of policy-rate increases. Crucially, this co-movement reflects cost-push (supply)

shocks rather than a demand-driven trade-off, so the raw correlation conflates movements along the Phillips Curve with shifts of it. The scatter therefore illustrates why the *unconditional* relation cannot be read as the structural slope κ, motivating the identification strategies pursued in the remainder of this part.

Setup. Start from the NKPC, writing $\hat{u}_t \equiv u_t - u_t^n$:

$$\pi_t = \beta\, \mathbb{E}_t\, \pi_{t+1} - \kappa\, \hat{u}_t + v_t. \tag{1}$$

Forward iteration. Substituting (1) into itself k times and using the law of iterated expectations,

$$\pi_t = \beta^k\, \mathbb{E}_t\, \pi_{t+k} + \sum_{i=0}^{k-1} \beta^i\, \mathbb{E}_t\big(-\kappa\, \hat{u}_{t+i} + v_{t+i}\big). \tag{2}$$

Imposing the transversality (no-bubble) condition $\lim_{k\to\infty} \beta^k\, \mathbb{E}_t\, \pi_{t+k} = 0$, valid since $\beta \in (0,1]$, and using $\hat{u}_{t+i} = u_{t+i} - u_{t+i}^n$,

$$\pi_t = -\kappa \sum_{i=0}^{\infty} \beta^i\, \mathbb{E}_t\, u_{t+i} + \underbrace{\sum_{i=0}^{\infty} \beta^i\, \mathbb{E}_t\big(\kappa\, u_{t+i}^n + v_{t+i}\big)}_{\omega_t}. \tag{3}$$

Long-run vs. gap. Add and subtract the long-run forecast $\mathbb{E}_t\, u_{t+\infty}$ inside the unemployment sum:

$$-\kappa \sum_{i=0}^{\infty} \beta^i\, \mathbb{E}_t\, u_{t+i} = \underbrace{-\kappa \sum_{i=0}^{\infty} \beta^i\; \mathbb{E}_t\, u_{t+\infty}}_{(a)} \; \underbrace{-\kappa \sum_{i=0}^{\infty} \beta^i \big(\mathbb{E}_t\, u_{t+i} - \mathbb{E}_t\, u_{t+\infty}\big)}_{(b)}. \tag{4}$$

Term (a). Since $\sum_{i=0}^{\infty} \beta^i = \frac{1}{1-\beta}$,

$$(a) = -\frac{\kappa}{1-\beta}\, \mathbb{E}_t\, u_{t+\infty} \;\equiv\; \mathbb{E}_t\, \pi_{t+\infty}. \tag{5}$$

Term (b). By assumption the unemployment rate follows a mean-μ AR(1), $u_t - \mu = \rho_u(u_{t-1} - \mu) + \eta_t$, hence $\mathbb{E}_t\, u_{t+i} = \mu + \rho_u^i(u_t - \mu)$ and $\mathbb{E}_t\, u_{t+\infty} = \mu$. Thus $\tilde{u}_t \equiv u_t - \mathbb{E}_t\, u_{t+\infty} = u_t - \mu$ is the demeaned rate, which inherits $\mathbb{E}_t\, \tilde{u}_{t+i} = \rho_u^i \tilde{u}_t$. Therefore

$$(b) = -\kappa \sum_{i=0}^{\infty} \beta^i\, \mathbb{E}_t\, \tilde{u}_{t+i} = -\kappa\, \tilde{u}_t \sum_{i=0}^{\infty} (\beta\rho_u)^i = -\frac{\kappa}{1-\beta\rho_u}\, \tilde{u}_t = -\psi\, \tilde{u}_t, \tag{6}$$

convergent because $\beta\rho_u < 1$.

Result. Collecting (a), (b) and ω_t,

$$\boxed{\pi_t = -\psi\, \tilde{u}_t + \mathbb{E}_t\, \pi_{t+\infty} + \omega_t, \quad \psi = \frac{\kappa}{1-\beta\rho_u}, \quad \tilde{u}_t = u_t - \mathbb{E}_t\, u_{t+\infty},} \tag{7}$$

with $\mathbb{E}_t\, \pi_{t+\infty} = -\frac{\kappa}{1-\beta}\, \mathbb{E}_t\, u_{t+\infty}$ and $\omega_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \big(\kappa u_{t+j}^n + v_{t+j}\big)$. ■

Question 2

Since $E_t\pi_{t+\infty}$ is unobserved, regressing π_t on $\tilde{u}_t$ alone places it in the error term, $e_t = E_t\pi_{t+\infty}+\omega_t$. The OLS estimator then satisfies

$$\hat{\psi} \xrightarrow{p} \psi - \frac{\mathrm{Cov}\big(\tilde{u}_t, \mathbb{E}_t\, \pi_{t+\infty}\big)}{\mathrm{Var}(\tilde{u}_t)}, \tag{8}$$

so consistency requires the orthogonality condition $\mathrm{Cov}(\tilde{u}_t, \mathbb{E}_t\, \pi_{t+\infty}) = 0$. This is not about the regressors being "perfectly" (un)correlated, but about the single regressor being uncorrelated with the omitted long-run expectations term; any nonzero covariance induces omitted-variable bias.

Question 3

We now assume a constant natural rate, $u_t^n = u^n$ for all t, and that the unemployment gap $\hat{u}_t \equiv u_t - u^n$ follows the AR(1)

$$\hat{u}_t = \rho_u \hat{u}_{t-1} + \varepsilon_t, \qquad 0 < \rho_u < 1, \quad \varepsilon_t \text{ i.i.d. mean zero.} \tag{9}$$

Forecasting the gap. Iterating (9) forward i periods by repeated substitution,

$$\hat{u}_{t+i} = \rho_u^i\, \hat{u}_t + \sum_{j=0}^{i-1} \rho_u^j\, \varepsilon_{t+i-j}. \tag{10}$$

Taking expectations conditional on time-t information and using $\mathbb{E}_t\, \varepsilon_{t+s} = 0$ for all $s \geq 1$ (the future shocks are unpredictable),

$$\mathbb{E}_t\, \hat{u}_{t+i} = \rho_u^i\, \hat{u}_t. \tag{11}$$

Since $0 < \rho_u < 1$, letting $i \to \infty$ gives $\rho_u^i \to 0$, so

$$\mathbb{E}_t\, \hat{u}_{t+\infty} = \lim_{i\to\infty} \rho_u^i\, \hat{u}_t = 0 : \tag{12}$$

the gap is mean-reverting and is expected to close in the long run.

Implication for the rate. With a constant natural rate, $u_t = u^n + \hat{u}_t$, hence

$$\boxed{\mathbb{E}_t\, u_{t+\infty} = u^n + \mathbb{E}_t\, \hat{u}_{t+\infty} = u^n.} \tag{13}$$

The expected long-run level of unemployment is therefore the constant natural rate u^n: it is deterministic and independent of current conditions. As a consequence $\tilde{u}_t = u_t - \mathbb{E}_t\, u_{t+\infty} = u_t - u^n = \hat{u}_t$ and $\mathbb{E}_t\, \pi_{t+\infty} = -\frac{\kappa}{1-\beta} u^n$ is constant, which is what collapses equation (2) into equation (4).

Question 4

With a constant natural rate $u_t^n = u^n$, write the NKPC in gap form and combine it with the AR(1) for the gap:

$$\pi_t = \beta\, \mathbb{E}_t\, \pi_{t+1} - \kappa\, \hat{u}_t + v_t, \tag{14}$$

$$\hat{u}_t = \rho_u \hat{u}_{t-1} + \varepsilon_t, \qquad 0 < \rho_u < 1, \quad \varepsilon_t \text{ i.i.d. mean zero.} \tag{15}$$

Forward iteration. Substituting (14) into itself k times and using the law of iterated expectations,

$$\pi_t = \beta^k \, \mathbb{E}_t \, \pi_{t+k} - \kappa \sum_{j=0}^{k-1} \beta^j \, \mathbb{E}_t \, \hat{u}_{t+j} + \sum_{j=0}^{k-1} \beta^j \, \mathbb{E}_t \, v_{t+j}. \tag{16}$$

The no-bubble (transversality) condition $\lim_{k\to\infty} \beta^k \, \mathbb{E}_t \, \pi_{t+k} = 0$, valid since $\beta \in (0, 1]$, removes the first term as $k \to \infty$.

Forecasting the gap. From (15), future innovations are unpredictable ($\mathbb{E}_t \, \varepsilon_{t+s} = 0$ for $s \geq 1$), so $\mathbb{E}_t \, \hat{u}_{t+j} = \rho_u^j \hat{u}_t$. Hence the unemployment term is a convergent geometric series ($\beta\rho_u < 1$):

$$-\kappa \sum_{j=0}^{\infty} \beta^j \, \mathbb{E}_t \, \hat{u}_{t+j} = -\kappa \, \hat{u}_t \sum_{j=0}^{\infty} (\beta\rho_u)^j = -\frac{\kappa}{1 - \beta\rho_u} \, \hat{u}_t = -\psi \, \hat{u}_t. \tag{17}$$

Result. Collecting terms and defining $\hat{\omega}_t \equiv \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j v_{t+j}$,

$$\boxed{\pi_t = -\psi \, \hat{u}_t + \hat{\omega}_t, \qquad \psi = \frac{\kappa}{1 - \beta\rho_u}.} \tag{18}$$

■

Question 5

Equation (4), $\pi_t = -\psi\hat{u}_t + \hat{\omega}_t$, is estimated by OLS. Writing the gap as $\hat{u}_t = u_t - u^n$ gives the estimable form

$$\pi_t = \underbrace{\psi u^n}_{\text{intercept}} - \psi \, u_t + \hat{\omega}_t \;\equiv\; a + b \, u_t + \hat{\omega}_t, \qquad b = -\psi. \tag{19}$$

Which assumption lets us use u_t in place of $\hat{u}_t$. The assumption that the natural rate is constant, $u_t^n = u^n$. Then $\hat{u}_t$ and u_t differ only by the constant u^n, so replacing the gap with the observed rate changes only the intercept: the slope is unaffected and the constant absorbs ψu^n (this is why an intercept must be included). The slope coefficient therefore identifies $-\psi$, and consistency further requires the orthogonality condition $\mathrm{Cov}(u_t, \hat{\omega}_t) = 0$, i.e. the unemployment rate must be uncorrelated with the discounted sum of future inflation shocks.

Table 1: OLS regression of PPI inflation on the unemployment rate (robust standard errors).

	Coefficient	Robust std. error	t	p
Constant	1.366378	1.202398	1.136	0.2593
Unemployment rate	−0.098209	0.146002	−0.673	0.5032
Observations $n = 79$; regressors $k = 2$.				
Residual std. error $\hat{\sigma} = 2.339626$; SSE = 421.486345.				

Estimate. The slope is $\hat{b} = -0.0982$, hence

$$\hat{\psi} = -\hat{b} = 0.0982 \quad (\text{robust se} = 0.1460). \tag{20}$$

The point estimate has the sign predicted by the NKPC (a positive unemployment gap lowers inflation), but with robust standard errors it is statistically insignificant ($t = -0.67$, $p = 0.50$): the estimate is indistinguishable from zero.

Comment. The result mirrors the flat scatter of Question 1. Two forces work against a clean estimate. First, identification: $\hat{\omega}_t$ is omitted and, if supply shocks v_t move both inflation and unemployment, $\text{Cov}(u_t, \hat{\omega}_t) \neq 0$, biasing $\hat{\psi}$. Second, the post-Covid cost-push episode dominates the variation, pushing the unconditional slope toward (and past) zero. The wide robust standard error confirms that a direct Phillips-curve regression is uninformative here, motivating the identification-based approach of Barnichon and Mesters (2021) in Questions 6–7.

Question 6

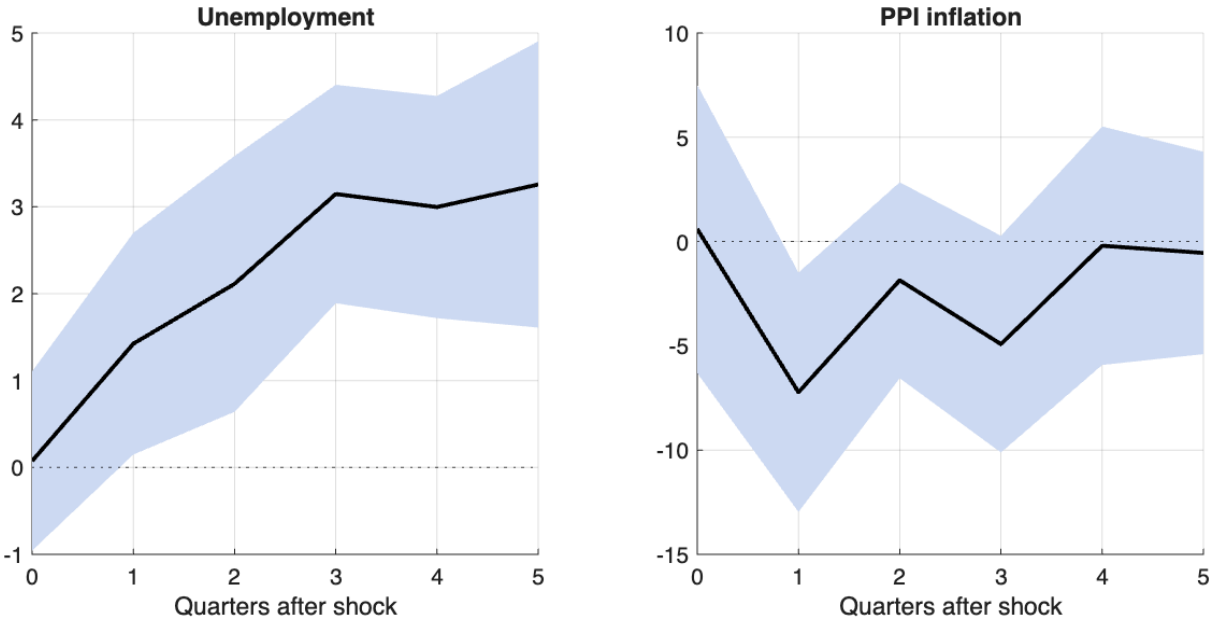


Figure 2: IRF though Local Projection

Part II

Question 1

Introduction to the Problem

We consider the intermediate goods firm j with the extended production function

$$Y_t(j) = K_t(j)^{\alpha} N_t(j)^{1-\alpha} \left(K_t^g\right)^{\mu}, \qquad \mu \geq 0, \tag{21}$$

where the public capital stock K_t^g is taken as given by the firm (it is external to the firm's choice set and the firm does not pay for it). The firm minimizes total real cost, taking the real wage W_t and the real rental rate of capital R_t^k as given. We (i) derive the first-order conditions and show that the capital–labor ratio is identical to that in Galí *et al.* (2007), and (ii) derive the real marginal cost as a function of W_t, R_t^k and K_t^g only.

Proof

(i) Cost minimization and the capital–labor ratio. The firm solves

$$\min_{K_t(j),\, N_t(j)} \; R_t^k K_t(j) + W_t N_t(j) \qquad \text{s.t.} \qquad K_t(j)^{\alpha} N_t(j)^{1-\alpha} \left(K_t^g\right)^{\mu} \geq Y_t(j). \tag{22}$$

The associated Lagrangian is

$$\mathcal{L} = R_t^k K_t(j) + W_t N_t(j) + \lambda_t \left[Y_t(j) - K_t(j)^{\alpha} N_t(j)^{1-\alpha} \left(K_t^g\right)^{\mu} \right], \tag{23}$$

where $\lambda_t \geq 0$ is the shadow value of one additional unit of output, i.e. the real marginal cost. The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial K_t(j)}: \quad R_t^k = \lambda_t \, \alpha \, K_t(j)^{\alpha-1} N_t(j)^{1-\alpha} \, (K_t^g)^{\mu}, \tag{24}$$

$$\frac{\partial \mathcal{L}}{\partial N_t(j)}: \quad W_t = \lambda_t \, (1-\alpha) \, K_t(j)^{\alpha} N_t(j)^{-\alpha} \, (K_t^g)^{\mu}. \tag{25}$$

Dividing (24) by (25), the terms λ_t and $(K_t^g)^{\mu}$ cancel, and we obtain

$$\frac{R_t^k}{W_t} = \frac{\alpha}{1-\alpha} \frac{N_t(j)}{K_t(j)} \qquad \Longleftrightarrow \qquad \boxed{\frac{K_t(j)}{N_t(j)} = \frac{\alpha}{1-\alpha} \frac{W_t}{R_t^k}} \tag{26}$$

exactly as in Galí *et al.* (2007). Intuitively, public capital enters (21) as a Hicks-neutral productivity shifter: it scales the marginal products of both private inputs proportionally, and therefore leaves the optimal input mix unaffected. □

(ii) Real marginal cost. Define $\phi_t \equiv (K_t^g)^{\mu}$ and $\kappa_t \equiv \frac{\alpha}{1-\alpha} \frac{W_t}{R_t^k}$. We derive the conditional factor demands by substituting (26) into the production constraint (which binds at the optimum). Using $K_t(j) = \kappa_t N_t(j)$,

$$Y_t(j) = (\kappa_t N_t(j))^{\alpha} \, N_t(j)^{1-\alpha} \phi_t = \kappa_t^{\alpha} \, N_t(j) \, \phi_t \qquad \Longrightarrow \qquad N_t(j) = \frac{Y_t(j)}{\phi_t} \, \kappa_t^{-\alpha}, \tag{27}$$

and, symmetrically, substituting $N_t(j) = \kappa_t^{-1} K_t(j)$,

$$K_t(j) = \frac{Y_t(j)}{\phi_t} \, \kappa_t^{1-\alpha}. \tag{28}$$

Both demands are linear in $Y_t(j)$, reflecting constant returns to scale in the private inputs. The cost function is therefore linear in output:

$$C_t(j) = R_t^k K_t(j) + W_t N_t(j) = \left[R_t^k \, \kappa_t^{1-\alpha} + W_t \, \kappa_t^{-\alpha} \right] \frac{Y_t(j)}{\phi_t}, \tag{29}$$

and marginal cost is

$$MC_t = \frac{\partial C_t(j)}{\partial Y_t(j)} = \frac{1}{\phi_t} \left[R_t^k \, \kappa_t^{1-\alpha} + W_t \, \kappa_t^{-\alpha} \right]. \tag{30}$$

Substituting $\kappa_t = \frac{\alpha}{1-\alpha} \frac{W_t}{R_t^k}$ into (30), both terms reduce to multiples of $\left(R_t^k\right)^{\alpha} W_t^{1-\alpha}$:

$$MC_t = \frac{\left(R_t^k\right)^{\alpha} W_t^{1-\alpha}}{\phi_t} \left[\left(\frac{1-\alpha}{\alpha} \right)^{\alpha-1} + \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} \right]^{-1 \cdot (-1)}, \tag{31}$$

where the bracket simplifies as

$$\left(\frac{1-\alpha}{\alpha} \right)^{\alpha} \left[\frac{\alpha}{1-\alpha} + 1 \right] = \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} \frac{1}{1-\alpha} = \alpha^{-\alpha} (1-\alpha)^{-(1-\alpha)} \equiv \Psi. \tag{32}$$

Hence

$$\boxed{MC_t = \Psi \left(R_t^k \right)^{\alpha} (W_t)^{1-\alpha} \, (K_t^g)^{-\mu}, \qquad \Psi \equiv \alpha^{-\alpha} (1-\alpha)^{-(1-\alpha)}.} \tag{33}$$

For $\mu = 0$, (33) collapses to the expression in Galí *et al.* (2007). Equivalently, solving (25) for λ_t after imposing (26) delivers (33) directly, confirming $\lambda_t = MC_t$ (envelope theorem). ■

Conclusions

The optimal capital–labor ratio is unchanged relative to Galí *et al.* (2007), since public capital raises the marginal products of capital and labor proportionally and thus cancels from the relative factor demand. Real marginal cost,

$$MC_t = \Psi \left(R_t^k\right)^{\alpha} (W_t)^{1-\alpha} (K_t^g)^{-\mu},$$

is common to all firms (it depends only on aggregate factor prices and K_t^g, not on $Y_t(j)$, by constant returns in private inputs) and is *decreasing* in the public capital stock: $\partial MC_t/\partial K_t^g = -\mu\, MC_t/K_t^g < 0$ for $\mu > 0$. A larger stock of public capital allows any output level to be produced with fewer private inputs, lowering real marginal cost. Through the New Keynesian Phillips curve this introduces a supply-side channel of fiscal policy: government spending accumulates into K_t^g via equation (6) and exerts downward pressure on inflation, partially offsetting the standard demand-side effect.

Question 2

Introduction to the Problem

We modify the reduced dynamical system of Appendix C in Galí *et al.* (2007) — equations (C.1), (C.3), (C.6), (C.7), (37) and (21) in the unknowns $(n_t, c_t, \pi_t, k_t, b_t, g_t)$ — to accommodate productive government spending. The only change in the primitives is the production function, whose log-linearization now reads

$$y_t = \alpha k_t + (1-\alpha) n_t + \mu k_t^g, \tag{35$'$}$$

where k_t^g denotes the log deviation of the public capital stock from its steady state. The guiding principle is therefore: *an equation of the system is modified if and only if its derivation uses the production function to substitute out y_t.* We add one variable (k_t^g) and one equation (the log-linearized law of motion (6)), obtaining a system of seven equations in $\mathbf{x}_t = (n_t, c_t, \pi_t, k_t, b_t, g_t, k_t^g)'$.

Derivations

Equation (C.1) — modified. From the market clearing condition (36), $y_t = \gamma_c c_t + \gamma_i i_t + g_t$ with $\gamma_i = 1 - \gamma_c - \gamma_g \equiv 1 - \tilde{\gamma}_c$, investment is

$$i_t = \frac{1}{1-\tilde{\gamma}_c}\left[\, y_t - \gamma_c c_t - g_t \,\right]. \tag{34}$$

Substituting (34) into the linearized capital accumulation equation (25), $k_{t+1} = (1-\delta)k_t + \delta i_t$, and replacing y_t using (35$'$) yields

$$k_{t+1} = \left(1 - \delta + \frac{\delta\alpha}{1-\tilde{\gamma}_c}\right) k_t + \frac{\delta(1-\alpha)}{1-\tilde{\gamma}_c}\, n_t - \frac{\delta\gamma_c}{1-\tilde{\gamma}_c}\, c_t - \frac{\delta}{1-\tilde{\gamma}_c}\, g_t + \frac{\delta\mu}{1-\tilde{\gamma}_c}\, k_t^g. \tag{C.1$'$}$$

Note that the steady-state shares retain their expressions from the paper: $K/Y = \alpha/[(\rho+\delta)\mu^p]$ follows from $R^k = \alpha Y/(\mu^p K)$ and $R^k = \rho + \delta$, both of which are unaffected by μ (public capital is Hicks-neutral), so γ_i and γ_c are as in Appendix B.

Intermediate expression (C.2) — unchanged. The price markup is $\hat{\mu}_t^p = mpn_t - w_t$. Since $(K_t^g)^{\mu}$ enters the technology multiplicatively, $MPN_t = (1-\alpha)Y_t/N_t$ retains its form, so $mpn_t = y_t - n_t$; combined with the wage schedule (30), $w_t = c_t + \varphi n_t$,

$$\hat{\mu}_t^p = y_t - c_t - (1+\varphi) n_t, \tag{C.2}$$

exactly as in the paper: (C.2) is written in terms of y_t and its derivation never invokes the production function. The effect of k_t^g is implicit in y_t and materializes only when y_t is substituted out.

Equation (C.3) — modified. Substituting (C.2) into the inflation equation (32) and then replacing y_t using (35′),

$$\begin{aligned}\pi_t &= \beta E_t\{\pi_{t+1}\} + \lambda_p \left[c_t - y_t + (1+\varphi) n_t\right] \\ &= \beta E_t\{\pi_{t+1}\} + \lambda_p c_t - \alpha\lambda_p k_t + (\alpha+\varphi)\lambda_p n_t - \mu\lambda_p\, k_t^g. \end{aligned} \tag{C.3′}$$

A higher public capital stock lowers real marginal cost and hence inflation: productive government spending acquires a supply-side channel in the Phillips curve.

Equation (C.6) — unchanged. The aggregate Euler equation is derived from (30), (27), (28) and (26), combined with the interest rate rule (18), the fiscal rule (20) and the spending process (21). The production function is never used to substitute out y_t, and the coefficients $\tilde{\sigma}$, ϵ_n, ϵ_t depend only on $(\gamma_c, \mu^p, \alpha, \lambda, \varphi)$ through the steady-state labor share $WN/Y = (1-\alpha)/\mu^p$, whose expression is unaffected by μ. Hence (C.6) carries over with no modification.

Equation (C.7) — modified. Two ingredients of the derivation must be checked. First, combining (C.2) with (34), $\hat{\mu}_t^p = (y_t - k_t) - r_t^k$, gives

$$r_t^k = y_t - k_t - \hat{\mu}_t^p = c_t - k_t + (1+\varphi) n_t, \tag{35}$$

where y_t *cancels*: the rental rate expression is unchanged. Second, the investment-to-capital ratio does change. Subtracting k_t from (34) and substituting (35′),

$$i_t - k_t = \frac{1}{1-\tilde{\gamma}_c}\left[(1-\alpha) n_t - \gamma_c c_t - g_t - (1-\tilde{\gamma}_c-\alpha) k_t + \mu k_t^g\right]. \tag{36}$$

Substituting (36), dated t and $t+1$, into the linearized Tobin's q/investment block and multiplying through by $(1-\tilde{\gamma}_c)$ as in the paper, the fourth equation becomes

$$\begin{aligned}&(1-\alpha) n_t - \gamma_c c_t - (1-\tilde{\gamma}_c-\alpha) k_t + (1-\tilde{\gamma}_c)\eta\varphi_\pi \pi_t + \mu k_t^g \\ &\quad = \left[\omega(1+\varphi) + \beta(1-\alpha)\right] E_t\{n_{t+1}\} + (\omega - \beta\gamma_c) E_t\{c_{t+1}\} - \left[\omega + \beta(1-\tilde{\gamma}_c-\alpha)\right] k_{t+1} \\ &\qquad + (1-\tilde{\gamma}_c)\eta E_t\{\pi_{t+1}\} + (1-\beta\rho_g) g_t + \beta\mu\, E_t\{k_{t+1}^g\}, \end{aligned} \tag{C.7′}$$

with $\omega \equiv \eta[1-\beta(1-\delta)](1-\tilde{\gamma}_c) > 0$ as in the paper. Since k_{t+1}^g is predetermined, $E_t\{k_{t+1}^g\} = k_{t+1}^g$, and one may equivalently substitute the law of motion (C.8) below to write the last term as $\beta\mu(1-\delta_g) k_t^g + \beta\mu\delta_g\, g_t$, in which case the coefficient on g_t becomes $(1 - \beta\rho_g + \beta\mu\delta_g)$.

New equation — log-linearized law of motion of public capital. Linearizing equation (6), $K_{t+1}^g = (1-\delta_g) K_t^g + G_t$, around the steady state (where $G = \delta_g K^g$) yields

$$k_{t+1}^g = (1-\delta_g)\, k_t^g + \delta_g\, g_t. \tag{C.8}$$

Equations (37) and (21) — unchanged. The debt accumulation equation and the autoregressive process for government spending involve no production-side variables and are unaffected.

Conclusions

The extended system consists of the seven equations (C.1′), (C.3′), (C.6), (C.7′), (37), (21) and (C.8) in the seven unknowns $\mathbf{x}_t = (n_t, c_t, \pi_t, k_t, b_t, g_t, k_t^g)'$, and can be cast in the matrix form $A\,E_t\{\mathbf{x}_{t+1}\} = B\,\mathbf{x}_t + \varepsilon_t$, where A and B are the paper's matrices augmented by (i) one column collecting the k^g terms in (C.1′), (C.3′) and (C.7′), and (ii) one row for the purely backward-looking equation (C.8). Setting $\mu = 0$ switches off every modification and k_t^g decouples from the rest of the system, recovering the original six-equation model — a useful consistency check. For $\mu > 0$, government spending operates through an additional supply-side channel: it accumulates into public capital via (C.8), lowers real marginal cost, and exerts downward pressure on inflation through (C.3′), while raising the productive capacity of the economy in (C.1′) and (C.7′).

Question 3

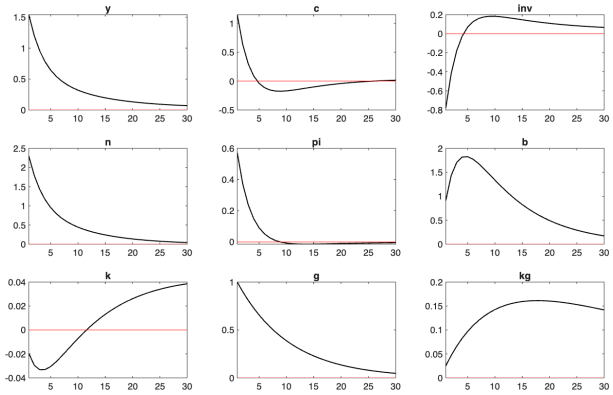


Figure 3

A MATLAB Code

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